

REPORT PRESENTATION ON
COST EFFECTIVE HOUSING TECHNIQUES

Mr.BANDI UPENDRA

(B.TECH-JNTUA, M.TECH-JNTUA)

MOBILE: 9100194630

FACULTY OF CIVIL ENGINEERING

Housing Scenario

Introduction

Housing is one of the basic human needs along with food, clothing and education. Mankind has been evolving different kinds of shelter with the changing civilizations and time. Human ingenuity has led to construction of structures as tall as 110 stores in recent times. High rise buildings dot the skyline of the large cities all over the world. Notwithstanding the rapid stride in the field of building technology, providing shelter to the teeming millions at affordable cost remains a distant reality in most of the developing countries across the globe. A World Bank study conducted in 1975 showed that 55% of the households in Mexico City, 35% in Bangkok, 68% in Nairobi, 47% in Bogota, 64% in Ahmedabad, and 63% in Madras were unable to afford the cheapest dwellings available in the open markets of those cities. Position in these cities has since further deteriorated.

Like any other developing country, India too is presently passing through a phase of acute housing shortage. As per National Buildings Organisation (NBO) estimates, in 1991, there was a shortage of 31 million dwelling units in the country out of which shortage of 10.4 million units existed in the urban sector and the remaining 20.6 million units in the rural sector. The backlog of housing is expected to mount up to 41 million by the turn of century. The picture is abysmally dismal at the lower end of the economic ladder. According to an UN estimate, over 33 per cent of the population in developing countries is houseless. A recent study undertaken by the UNCHS reveals that over 100 million people live in a state of absolute homelessness, while in excess of one billion people are forced by circumstances to reside in desperately inadequate housing conditions which threaten their health, security, safety and dignity. The Economic Survey of India for 1991-92 indicates that in 1981, there were about 0.6 million absolutely homeless households in the country and the number has further grown in the intervening period. As per one estimate, in 1993, there were three million absolutely homeless households in the country.

It is significant that against the annual population growth rate of 2% in the country, the number of residential buildings has increased at an annual rate of only 1.5 percent. One Australia or Nepal, equivalent to a population of 17 million people, is being added to the Indian population annually. With the uncontrolled population explosion and rapid pace of urbanisation, the problem of providing shelter to the poor is bound to accentuate in coming years. According to the Federation of Indian Chambers of Commerce and Industry (FICCI), keeping in view the existing housing crisis in the country, the present addition of 2.5 million units a year is hopelessly inadequate. The norm of five dwellings per annum per thousand heads of population set down for developing countries has not been fulfilled even up to the halfway mark, except in a few states like Andhra Pradesh, Assam, Tamil Nadu and Kerala. Some

110 million housing units are required to be constructed annually for meeting the housing requirements of the country.

The gravity of the housing situation in India's urban as well as rural sector is discussed at length in the following pages.

Status of Urban Housing

Urban housing in India presents a very complicated picture. On the one hand, there is semblance of modern housing on the lines of Western development in the urban agglomerations, whereas one also happens to perceive the gloomy aspect of housing, on the other. Most of the towns and cities in the country are characterised by the large-scale migration of poverty-stricken people from villages and smaller towns, overcrowding, encroachment of available open space by squatter resulting in environmental degradation and crime galore. With rapid urbanisation and population, the number of slums and shanty towns have been proliferating in the major Indian cities at an alarming rate. Over 46 million people live in slums and squatter settlements in the cities. Most of these people cannot hope to afford the cheapest house available in the urban market. And the spiralling costs of land and building materials have further aggravated the problem of housing by widening the gulf between the demand and availability of housing units at affordable cost.

During 1971 to 1991 there has been the fastest rate of urban growth of the century, in India both in terms of the number of urban centres which have come up as also the urban population. The number of metropolitan cities has gone up from 12 in 1981 to 23 in 1991. The cities like Ahmedabad, Bombay, Bangalore, Calcutta, Delhi, Madras, Kanpur and Pune are experiencing population explosion. An alarming trend of migration to these cities from small towns and rural areas has been observed. According to the 1991 census out of the total national population of 844 million, 217 million lived in 3,768 urban agglomerations and towns. One-sixth of the total urban population of the country lived in just four mega cities namely, Bombay, Delhi, Calcutta and Madras, and the proportion of this population living in class I (1 million or above) cities is as high as 65 per cent. On the other hand, the share of the urban population in small towns has been drastically reduced. In 1951, some 13% urban population lived in class V (5,000 to 10,000) settlements. The figure came down to meagre 2.5% in 1991. Similarly, in case of class IV (10,000-20,000) towns, the figure has come down from 3.09% in 1951 to as low as 0.29% in 1991. The concentration of settlements in the country has, thus, become lopsided. This phenomenon shows excessive concentration of population in the large and metropolitan cities and decay of small and medium towns. This factor alone has led to the overstraining the existing infrastructures in the mega cities.

Interestingly, the pattern of population growth of the metropolitan cities exhibits a characteristic feature. The annual growth rate of the population of metro centres, as a result of influx of the rural poor to these centres, exceeds the natural annual growth rate of the population. The gravity of the situation arising out of the influx of the people from rural areas driven by continuing socio-economic disparity in villages may be gauged from the fact that about 2 lakh people migrate to Delhi every year. As for Bombay, migrants in similar number

arrive in the city every day in search of livelihood. In most of the metropolitan cities about 50% people live in make-shift shelters, slums and squatter settlements in conditions which are affront to human dignity. Some 46.62 million people in India live in urban slums. In Bombay alone, 5.5 million people stay in inhuman conditions in the 2641 slums scattered all over the city. Situation in other major cities is equally horrifying. Bombay's 'jhonpadpatties', Calcutta's 'bustees', Delhi's jhuggi jhonpries' (JJ), and the 'cheries' of Madras have all emerged in a big way.

Rural migration to cities, especially by farming households and tribal minorities, has been caused by widespread poverty in the villages. Poverty, on the other hand, has often been off-shoot of tremendous socio-economic disparity prevailing in our villages. In some states, continuance of semi feudalistic pattern of society arising out of non-implementation of the norms of land reforms has aggravated the rural poverty, and has forced the rural landless, devoid of any means of livelihood in their villages, to migrate to cities for their subsistence. Larger employment opportunities in cities are also a major contributing factor in attracting the destitutes from rural areas and smaller towns to larger towns and cities. A large city is perceived as an engine of economic growth.

According to a recent estimate, in the national capital, Delhi, about 2 million people live in shanty clusters squatting on 1,100 acres of prime government land with a market value of Rs 500 crore, and another 1.8 million people live in slum-designated areas. In 1994, there were as many as 1080 Jhuggi-Jhonpri (JJ) clusters dotted all over Delhi U.T. Apart from mushrooming of squatter settlements in the cities, unauthorised colonies are also now a permanent feature of urban housing. There are some 1,500 unauthorised colonies in Delhi, wherein there are no roads, no electricity and water connections and no basic amenities worth the name. These colonies cover 12,000 hectares of land, 10 times the area occupied by slums and squatter settlements in the city. More than 2 million people live in such colonies. Ineffective land policy and the excessive demand for land for housing are primarily responsible for the growth of such unauthorised constructions. Thus over two-thirds, of Delhi's housing stock is presently under occupation of slums, squatter and unauthorised settlements.

According to a World Bank paper, many big cities in developing countries are experiencing shanty town population growth in excess of 20% a year, and the problem of a doubling of slums and shanty towns within the next four to six years.

A qualitative study of housing shortage makes the position with regard to urban housing more disquieting. At present, there is an estimated number of 15 million kutcha unserviceable housing stock (units with thatched walls and thatched roofs) in the urban and rural areas in the country. According to NBO estimates, about 68% of the housing stock in urban areas is 'pucca' with walls generally made of burnt bricks, stone, GI sheets or other metal sheets, and roofs of tiles, G.I. sheets, corrugated iron/zinc sheets and reinforced brick/concrete. 24.7% is 'semi-pucca' (units which do not belong to the classified categories namely pucca, serviceable kutcha and unserviceable kutcha) and the balance is unserviceable kutcha (serviceable kutcha units comprise of mud walls and thatch roofs). Nearly 50% of the

urban households live in one-room units and 28% in two- room units. The position is much worse in the metropolitan cities reflecting the tremendous overcrowding resulting from urbanisation and population growth, as also the lack of affordable shelter on rent or ownership. The percentage of households living in one-room units is 77.4% in Bombay, 57% in Delhi, 67.6% in Calcutta, 65% in Pune, 59.6% in Ahmedabad and 53.6% in Madras. About a million people live precariously in 90,000 old and dilapidated buildings in Bombay awaiting imminent collapse.

Status of Rural Housing

The scenario on the rural housing front in the country is far worse. According to NBO estimates, the housing shortage in the rural areas was of the order of 20.6 million dwelling units in March, 1991. The findings of the National Sample Survey reveal that the majority of the rural houses are kutchas with walls made of unburnt bricks, mud, bio-mass and roofs mostly of bio-mass. Only a small portion could be categorised as pucca. As per another estimate, about 75% of houses in the countryside belonged to the category of semi or non- permanent construction. Besides, 70% of the rural houses do not have sanitation, and adequate supply of drinking water and electricity.

About 74% of the national population (843 million in 1991) live in about 5.83 lakh villages. According to the latest official information, 34.7% of the country's population happen to live below the poverty line, and the majority of the people living below the poverty line is to be found in the rural agglomerations. This alone speaks of the poor affordability of the rural population. As a result of their inability to generate the needed resources, the rural poor are compelled to live in conditions of squalor.

In rural areas, 44.38% of households lived in one-room units, thus accounting for about 73% of the total housing stock in the country side as per the 1981 Census. The one-room dwelling units had an average population of 4.61 per room, against 2.84 of two-room units.

As per the study conducted by the Institute of Development Studies, nearly half of the rural houses are more than 50 years old requiring urgent renovation on war footing. It is disgusting to observe that the construction of pucca houses declined from 67% in 1961-71 to 32.4% in 1971-81. Simultaneously, the number of kutchas construction has increased testifying the fact that the rich get richer and the poor still poorer in the country.

CHAPTER 2

Housing Finance

Introduction

The grim housing situation in the country warrants some urgent and effective steps to combat the ever-growing demand of housing. It calls for huge investment in housing sector. Non-availability of adequate funds with the Government exchequer is, however, a major stumbling block in providing shelter to the needy poor in the country. Though in absolute terms, the investment in housing sector has increased substantially over the years, from Rs. 1,150 crores (total of public and private sector) in the First Five-Year Plan (1951-56) to Rs. 31,458 crores in the Seventh plan (1985-90), the relative sharing of housing sector in the total plan investment has declined drastically from an impressive 34% in the First Plan to a meagre 9% in the Seventh Plan (1985-90) (Table 1). The decline has been precipitous-down to 10% in the Second, 15% in Third, 12% in Fourth, 10% in Fifth and 7.5% in the Sixth Five-Year Plan. The comparative decline of investment in housing and supply constraints in land have been major irritants in solving the gigantic problem of housing in the country.

Table 1

Housing Investment in Five-Year Plans

Plan Period	Total investment in the economy (Rs. In crore)			Investment in housing (Rs. In crore)			% of Housing Investment to total Inv. In economy
	Pub.	Pvt.	Total	Pub.	Pvt.	Total	
1 st	1,560	1,800	3,360	250	900	1,150	34
2 nd	3,650	3,100	6,750	300	1,000	1,300	10
3 rd	6,100	4,300	10,400	425	1,125	1,550	15
4 th	13,655	6,980	22,635	625	2,175	2,800	12
5 th	31,400	16,161	47,561	1,044	3,636	4,680	10
6 th	97,500	74,710	1,72,210	1,491	11,500	12,991	7.5
7 th	1,68,148	1,80,000	3,49,148	2,458	29,000	31,458	9
8 th				7,750	69,746 (proposed)	77,746	

Source: Draft National Housing Policy, May 1990

The problem with housing and real estate sector in India lies in the fact that so far housing has not been accorded the status of an industry. In the USA, housing along with the automobile industry forms a strong pillar of nation's economy. The US Administration spends 5.2% of the GNP on shelter. India with several times more of the homeless, however, spends about 2%. Moreover, in the West and even in many of the relatively better-off developing countries, the

Government provides affordable housing finance through various channels. Rent supplements are provided loans at nominal interest rates and more important are the tax breaks provided to make housing cheaper. In the erstwhile West Germany, for instance, little or no income tax was levied on 30% of the corporate profit, provided the companies used the money to extend interest free loans to employees. People were given incentives to invest in savings and loans, associations and house building societies, and loans from these institutions were generally kept outside the tax net. This was supplemented by a network of housing finance institutions.

Countries like the UK and the USA too, have a well-organised housing finance system for mobilising the savings of common man for lending to the housing sector. The deposits of the building societies and the savings and loan associations of the UK and the USA are nearly equal to the resources mobilised by the commercial banking system of those countries. These have been largely made possible by a judicious provision of fiscal and monetary relief by the government to these institutions. Germany, which was devastated by the two world wars, devised its own housing plan known as "Bausparkassen" which has played a significant role in rebuilding its devastated cities. The basic feature of such savings plan is that people contract to save a certain sum after which they are entitled to receive a housing loan, the amount of which is related to the sum saved. Both the savings and the loan attract an interest below the market rate. It is the saving of potential home buyers which is used to provide loan to purchase a house.

Successful examples of the aforesaid countries could be a trend setter for our future strategy to combat the problem of housing in India.

Existing Housing Finance System in India

Compared to the Western scenario, the existing pattern for housing finance in India presents an entirely different picture. According to an estimate, only about 10% of the houses purchased in India are financed through institutional means. The development of the housing finance system in India in the form of an institutionalised and formal superstructure is of relatively recent origin. Prior to the setting up of the Housing and Urban Development Corporation (HUDCO) in 1970, as an apex techno-financing government company entrusted with the task of financing and undertaking housing and urban development projects with prime focus on Economically Weaker Section (EWS) and Low-Income groups (LIG), the organised institutional housing finance was non-existent in the country.

The housing finance system, as it exists now, consists of two distinct components: the formal sector and the informal sector. The formal sector includes budgetary allocations of Central and State Governments, assistance from financial institutions like Life Insurance Corporation (LIC), General Insurance Corporation (GIC), Unit Trust of India (UTI), National Housing Bank (NHB), Housing Development Finance Corporation (HDFC), commercial banks, provident funds, specialised housing finance institutions like housing finance companies and cooperative housing finance societies and employers in organised public and private sector extending housing advances and staff quarters facilities to their employees.

The informal sector, on the other hand, contributes to the housing finance system through various sources. These include sale of personal assets such as personal saving in cash and kind, sale of personal assets such as jewellery, land and agricultural property, and borrowings from friends, relatives and informal money lenders/credit unions.

A brief description of the functioning and achievements of the formal financing institutions is given below:

i) Housing and Urban Development Corporation (HUDCO)

The role of HUDCO as a major public sector agency channelising investible funds to the State Housing Boards, Development Authorities, Improvement Trusts, and Co-operative societies, etc., has been quite significant. The HUDCO earmarks 55% of its sanctions for EWS and LIG housing projects and balance 45% for MIG, HIG, rental and commercial projects. Concessional rates of interest is charged from EWS and LIG categories (8% and 10% respectively) with some cross subsidisation and equity support share. In terms of number of dwellings, the share of EWS and LIG categories works out to over 90% of total houses financed by HUDCO. The resources of HUDCO include equity support from the Government of India, loans from LIC, GIC, UTI, NHB and market borrowings. The corporation has so far financed over 55 lakh houses under 9556 schemes involving a financial outlay of Rs. 12,660 crores. It is also a key financing institution catering to the needs of the rural population. It has constructed over 25 lakh rural dwelling units over the past 22 years.

Of late HUDCO has decided to launch an individual housing scheme for prospective house builders shortly.

ii) Life Insurance Corporation (LIC) and General Insurance Corporation (GIC)

The LIC and GIC are playing a very important role as catalyst as they are now confining themselves largely to refinancing operations. In fact, in the field of refinancing, both the LIC and GIC have channelised large volume of advances through the HUDCO, the Housing Development Finance Corporation (HDFC), State Governments, Housing Boards, Development Authorities and Apex Co-operative Housing Finance Societies which are the leading institutional investors in the existing housing finance set-up. The recipients of LIC's financial assistance are the crucial agencies which can cater to local requirements and help the private sector housing activities (with a major thrust on housing the EWS and LIG families) to grow in a big way.

The LIC has emerged as an important contributor to the resources for the housing sector. Up to March 31 1991, it had contributed over Rs. 4,400 crores to the housing sector and helped generate 30 lakh dwelling units. The flow of LIC investment to the housing was of the order of only Rs. 146 crores in 1992-93 (including loans of Rs. 300 crores to NHB under Central Government's quota).

The GIC loans are now also available for social housing schemes in the rural sector. Under the existing guidelines of the Government of India, the GIC earmarks 36% of its annual accretion to investible funds as loans to State government for rural housing schemes and also loans to HUDCO.

With the setting-up of National Housing Bank, the LIC has jumped into housing finance scenario in a big way. In June 1989, LIC Housing Finance Ltd. (LIC-HFL) was set-up in association with UTI, ICICI (Industrial Credit and Investment Corporation of India) and IFCL. The LIC-HFL has since established itself as a leading housing finance institution in the country. The housing loan schemes offered, are usually backed up with LIC policy as collateral security. This special feature helps the family of borrower to inherit the house free of loan in case of untimely death of the borrower during the term of the loan.

For individual housing, LIC-HFL offers two schemes namely 'Grih Prakash Scheme' and 'Grih Lakshmi Scheme'. Under the former, the loan offered is in the range of Rs. 25,000 to Rs. 20 lakhs at the annual rate of interest varying between 12.0 and 16.5%. Under the latter scheme, the loan provided is in the range of Rs. 11 lakhs to Rs. 20 lakhs at an annual rate of interest of 16.5 to 17.5%. The LIC policy of the borrower provides collateral security under the "Grih Prakash Scheme". Such security is however, not insisted upon in case of the loans sought under the 'Grih Lakshmi Scheme'. LIC has also recently launched a 'Grih Sudhar Scheme' for making available loans to individuals for undertaking repairs of their houses. Presently, the scheme is available only in the metropolitan cities of Delhi, Bombay, Calcutta and Madras. It is, however, likely to be extended to other towns shortly. Under this scheme the loan to be given to individuals ranges from Rs. 50,000 to Rs. 2,00,000.

iii) Scheduled Commercial Banks (SCBs)

Prior to 1989, the Reserve Bank of India made annual credit allocation to commercial banks for lending to housing sector. These guidelines were subsequently revised, and SCBs are now required to allocate 1.5% of the incremental deposits in the previous year towards lending to the housing sector. The RBI guidelines have stipulated the utilisation pattern for the allocations made by SCBs for the housing sector 30% of the allocation is for direct lending of which half should be in rural or semi-urban areas, a further 30% by way of indirect lending and the balance 40% to be deployed in government guaranteed bonds and debentures of HUDCO and NHB.

Though banks mobilise a major portion (over 39%) of the total financial saving of the household sector, there has been a perceptible shift in the pattern of financial saving of household with investments in capital market instruments showing a rising trend. In such a scenario, the SCBs contribution to the housing sector, which has grown from Rs. 186 crores in 1986-87 to Rs. 387 crores in 1990-91, is likely to decrease unless existing allocation of deposits is raised. It is also relevant to note that banks have a tendency to opt out of housing finance since it represents a high risk, low volume business which poses more difficulties in servicing as compared to industrial and other low risk, high volume ventures.

iv) Provident/Pension Funds

Provident and pension funds mobilise over 17% of the financial saving of the household sector. In terms of absolute volumes, they rank next only to bank deposits. The contribution of provident/pension funds to the housing sector at present is mainly by way of loans extended to members. In absolute terms, the housing advances from the provident fund increased from Rs. 129 crores in 1983-84 to Rs. 478 crores in 1990-91.

Besides being a repository of huge investible resources, the provident fund resources are eminently suited to housing sector financing as a) they are available as medium to long term capital, b) they are held as financial assets and provide a steady flow of capital over time and c) the investment is closely controlled by government regulation to ensure security of the investment.

v) National Housing Bank (NHB)

With a view to enlarging the scope of housing in public sector, the NHB was established in 1988, which is wholly owned by the Reserve Bank of India. The establishment of NHB is deemed to be a positive step towards meeting the long-felt need for an apex organisation which could give impetus to the process of building up a decentralised housing finance system, promote housing finance institutions and facilitate proper co-ordination among the various financing agencies associated with the housing activity. The new National Housing Policy announced in May, 1992 is aimed at enabling environment for housing activity by eliminating its existing constraints, and shifting the role of the government as builder to that of the "facilitator." The NHB could play a major role to achieve this object. It provides short-term loans to housing agencies for development and supply of land, and long-term loans to individuals for buying a developed plot and undertaking house construction.

The NHB introduced the Home Loan Account (HLA) scheme in July, 1989 to mobilise resources for housing. This is a loan-linked saving scheme on the lines of Germany's 'Bausparkasen.' As per this scheme, any individual, major or minor not owning a house anywhere in India, can open the HLA with any designated scheduled bank which takes responsibility of dealing with him or her on its behalf. The scheme is operated through the branches of public sector banks, scheduled private sector banks, state co-operative banks, urban co-operative banks and recognised Housing Finance Companies (HFCs). The minimum monthly deposit permissible under the scheme is only Rs. 30 (three years for the NHB financial housing projects). This would make the person eligible for a housing loan after five years. The person maintaining an HLA earns return at the rate of 10 per annum, and he/she receives the loan for housing in multiples of accumulated savings including interest at the interest rate varying from 10.5 per cent to 14.5 per cent. The loan amount ranges between 1.5 and 4 times the accumulated savings including the interest earned. Savings in this account and interest accrued qualify for income tax concession under section 80(c). Saving, thus mobilised, would be transferred to NHB to be used by the bank for advancing loans to the public housing agencies, industries manufacturing materials and to individuals.

The NHB extends facility of refinancing to scheduled commercial banks, scheduled state co-operative banks, scheduled urban co-operative banks, state level apex co-operative housing finance societies and housing finance companies in respect of the loans extended by them to (a) individuals (including co-operative societies) for construction/acquisition of new housing units or for upgradation/repair of existing units, and (b) public agencies, i.e., State Housing Boards and Development Authorities, co-operative Housing Societies, and professional developers for undertaking land development and shelter projects.

The maximum amount of loan sanctioned by the HFCs is normally up to Rs. 10 lakhs. The loan amount is required to be repaid in equal instalments stretching over a longer period of time ranging from 15 to 20 years. The NHB has so far approved its financial support to 19 HFCs having a network of more than 300 branches all over the country. The significant players in this field besides HDFC include LIC Housing Finance Ltd., Can Fin Home Ltd., (a joint undertaking of Canara Bank and UTI), and Dewan Housing Finance Corporation Ltd. Other new entrants are Akshanya Awas Nirman Vitta Ltd. (Bank of Baroda), Fair Growth House Finance Ltd., GIC Griha Vitta Ltd., Ind Bank Housing Ltd., Indian Housing Finance and Development Ltd., Punjab National Bank (PNB) Housing Finance Ltd., State Bank of India (SBI) Home Finance Ltd., Citi Home (Citi Bank undertaking) and Vysya Bank Housing Finance Ltd. Under the Home Finance Scheme of SBI, loan up to Rs. 3 lakh is sanctioned to the prospective builder at annual interest rate of 12.5 to 17.25%. Only under exceptional circumstances, the loan amount is raised to Rs. 5 lakhs. The PNB Housing Finance Ltd. and the Dewan Housing Finance Corporation Ltd. make available loan up to a ceiling of Rs. 10 lakhs with the annual rate of interest chargeable by the former ranging from 15% to 17%. The Citi-Home provides the largest loan quantum to the prospective builders. The loan size offered by Citi Home varies from Rs. 2 lakhs to Rs. 30 lakhs.

Till June 1994, the NHB had disbursed over 2,000 crores to banks and eligible housing finance institutions to benefit over one million families.

As a result of recent revision in the interest rates of housing loans sanctioned by State Level Apex Cooperative Housing Finance Societies, the current interest rates in respect of the NHB refinance schemes range between 10% to 13.5% on the loan amount ranging between Rs. 25,000 to Rs. 3 lakhs. Under the scheme, loan is now also made available for undertaking upgradation and repairs of existing buildings. The higher limit of the loan amount sanctioned for the purpose of repairs and restoration is Rs. 60,000 at the interest rate of 13.5%.

The NHB has identified suitable institutions for directing its credit and other assistance to the people in rural areas too. The State Land Development Banks (SLDBs) and Agricultural and Rural Development Banks (ARDBs) in various States are extended assistance by way of subscription to the Special Rural Housing Debentures floated by them for the specific purpose of providing finance for housing in rural areas.

vi) Primary and Apex Co-operative Housing Finance Societies

The co-operative housing movement is a strong and well-spread movement comprising some 70,000 primary societies, 25 State-level apex institutions and the National Co-operative Housing Federation (NCHF) of India. Primary housing co-operatives have a total membership of 45 lakhs, and over 13 lakh houses and flats have been constructed through housing co-operatives and an equal number are under construction. A part of funds for construction is raised by primary co-operative societies. The state-level Apex societies had advanced a loan of Rs. 2958.44 crore to the Primary Co-operative Societies for the construction of 13,33,316 houses up to March, 1993.

The major source of funding of the Apex Societies is LIC, which accounts for 90% of the loans available to them. The HUDCO also extends financial assistance directly to both primary

and Apex Societies for specific housing projects. Other sources are State Governments, floating debentures and banks which have also launched schemes to help co-operatives.

Primary housing co-operatives are characteristically oriented towards achieving the objective of affordable housing, and as such deserve greater financial support from the financial institutions such as HUDCO, HDFC and apex Co-operative Housing Finance Societies at the State level.

At present, the housing cooperative societies make available the cheapest housing finance to the prospective house builders the chargeable interest rate being only 13%. As a thumb rule, a borrower is entitled to a loan amounting to 48 times his/her monthly salary from the society. Of late the Delhi Co-operative Housing Finance Corporation (DCHFC) has raised the loan limits for individuals to Rs. 3 lakhs.

vii) Housing Development Finance Corporation (HDFC)

The need for housing finance for individuals in the country was fulfilled with the advent of the Housing Development Finance Corporation (HDFC) in 1977. The Corporation has pioneered long term housing finance system in the country, and has sanctioned loans aggregating over Rs. 3,500 crores, and financed more than 6 lakh families to help them acquire their own houses. It has spread its activity through its 28 branches located in different regions of the country.

The HDFC gives loans under various schemes to individuals, associations of individual groups of individuals and individual members of co-operative societies. An individual can avail loan up to 85% of the cost of house construction subject to a ceiling of Rs. 25 lakhs. The lower limit for the loan is Rs. 25,000. The rate of interest charged is related to the loan amount which, in turn, is determined by the repaying capacity of the borrower. In determining the repayment capacity, factors such as age, income, qualification, number of dependents, spouse's income, assets, liabilities, savings history of the borrower are also taken into consideration. It, ranges between 12.0% to 16.5% per annum. The principle generally adopted is that the instalment should not exceed 30% of the monthly income of the household. The borrower can choose the repayment period in the range of 5 to 20 years, not exceeding the age of retirement of the beneficiary. Of late the HDFC has also launched a scheme for providing loan up to Rs. 10 lakhs for undertaking repairs and renovation of the existing buildings.

Of late, the HDFC has been actively involved in promoting new institutions for housing finance, namely, the Gujarat Rural Housing Finance Corporation Ltd. (with support from the International Finance Corporation, Washington; the Aga Khan Fund for Economic Development, Geneva, and the Government of Gujarat). the Housing Promotion and Finance Corporation Ltd. (with the State Bank of India and capital markets), Infrastructure Leasing and Financial Services Limited (with Central Bank of India and UTI) and the Can-Fin-Homes Ltd. (with Canara Bank and UTI). In the case of last three institutions the HDFC holds 20% of their shares.

Governmental Role as Facilitator

Realising the exigency of enhancing the investment in housing sector, the National Housing Policy (1992) has envisaged that 20% of the requirement of investment in housing would be met by specialised housing finance institutions, insurance and banking sector, provident fund, mutual fund, etc., and additional mobilisation of household savings. There are also some suggestions to channelise the pension fund and gratuity towards the above cause.

The role of the government during the Eighth plan (1992-97) is being viewed as that of 'facilitator' except in case of Scheduled Castes, Scheduled Tribes, women and other poorer sections of the society. Above all, a fully subsidised rural housing programme, Indira Awas Yojana is presently being operated by the Ministry of Rural Development as part of rural development programme, and the Jawahar Rozgar Yojana for providing houses to SCs/STs and freed bonded labour.

The governmental involvement in the housing sector will henceforth be only by way of removing legal and procedural constraints, enhancing flow of credit from formal institutional sector and popularisation of locally available building materials. Besides, expansion of basic infrastructural facilities to facilitate housing would lie with the government. Direct assistance from the government is to be limited to environmental improvement and upgradation of slums and squatter settlements in the urban areas and support both in cash and kind to the poorer segments. To meet the above goal, the government's assistance will have to be supplemented by a massive dose of assistance from banks.

Status of Rural Housing Finance

Rural communities languish in a near total absence of institutionalised housing finance support. Housing in rural areas has been mainly through private efforts as per the study of National Sample Survey (NSS). The 44th Round of NSS (1988-89) reveals that only 5.18% of total finance for rural housing comes from formal agencies. Bulk of the resources for rural housing comes from savings - 42.75%, borrowing from friends and relatives - 9.36% and other resources - 22.93%. The HUDCO is a premier financing institution of the country to reach housing finance to the rural areas where it has provided a loan of Rs. 907 crores to enable construction of over 25 lakh dwelling units during the past 22 years.

The LIC and GIC are the other two financing institutions in the formal sector catering to the rural housing needs. During 1993-94, the LIC had earmarked Rs. 50.44 crores for rural housing. The GIC loans are also now available for social housing schemes in the rural sector. During 1993-94, a sum of Rs. 557.83 crore was allocated for the purpose.

The National Co-operative Agriculture and Rural Development Banks' Federation Ltd. has planned to earmark Rs 500 crore of lending through Agriculture and Rural Development Bank for rural housing during the Eighth Plan.

The Rural Housing Co-operatives could play an important role in mobilising savings and channelising community action for construction of houses. The Maharashtra Co-operative

Housing Finance Society provides long-term credit to prospective builders on priority basis. Similarly, in Andhra Pradesh, the Scheduled Castes and Scheduled Tribes Co-operative Housing Federation has provided a large number of houses to the rural poor requiring urgent renovation on war footing.

The NHB has identified suitable institutions for directing its credit and other assistance to the people in rural areas. State Land Development Banks (SLDBs) and Agricultural and Rural Development Banks (ARDBs) in various states are extended assistance by way of subscription to the special Rural Housing Debentures floated by them for the specific purpose of providing finance for housing in rural areas.

Impediments in Housing Finance and Related Issues

While functioning as the mechanism for promoting both direct and indirect investment, the existing housing finance system comes across some endogenous and exogenous factors. Among the endogenous factors, there are various legal, fiscal administrative and operational constraints which need to be removed. Among the exogenous factors, particular mention may be made of lack of adequate measures to mobilise private savings to considerable extent by providing stimulus to long-term investment in housing and creating mass awareness through publicity and contact.

The expensive and time-consuming procedural formalities involved in the registration and transfer of land are the results of complex laws and regulations relating to real estate transactions. Availability of land, though an indispensable prerequisite for housing, is full of legal formalities. This situation indirectly leads to land speculation and skyrocketing of land prices by creating impediment to the increase of the supply of developed land. Positive steps are, therefore, essential to simplify these legal procedures and bring down the expenses to be incurred over registration land. Removal of the existing stumbling blocks in the shape of Rent control Act, Land Acquisition Act, Under Land Ceiling Act and Agricultural Land Ceiling Act, and suitable modification of Municipal Bye-laws will go a long way in achieving the object of accelerating of Urban Land Ceiling Act is presently under consideration of the Union Government. This might away with some of the hurdles presently being experienced in the existing Act. The Delhi Rent Bill (1994) introduced in the Parliament is also aimed at making the investments in housing a profitable proposition. It would encourage supply of rental housing through removal of at making the investments in housing a profitable proposition. It would encourage supply of rental housing market.

At the national seminar on Mass Housing Development, held in July 1993, the need for larger participation of private builders in mass housing programmes was greatly emphasised. The National Housing Policy also encourages involvement of private developers in development, construction and provision of infrastructural facilities. Some of the major housing agencies, like the Delhi Development Authority (DDA) have already announced their decision to involve private entrepreneurs in their large-scale housing programmes. Some apprehensions, however, have been expressed with regard to the evil effects of such participation in the mass housing programmes for the lower income group of the population.

The fear arises out of the feeling that the private firms will be inclined to scrutinise beneficiaries stronger on their credit-worthiness. The Poor are not credit worthy Moreover, the private firms will be supposedly reluctant to make use of the cross-subsidy being more inclined to ask a price exceeding the cost price, thus closer to the market.

It is inferred from the above discussions that in the present scenario, the existing housing deficits are enormous and the economy, although improving, is not in a position to support the level of housing investments which could actually overcome the shortages or even keep up with the new demand.

Compared to the situation prevailing some 5 years ago, more facility for housing finance is available now. However, the plight of borrowers has not reduced considerably. Interest rates on loan provided by the financial institutions being high, repayment is definitely a difficult proposition as far as the affordability of the urban poor is concerned. Besides this, 0.8% application fee, post-sanction costs like processing charges at 1%, stamp duty and registration fee, etc., add up to about 10 to 15% of the house cost. (The stamp duty for transfer of immovable property in Delhi is 8% of the consideration amount. Ironically, there is no stamp duty on transfer of such property in Germany and the USA). Thus, the housing still appears to be unaffordable proposition for many under lower income bracket.

Cross-subsidy has been used more and more as a way to increase affordability. Cross-subsidy means that the plots for commercial purposes and those for higher income groups are sold at the market price above the cost price. The difference between the two is used to decrease the price of plots for poorer households.

CHAPTER 3

Land Use and Physical Planning for Housing

Introduction

Land is one of the important prerequisites for planning. No housing is conceivable without land. In most of the countries in Asia, Africa and Latin America, including India there has been increasing pressure on the urban land over past three decades. Population in these countries has increased as much as six-fold resulting in a situation wherein the demand for housing is far in excess of supply. Large scale influx of rural population into urban agglomerations, and rapid urbanisation have enormously increased the pressure on demand for housing in cities and towns. In 1991, about one-fourth of the country's population was concentrated in 1.5% of its total geographical area covering 3768 urban settlements, out of which class I towns numbering 300 alone accounted for 65% of the urban population. In metropolitan cities, there are large areas where density of population is as high as 1,000 to 2,000 persons per hectare. Due to the increasing population pressure, the precious agricultural land on the outskirts of the urban agglomeration is being encroached upon to meet the growing demand of housing. On an average, 75,000 hectares of agricultural land are being lost to cities and towns annually.

Supply of serviced land at affordable prices is a critical input for housing activity. It is widely recognised that the effective operation of land market has been affected by the existing legal and regulatory framework, lack of infrastructure, and the slow pace of release of serviced land by public agencies. This, along with other economic factors, has led to unwarranted increase in land price and housing costs, and widespread speculation and profiteering specially in larger cities. The exclusion of the majority of the poor from the formal market, and their inability to build or require legal shelter has led to the proliferation of settlements and unauthorised colonies.

The demand for land has led to innumerable problems of urban land development, transformation of rural and agricultural land into sites of urban expansion, monopoly value, escalation in prices of land, pressure on infrastructure, and haphazard growth of towns and cities, and land is locked up in speculative pursuits. Because of the element of speculation, land competes with the investment of surplus funds in the formal and parallel economy along with other assets like gold. There is very little land in large cities which can be earmarked for housing the poor in planned manner. Housing costs, especially for migrants, are raised by the increasing values of land components, apart from its scarcity. This leaves them with little choice other than to squat illegally on vacant land, or to occupy sites on the urban periphery far away from their place of duty.

Planning of Urban Land

Concern for planned development of urban land in modern India can be traced back to 1894 when the British Government introduced the Centre Land Acquisition Act, primarily for the

purpose of housing its military and civilian population in settlements adjoining a few existing urban centres and establishing new towns. The Act provided the basis for legislation in several States in the post-Independence period. The urban planning system, during the British period, however, envisaged mainly private site development, while public development was restricted to the provision of infrastructure and social amenities.

The inadequacy of serviced land in most of the large cities, the existence of high-density pockets along with chunks of vacant land, the growth of unhygienic bustiest and congested 'chawls' in and around industrial and commercial areas and such other features drew the attention of planners to the problem of land use in the post-independence period. In 1958, the Town and Country Planning Organisation (TCPO) envisaged a scheme of land use classification to achieve a standardisation in the usage of terms and categories. Subsequently, in 1966, it conducted study of land use pattern of 103 cities and towns belonging to different sizes and classes. In 1979, the TCPO published a follow-up study covering 403 urban centres based on surveys conducted by the State Town Planning Department within a normative framework and provides the key and the most effective instrument of intervention in urban planning. This thinking dominated the urban scene during the 'fifties and the 'sixties when the country went in a big way for preparation of City Master Plans with the objective of controlling and directing the growth of large cities. These plans were prepared on the basis of population projection, qualitative assessment of the trends in the sectoral composition of the city economy and certain norms with regard to land requirements for various activities. Thus, a given socio-economic profile of a city was sought to be achieved primarily through the control over the use of land. Master Plans for more than 500 existing cities and towns like Chandigarh, Bhubaneswar, Gandhinagar, etc., were also planned and built during the 'sixties and 'seventies within this framework. (As per TCPO's estimates, development plans now exist in respect of about 1,100 urban settlements in the country.) Unfortunately, not many of these plans could become fully operational in the long run. This was due to the inadequacies in the organisational and legislative support, beside the paucity of funds. In the cities where the plans were implemented, violation of the prescribed land use became a rule rather than an exception. This was often attributed to the way the Master Plans were drawn. It was also argued that the land policies, in general, and the Master Plans, in particular, have discouraged high residential densities and growth of employment opportunities in large cities. Some Critics even called the above initiative as having incorporated an "anti-poor and pro-rich" bias in the structure of the cities.

In the absence of a clear-cut policy on land use, values of land have been skyrocketing in large cities. An interesting feature of the land prices in the urban centres is, they have risen at steady but different rates in different urban centres as also in different localities in a city. The increase in land values has been attributed to inflation, population growth, lacunae in land legislations, inefficiency of the concerned organisations, and above all, the speculative factors.

There are two streams of thought about the issue pertaining to land. There is the planner's perception that the pressure of population in a limited area is the main explanation for the deterioration in living standards and the housing shortages. On the other hand, the

proponents of people's right to land believe that within a city conceived as business enterprise, land emerges as the most precious commodity, both as a means of production and as a prime consumer goods in short supply. Around land use and redevelopment there ensues an endemic conflict of interests among the rich, the middle class and the poor. According to them, the problem is not so much the non-availability of land as the inability of the urban land market, as it is constituted to supply adequate land for housing the poor. Through the maze of urban planning and land use regulations, the conflicting relationship between the needs of land of the poor and the speculative urge of the propertied class and the promoters in being allowed to perpetuate the housing scenario.

In the present context, land has lion's share in housing finance. Its non-availability in ample measure is a major hurdle in house delivery system. The concern with land prices mainly stems from its impact on residential property prices especially its impact on the lower income groups. An affordability analysis based on the income groups and savings rate of 10% to 15% and the affordable house price calculated according to cheaper HUDCO finance, indicates that 40% of the households in the EWS category could not afford a one-room house of 12.5 sq.m. for land values higher than Rs. 225 per sq.m. of gross plot area. As the land price goes in to the range between Rs 1,000 and Rs 3,000 per sq.m., the proportion of the households, who cannot afford even a one-room pucca house, rises to 63% to 77% respectively. Considered against the prevailing land prices in different parts of Bombay, this explains the extent of the city's population living in squatter settlements and overcrowded tenements.

Land prices can be kept in check by increasing the supply of developed land amongst the people of different income groups. At present, nearly 1,300 acres of DDA land is allegedly under encroachment in Delhi out of which 459 acres is under occupation of JJ dwellers. Unscrupulous element is also in possession of large chunk of DDA land in Delhi. Appropriate vacant land-taxation policies, suitable amendments in the rent control acts and encouragements to private sector to supplement the efforts of the Government can be useful in increasing the supply of developed land and enriching the depleted housing stock. Similarly, the Urban Land Ceiling and Regulation Act, (ULCRA), 1976-a very useful Act aiming to socialise the developed urban land-needs to be suitably amended so as to eliminate the existing loopholes, and make it a more positive Act so that the urge to circumvent it is minimised.

Urban Land Ceiling and Regulation Act, 1976

Faced with the problems of dearth of resources and high land prices, most public agencies have tried to acquire land through legislation. Urban land being a State subject, several States, have passed laws similar to the Land Acquisition Act of 1894, so that the compensation paid to the land owners could be less than the market price. The low rate of compensation in the initial years of land acquisition, however, encouraged the farmers, apprehensive of their land being acquired by the government, to sell the same to private colonisers with the motive of profiteering. Studies undertaken to review acquisition of land by public agencies in different States have indicated that their performances have always remained below targets.

With a view to augmenting availability of land for development purposes, the Govt. of India enforced ULCRA, which was basically aimed at preventing concentration of urban land in a few hands, with a view to bringing about an equitable distribution in favour of the urban poor and to prevent speculative transactions relating to scarce and precious urban land. Under the Act, the owner had to surrender in excess of 500 sq. meters to the government at pre-determined rates and the same land was to be used for housing the poor. The intention was, indeed, laudable. However, the idea boomeranged into becoming a major impediment in the housing activity. Thousands of hectares of the surplus land that the Government proposed to acquire under the Act was caught in the never-ending rigmaroles of court litigations. This drastically reduced the availability of land in most of the big cities, causing speculation in urban land and rapid escalation in the property rates. In 17 years of its existence, the government has reportedly been able to acquire barely 14,816 hectares of the total 2.20 lakh hectares declared surplus all over India. During the subsequent amendments, the holder of excess land was allowed to either utilize the land for building houses for the EWS within a period of five years or pay hefty amounts towards 'shelter fund' to gain exemption. The contribution was fixed at 30% of the notified land value in respect of the floor area ratio (FAR) being 100, with rate of contribution increasing proportionately if higher FAR is allowed or land remains vacant beyond five years of exemption period. This, too, failed to move any appreciable impact. The builders, in fact, opposed it tooth and nail saying it they were forced to shell out hefty sums towards 'shelter fund' and penalties, the property rates would soar higher as the burden of higher costs would be passed on to the ultimate buyers.

The National Urbanisation Commission in its interim report (1986) had mentioned that, because of the manner in which it has been drafted and implemented, the ULCRA has not been able to achieve any of the objectives for which it was enacted. The Act has failed in its objectives of stabilising the land price allocating land for priority use and encouraging low-income housing. Failure of the ULCRA is supported by the fact that in 1984 a mere 27 landowners, or 3% of the total, owned fully 70% of all exploitable vacant land in Greater Bombay, and about 20 builders monopolised over two-thirds of all flat construction in the city. The enactment of the ULCRA has two serious repercussions—imposition of restriction on buying and selling of frozen land and skyrocketing of land values. The slow freezing of large facts of land, and the slow development and marketing of land has had the adverse effect on the poor contrary to the objective of urban land policy. Most of the urban housing problems can be attributed to the problems relating to the urban land market. There is almost a freeze in the availability of urban land for housing due to the existing Urban Land Ceiling Act. The act deserves amendment on rational lines so as to encourage larger participation of private builders in mass housing programs for the teeming millions in the country. Of late the Government of India is contemplating amendments in the existing ULCRA and Rent Control Act which might help augment supply of land for undertaking housing programmes for masses in the country.

In recent years, emphasis is increasingly being laid on greater involvement of private sector in meeting the challenge of housing in the country which is necessitated by the evergrowing demand for housing coupled with the inability on the part of public sector to meet the housing requirements of masses. Whatsoever, an easy availability of land through

the necessary amendments in the ULCRA will encourage larger participation of private builders in mass housing programme for the people with poor affordability.

Effectiveness of Building Bye-laws

Despite formulation of town planning regulations and building bye-laws for major urban centres in the country, an effective control on building activities in cities has not been possible. Violation of the building bye-laws and the provisions of the Master Plan has become order of the day in most of the metropolitan cities. This fact is supported by coming-up of a large number, of unauthorised colonies and proliferation of slums and squatter settlements in large cities. Smaller towns, too are no better in terms of violation of building bye-law and the Master Plan provisions.

Building bye-laws are intended to regulate the orderly growth of towns and cities, and to ensure safe and sound building construction. As a rule, any housing plan should conform to the building bye-laws of the area. In certain cases, building bye-laws do not cater to the increasing needs of the community. For vast peripheral rural areas around metropolitan cities and large towns (rural-urban continuum), there has been no proper town planning legislation or bye-laws to govern the planning and construction activities. This has resulted in the simultaneous growth of regulated and haphazard construction activities in most of the large cities. In course of time these peripheral constructions get accommodated in urban agglomeration. Such constructions pose the problem of maintenance and provision of environmental facilities. According recognition to the rural-urban continuum would help the city/town administrators in easing the problem on this account.

The building bye-laws framed by the municipal authorities and other civic bodies, which are in-charge of their implementation, are by and large outmoded resulting in uneconomical use of land and constructions. The inhibits incorporation of recent innovations in the fields of architecture planning and building technology in new constructions through application of which not only economy in construction could be achieved but also could be possible to make structures more efficient in terms of their performance. Due to pressure of modern urban requirements, violation building bye-laws has become a common practice in urban centres. Moreover, involvement of lengthy procedures and restrictive nature of the bye-laws have often led to their bypassing and undertaking construction work in an unauthorised manner taking advantage of the inadequacy of urban administration to cope with their responsibilities. The inability of authorised to control such development, coupled at times with the presumed authorities housing colonies and squatter settlement are able to muster for regularisation of such settlements at a future date, has resulted in many flagrant violations of the building bye-laws.

By and large the municipal administration and urban development authorities have not given adequate attention to the work of control of urban building activities and this, in a way, has allowed the problem to grow to such an extent in certain urban centres that remedial measures are either not possible or very costly and difficult to implement. In dealing with such problem which is vital for the proper urban development, it is essential for the developing

countries like India with high rate of urban growth to identify the inadequacies in the building regulations, and to take remedial measures. Some of the important remedial measures are given below.

Measures such as realistic review of Master Plans, Standards, land use plans and regulations, rationalisation of building bye-laws based on the periodical assessment of the practical implications of the existing bye-laws and regulatory practices framed by the municipal administration/local authorities backed up by feed-back of experiences of house builders, architects, engineers and town planners; permitting innovations in buildings as per the provisions of the National Building Code which lays emphasis on performance concepts of buildings along with their structural, constructional, fire and healthy safety considerations, guidelines for low cost housing as evolved by the Bureau of Indian Standards (BIS) and research institutions like BMTPC, CBRI, Roorkee and HUDCO, etc., from time to time; making available type designs in conformity with the existing bye-laws to builders in low income groups; and evolving new concepts of urban development with joint participation of architects, engineers, planners and builders to ensure economical use of land and its development to provide necessary urban services may be helpful in evolving the minimum norms which are compatible with both health and safety as well as the means of the poor.

Residential Densities

The following densities are normally considered as optimum:

- (i) single storeyed houses : 50-65 houses per gross hectare;
- (ii) two-storeyed houses : 75-100 houses per gross hectare;
- (iii) three-storeyed houses : 100-125 houses per gross hectare
- (iv) four-storeyed houses : 125-150 houses per gross hectare
- (v) five-storeyed houses : 150 houses per gross hectare

If the above densities are adhered to and layouts are prepared by qualified town planning agencies, it may be feasible to ensure adequate open spaces, wide roads, parks and other sites for communal facilities consistent with land use economy.

Concluding Remarks

To conclude, inadequate availability of land and its high prices due to a variety of legal and administrative constraints is leaving vast numbers of households with no alternative but to illegal settlement on poorly-serviced land, or overcrowding in poorly-serviced dilapidated structures. Moreover, the rising values of land under pressure of urbanisation and population growth make it difficult for the public authorities to acquire land for social housing programmes. It is hoped, the intended amendments in the existing ULCRA and the Rent Control Act might help accelerate the pace of housing activity in the country.

CHAPTER 4

Housing the Urban Poor

Introduction

Urban poverty is synonymous with the growth of slums and squatter settlements in cities and town. Fast expanding slums are a common feature of the cities of the Third World countries. In India, in 1991, over 46 million people lived in slums, and twice that number live in substandard dwellings. Rapid growth of slums in big cities is an indicator of the inadequacy of the housing in the major urban centres. Supply of housing in most of the developing countries including India falls much short of the present needs of the population in terms of location, size and tenure. Moreover, housing inadequacies in these in these countries have both quantitative and qualitative aspects. On the quantitative side, the problem of housing has been worsening over the years due to (i) exponential population growth; (ii) fast rate of urbanisation; (iii) inadequate addition to the existing housing stock (iv) large-scale migration of the poor from villages and smaller towns, and (v) obsolescence of existing housing stock on account of lack of its proper maintenance.

The statistics of housing shortage in major cities is mindboggling. The city of Bombay had a backlog of 9,00,000 houses in 1991. The present rate of construction of housing units in the city is, however, a meagre 5,000 units per year, whereas private housing including co-operative efforts take care of another 20,000 units. The total effort thus is far too short of demand. In effect, shortage increases by about 30,000 units every year. Situation is equally horrifying in case of Delhi too. According to one estimate, 16.2 lakh dwelling units will be required in Delhi by the year 2000 AD. Against this, the DDA is, as of now, in a position to build at the most 10,000 dwelling units per year.

It is now well-established that the growth of slums takes place mainly because of the supply of houses at affordable cost being far too short of the demand. Attempts to increase the stock of low-cost housing in urban areas are thwarted by soaring prices of land and building materials, and inadequate financial resources of the government engaged in the exasperating task of improving the quality of living environment of millions of the urban poor condemned to the sub-human conditions of slums.

Gravity of the situation arising out of the proliferation of slums in metropolitan centres is gauged from the fact that in 1956 only 22% of Bombay housing prospect! According to a recent estimate (1994), 45 lakh of Bombay's one crore plus population live in 1,686 notified slum pockets occupying a 12.52 sq. km area of municipal, government and privately owned land. Bombay has the dubious distinction of having Asia's largest slums, a sprawling conglomeration of shanties called 'Dharavi'. An estimated population of 4,00,000 live in Dharavi. The housing situation in other metropolitan cities is not much different. The slum population in Delhi has been increasing at a rate that is nearly twice the rate of natural growth

of the population of the city. Nearly 50 lakh people out of Delhi's population of 93 lakh in 1993 lived in slums or in slum-like conditions. The appalling housing condition in metros is also gauged from the fact that about 6,00,000 people live on pavements in Bombay and Calcutta.

Living Conditions in Slums

Almost all the houses in slums in large cities are in bad and dilapidated condition. The houses are usually inadequately ventilated. The designs are such as to afford bare shelter, leading to acute congestion. Hutments in slums are generally constructed with materials ranging from polythene sheets, mats, asbestos sheets, mats, tarpaulin, gunny bags to bamboo, recycled tin sheets, broken bricks in mud mortar and plaster, clay roof tiles, thatch, etc. They are usually inadequately ventilated and mostly single-room units with inadequate space standards.

Private toilets do not exist in a majority of slum areas and even common toilets are available only in a few. Where there are no toilets, people defecate in the open which, besides being an environmental nuisance, poses hardship for especially women. Community latrines, even where available, generally remain in filthy and insanitary conditions.

Most of the houses in slum areas do not have individual water supply. In some slum pockets, public taps do exist but the number of persons using such taps is generally very high. Long rows of people with buckets before public hydrants are a common feature in squatter settlements.

The drainage system in slums is almost non-existent; situation gets worse during monsoon season when storm water finds its way into low lying settlements, making the conditions in slum areas extremely unhygienic. Besides, roads and lanes there are narrow and often unpaved. Road and street lightings are generally inadequate. This adversely affects the safety and security of residents, and is generally believed to be responsible for high rates of night-time crime.

There is a close relationship between environment and health. A number of studies have shown that the unhygienic slum environments make slum dwellers more susceptible to diseases like respiratory diseases, gastro-intestinal disorders, skin diseases, malarial fever, tuberculosis, etc.

A majority of the slum dwellers are employed in the unorganised or informal sector. A large proportion of them work on a daily wage basis, and therefore, have little or no job security. Due to low income, their standard of living is also low, and hence they are left with little income for housing after meeting basic needs of food and clothing of their families.

Approaches and Strategies for Housing Urban Poor

More than 70% of the urban population in India lives in dilapidated environment. Rehousing the entire slum population is beyond the means of the nation. Considering large proportion of population in cities living in slums, the priority areas of government intervention in the

mammoth task of providing shelter to the millions of slum dwellers in the country include environmental improvement, provision of basic amenities, community development and prevention of proliferation of slums and squatter settlements. Several strategies have been adopted by the government in the past to meet the challenges of urban slums and squatter settlements in the country with the shifts envisaged in policy and strategy to combat the problem from time to time. Some of the noteworthy approaches adopted by government in the past in this direction are described below:

a) Move to Decongest Large Cities

The Government of India has, through its successive Five-year Plans for economic development, always emphasised the need for rural development. Simultaneously, it has also pushed through the policy of restricting the growth of large cities, particularly metropolitan cities which alone account for nation's one-third urban population and simultaneously encouraging small and medium size towns where at least the available land for housing is relatively cheaper within reasonable distances from work centres. With shorter commuting distances, the urban poor can save on the cost of transportation, increasing their ability to channelise larger portions of their income towards housing. The Integrated Development of Small and Medium Towns (IDSMT) is one of the governments sponsored schemes launched in 1979 on the recommendations of the Task Force on Planning and Development of Small and Medium Towns and Cities constituted by the erstwhile Ministry of Works and Housing (now Ministry of Urban Development), Government of India, with an allocation of Rs. 96.00 crore. The scheme is aimed at reducing pressure on large cities where problems of slums and squatters are relatively more intense. Funds are allocated under the Scheme by the Central Government for creation of infrastructure that would improve the living conditions in smaller towns and, hence, will make it more attractive to a prospective migrant who may otherwise be heading towards a larger city. During the Eighth plan (1992-97), only towns with a population of up to three lakhs, as per the 1991 Census, are eligible for financial assistance under IDSMT. In all, 200 towns are to be covered during the aforesaid plan-period with a financial outlay of Rs. 155 crores.

Under the IDSMT Scheme, a 50% Central subsidy is provided for conversion of dry latrines into pour-flush latrines. All other services targeted towards the urban poor are to be met by the States under the State-sector component of the IDSMT. The funding pattern of the Scheme has since been revised from 50:50 to 60:40 between the centre and state Governments during the eighth plan period. Actual experience, however, indicates that due to inadequate budgetary allocation or delays in release of funds by the State Governments, there has been tardy implementation of the State component of IDSMT. Additionally, the States have also not been able to take full advantage of the Central subsidy for low-cost sanitation, due to their own budgetary sources or through beneficiary participation,

Slum improvement or upgradation programmes are also an integral part of the Scheme, which are deemed to be funded through state Plans. The IDSMT Scheme can be described as marginally successful since finances allocated are too meagre in relation to the projects and the number of settlements taken up under the scheme.

The Integrated Rural Development Programme (IRDP) and a host of other programmes initiated in rural areas such as the Rural Landless Employment Guarantee Programme (RLEGP), the Training of Rural Youths for Self-Employment (TRYSEM), the Rural Employment Programme for Women (REPW), the Rural Water Supply Programme and other similar programmes are, in fact, supplementary to the IDSMT Scheme as they aim to improve the lot of the poor in rural areas and contain them there rather than let them migrate to cities. But, like the IDSMT, the programmes for rural development are beset by severe resource crunch and mismanagement of funds. The schemes, therefore, have not been successful in stemming the migration tide to the desired extent. Another notable governmental effort to decongest large city has been the conception of the National Capital Region (NCR). It was mooted to decongest Delhi and regulate its population within manageable limits.

b) Government-Sponsored Programmes for Slum Upgradation

The main programmes undertaken by the government to mitigate the housing problem of the urban poor include the following:

- 1) Slum Clearance and Relocation Schemes (SCRS)
- 2) Environmental Improvement of Urban Slum Scheme (EIUS)
- 3) Sites-and-Services Scheme (S&S)
- 4) Urban Basic Service Programme for the Poor (UBSP)
- 5) Scheme for Housing and Shelter Upgradation (SHASU)

1) Slum Clearance and Relocation Schemes

Under the Slum Clearance and Relocation Scheme, launched in 1956, it was envisaged to get the existing slum areas cleared off and settle the slum dwellers at alternate sites. They were given highly subsidised tenements. However, these schemes did not succeed in the sense that these could provide shelter only to a marginal segment of slum dwellers. For instance, in Delhi where these programmes were implemented vigorously only 20.66% of the target population could be settled up to 1977. The programme could not benefit the target groups. The majority of those who were settled in resettlement colonies were replaced by comparatively higher income groups. The failure of this scheme could be attributed to the following three factors:

- i) The resettlement colonies were brought up in peripheral areas of the cities which were quite distant from their earlier place of living and work. This not only disturbed their social fabric but also their source of income. Since, for the poor, employment is more important than shelter, that is why in order to maintain access to earning opportunities most of them perforce require residence close to the place of work. threatened
- ii) Resettlement colonies being far from work places raise the transport cost and reduces the working hours which, in turn, further depletes their income. This acted as a stimulus to sell whatever shelter they had, and to move closer to their work places.
- iii) Shortage of supply of dwelling units meant for low and

lower-middle highly income groups push these groups to get accommodation which are otherwise meant for the urban poor. They appropriated the highly subsidised shelter either by registering the poor or by offering handsome amounts to the intended beneficiaries. The implementation of the scheme, nonetheless, improved the insanitary condition in the slum areas of cities like Delhi. But the formation of slums and squatter settlements continued unabated, posing enormous problem for the government.

2) Environmental Improvement of Urban Slum Schemes (EIUS)

In the light of the experience of Slum Clearance and Relocation Schemes, the approach was changed with the implementation of the Environmental Improvement of Urban Slums (EIUS) Scheme in 1974. The scheme was a part of the Minimum Needs Programme (MNP) launched the same year. The MNP emphasized the need for slum improvement in larger towns. The scheme is at ameliorating the living conditions of urban slum dwellers, and envisages provision of drinking water, drainage, community baths, community latrines, widening and paving of existing lanes, street lightening and other community facilities. The approach of tackling the problems of slums through environmental improvement was a significant improvement over the earlier efforts of slum clearance. The Bustee Improvement Programme implemented in Calcutta clearly demonstrates the kind of result that can be achieved with the minimum investment. The programme had benefitted some 1.6 million people living in 1,500 locations over a period of about 9 years (1971-79).

The Central Government moves plan allocations for undertaking slum improvement under the EIUS Scheme. The said scheme is taken up by agencies operating at State and city level with additional support from the State Governments for housing improvement and other socio-economic programmes. The scheme to be taken up by State Government agencies as a grant scheme, provides for taking up physical infrastructures like widening and paving of existing lanes, community bath, community latrines, street lighting etc. The element of subsidy is reduced compared to the one provided under the earlier scheme of Slum Clearance and Relocation. Grants are provided under the EIUS Scheme at the rate of Rs. 525 per capita. Shelter norms were also lowered down on the plea, that since the affordability of the poor has not improved, a greater element of subsidy would mean the transfer of home ownership to higher income groups. This centrally-sponsored scheme (EIUS) is taken up by agencies operating at State and city level with additional support from State Governments for housing improvements and other socio-economic support programmes.

The position with regard to recovery of capital costs and service charges under this scheme are far from satisfactory. The scheme when implemented in slums situated on public lands provide de facto security leading to complications of non-recovery of even service charges. The slums situated on the privately-owned lands seldom get covered under the scheme, and wherever they are covered occasionally, leads to exploitation by slumlords and land owners. Moreover, under the EIUS Scheme, the process of decision-making and project implementation is extremely controlled leading to indifferent community response. This results in poor maintenance and worsening recovery of dues. On the whole, the scheme leads

to an inflexible process which can only, to a limited extent, take into consideration the needs, aspirations and priorities of the community.

In Delhi, presently two slum upgradation schemes are being enforced namely the EIUS targeted at notified slums and the Environmental Improvement of Jhuggi-Jhonpries (EIJJ) targeted at squatter settlements. The EIUS Scheme aims at providing the living environment in notified slums through provision of civic facilities including water supply (one tap for 150 persons), sewerage, storm water drains, community latrines (one lavatory seat for 20-50 persons), community baths (one bath for 20 to 50 persons), widening and paving of existing lanes to make room for easy flow of pedestrians, bicycles and handcarts and to avoid mud and slush, street light (one pole per metre), community halls baratghars) open air theatres, akharas, etc. Under the EIJJ, Scheme, which is operational since 1987, pay and use community toilets/baths, street lights pole, water supply, dhalos/dust bins, paved pathways, drains, etc. are provided in jhuggi jhonpri clusters (under the parallel plan of EIUS). The quality of life of about 7 lakh people has improved substantially with these facilities. Under the EIJJ Scheme, a massive programme of constructing pay and use toilet-cum-bath (Jan Suvidha) complex in the JJ clusters has been launched. The complexes are available to the women and children free of charge. However, the men folks have to pay a nominal amount each time they use the complex. In addition to EIUS and EIJJ schemes, a massive crash programme for providing 50,000 serviced sites in well-planned and laid out colonies to the slum dwellers is to be launched by the DDA in near future. Moreover, the Slum Wing of DDA has recently launched a Registration Scheme for allotment of flats to the residents of slum/jhuggi settlements, resettlement colonies, unauthorised colonies and urban villages, and 27,000 persons have registered themselves under this scheme. So far, 3,000 flats are being constructed out of which more than 1,000 have already been allotted.

The investment in Bombay's shelters, too, has been recently augmented substantially through the sanction of Rs. 1,000 million for the re-development of 'Dharavi, urban renewal in the island, and upgradation of additional slums.

3) Sites-and-Services Scheme (S & S)

The Sites-and-Services (S&S) Scheme is yet another experiment undertaken in respect of improving the housing condition of the urban poor. Under this scheme the slum dwellers were provided small plots (25 sq.m.) along with necessary infrastructure. The smallest plots under the scheme measured with 21 sq.m in Bombay. The owners of the plots were supposed to construct the structures in future depending upon their affordability. The cost of antistructure was borne by the government. The element of subsidy was again reduced under the scheme. However, the scheme has been widely supported by various government agencies, and international financing agencies have been increasingly willing to fund such programmes. Though of recent origin, it has gained considerable popularity in Asia, Africa and Latin America. In all these continents the scheme is an important component of housing policies. Like the Slum Clearance and Relocation Scheme, this scheme has also suffered from the location and siting considerations. Cost recovery records were, however, better in sites-and- services than the upgrading projects. As in the case of other social housing schemes, in respect of the Sites-and Services Scheme also, the HUDCO has been designated as national level nodal

agency for routing central allocations. The HUDCO has, in turn, put the stipulation that all the composite housing schemes that it finances, one-third of loan provided has to be spent by the receipt (State) housing agency on construction of houses for EWS. It could either be in the form of sites-and-services or complete dwelling units.

Under Jhuggi-Jhonpri (JJ) Resettlement Programme, initiated prior to the inception of the Seventh Five-Year Plan, about 2.20 lakh jhuggi jhonpri families were re-settled in 46 JJ Colonies, where these families were provided plots of 60 sq.m to 21 sq.m minimum size from time to time with basic infrastructural facilities. The plot size had been varying from time to time depending upon the magnitude of the problem and physical/financial constraints. The scheme of JJ Resettlement was, however, discontinued with the termination of the Sixth-Five- Year Plan (1980-85), and emphasis shifted from resettlement of jhuggi jhonpri dwellers to improvement of slum/JJ Clusters on 'as is where is' basis.

Among other notable projects undertaken under the Sites-and-Services scheme, names of Arumbakka Sites-and-Services Project of Madras, and Vasna slum Rehabilitation Project of Ahmedabad are worth mentioning.

4) Urban Basic Service Programme for the Poor (UBSP)

One of the identified deficiencies both under the EIUS and sites-and-services is that clients are not well-equipped to avail full benefits of the services provided. In this connection it is felt that there is a need to educate the prospective clients and prepare them for their participation in community work. Studies in the past have shown that whenever there is active involvement of the clients in the design and implementation of the development scheme, it not only leads to substantial reduction in capital costs but also results in ensuring efficient operation and maintenance of the newly-created facilities. Once a commitment of the beneficiary is accomplished, cost recoveries posed no serious problem. Keeping this finding in view, the seventh plan proposed the revival of the Urban Community Development Scheme of the earlier plans in a slightly modified form. The new scheme known as Urban Basic Service Programme for the poor (UBSP), launched in 1990, was evolved by integrating Urban Community Development Programme, Programme, UNICEF-aided Small and Medium Towns Development and Low-Cost Sanitation. The scheme's emphasis is on social inputs to go Side by side with physical improvements, in construction of houses and other infrastructural facilities. The social inputs include primary health and nutrition & extension, hygienic and civic responsibility, education, pre-d primary and primary school facilities, and vocational training for the women folk.

Since the inception of the Seventh Five-Year Plan, the programme is being implemented on a 60:40 sharing basis between the Central and the State Government. A total outlay of Rs 100 crore has been provided for the scheme for the eighth plan period. The scheme is being implemented in association with the State sector scheme of EIUS and the Centrally-sponsored-Nehru Rozgar Yojana.

5) Scheme for Housing and Shelter Upgradation (SHASU)

The Scheme for Housing and Shelter Upgradation (SHASU) is a part of the Nehru Rozgar Yojana (NRY) launched by the Government of India in October 1989. The NRY was recast in March 1990. The entire expenditure on the NRY of which SHASU is a part is to be shared on a 60:40 basis between the Central Government and State Government with effect from the eighth plan. The SHASU seeks to provide assistance for housing and shelter upgradation to economically weaker sections of the urban population as well as to provide opportunities for wage employment and upgradation of construction skills. A loan up to a ceiling of Rs 3,000/-at 10% rate of interest is provided under the scheme to the entitled beneficiaries for Housing/Shelter Upgradation. In case of enhanced financial requirement beyond Rs 4,000/-, an additional loan can be procured from HUDCO under its scheme for EWS Housing. This scheme is applicable to urban settlements having a population between one and twenty lakhs. Requirements for institutional finance for the scheme are met by HUDCO. The Central Government had made an allocation of Rs 10.45 crore as subsidy for shelter upgradation, and Rs 2.61 crore for training for skill upgradation in construction trades during 1993-94. HUDCO has sanctioned 420 schemes for Housing/Shelter Upgradation up to December 1993 since the inception of the NRY.

c) National Housing Policy

The National Housing Policy announced in May 1992 envisages a separate clause from slum and squatter settlements in urban areas and housing for the urban poor. Keeping in view the policies of planned growth of urbanisation, income support and poverty alleviation, and together with steps to arrest the growth of slums in urban areas, the Central and State Governments would take steps to:

- avoid forcible relocation or dis-housing of slum dwellers
- encourage in-situ upgradation, slum renovation, and progressive housing development with conferment of occupancy rights wherever feasible, and to undertake selective relocation with community involvement only for clearance of priority sites in public interest
- expand provision of water supply, sanitation and other basic services in slum and other settlements occupied by the poor
- ensure proper maintenance of amenities through community involvement and decentralised institutional arrangements
- integrate the provision of physical amenities with basic services including maternal and child welfare services and health care, structured on community participation and involvement of voluntary agencies and management by local bodies
- promote incremental construction and upgradation by poorer households through access to land and services, through technical support, outlets for low-cost technology and materials, opportunities for skill upgradation and access to housing finance on flexible terms
- provide night shelters and sanitary facilities for the footpath dwellers and the homeless.

d) World Bank Aided Programmes for Urban Poor

The World Bank has played a significant role in providing financial assistance to the programmes for improving the plight of urban housing in the country. It has extended its support to deal with the problem of slums in bigger cities like Bombay, Calcutta and Madras in a noteworthy manner. It considers its urban development project in Madras as a showcase. The World Bank provided its first loan to India for the purpose of Urban Development in 1973. In contrast to the Centrally-sponsored EIU Scheme, with its origin in a 'welfare state' approach with subsidies, the World Bank funded projects aim at recovery of costs to an extent. These projects also aim at considering and maintaining the improvement works carried out by way of granting tenurial rights and enforcing recovery of service charges and developed plots. In case of the World Bank funded projects, cost recovery records have been more encouraging in sites-and-services than upgrading projects.

At present, the World Bank aided projects are under implementation in the States of Maharashtra, Gujarat, Uttar Pradesh and Tamil Nadu. The projects cover several sectors/areas such as Shelter/Slum Upgradation, Sites and Services/Area Development, Transport, Low-Cost Sanitation, Solid Waste Management, Municipal Development Fund, etc. The Ministry of Urban Development (Government of India) monitors the implementation of the World Bank aided projects in the States.

e) Involvement of Co-operatives in Resettlement Programmes

Recently, a new strategy for resettling jhuggi households has been embarked upon by DDA. Under this approach of sites and services, plots of 18 sq.m with share-in-open courtyard under 'Cluster-court-town-house, planning concept are to be provided for resettlement of these squatter families whose jhuggies had been in existence prior to 31.1.1990. The resettlement complexes are to be the integral part of the new residential schemes of DDA. Sites measuring about five hectares are to be utilised for provision of 1,000 plots/residential units by achieving a density of 200 units per hectare, and in layouts, one hectare of land is utilised for provision of community facilities such as primary schools, open spaces, 'Shishu vaticas', 'basti vikas kendras', community facilities complexes, and 'dhalavs, etc. The layout plan is around open for court-yard with 4 to 6 dwelling units about 7 sq.m. each in open courtyard for the group. Full coverage of 18 sq.m. plots is permitted to the squatter families on the ground floor, and subsequently, when the affordability of allottees improves, additional floor can be added by the beneficiaries, approachable by a ladder/staircase provided added within the plot on the ground floor. Independent Water Closet (WC) units and bathrooms with cooking shelves are to be provided to make the residential units as self-contained.

The squatter families are organised into multi-purpose cooperative societies with the association of NGOs/voluntary organisations and the slum wing of DDA as per the approved byelaws. For shelter construction, Delhi Co-operative Housing Finance Society (DCHFS) is designated as funding agency for providing a loan of Rs 15,000 per jhuggi households in three phases through procurement of bulk finance from HUDCO. The loan is to be recovered from slum dwellers in 261 equal instalments. As a part of the project before shifting of the jhuggi

households to resettlement complexes, construction of foundation up to plinth level along with WC seats and baths is to be undertaken at the sites of alternative plots either by the Slum Department of DDA or by Co-operative Societies, as the case may be.

f) UCD Approach

The Slum upgrading programme successfully taken up by Municipal Corporation of Hyderabad and Vishakhapatnam in 1983-84 and 1988-89 respectively under the financial assistance from Overseas Development Administration (ODA) of UK, is widely known as 'UCD' approach. UCD stands for 'Urban Community Development'. The programme is geared up to pursue the objectives of achieving improved quality of life for the residents by making the community aware of its potentials and providing the required inputs. The inputs include provision of physical infrastructure, social facilities, economic support programmes and at some places include financial assistance for rebuilding or improving the house structure. The focus of the programme is on integrating marginalised communities into the city fabric by extending access to services and infrastructures as also legalised tenure security. The success of the UCD approach at Hyderabad and Vishakhapatnam is comparable to the popularity of 'Kampung Improvement Programme' (KIP) in Indonesia. The KIP is considered as one of the world's outstanding slum improvement programmes undertaken with the active participation of the community.

In Dharavi (Bombay), groups representing 4,00,000 people have also joined together to form people's Responsible Organisation for United Dharavi (PROUD). It is a highly democratic body, with each member organisation electing its own representative to the PROUD which is itself organised to address specific issues. The PROUD has its own team of professionals that prepare schemes for the upgrading of squatter communities and for the regularisation of land.

The Indore Habitant Improvement Project (IHIP) for slum upgradation with assistance from ODA in January, 1990 is another worth mentioning/example of the co-ordinated efforts to improve the living conditions of city slums through community participation.

The Lakshmi Puram Slum Rehousing Project undertaken in Bangalore is yet another successful story of community participation settling the problem of rehousing the slum dwellers without their eviction from the site of their squatting. For the first time in Bangalore, secure land tenure has been offered as an in built provision in a slum rehousing project.

g) CD Wings

The other approach attempted is integrating community development activity with organisations like the Slum Boards and Housing Development Authorities. The Tamil Nadu Slum Clearance Board and the Maharashtra Housing and Area Development Authority (MHADA) have community wings to undertake this function. In this structure, Community Development (CD) Wings perform a supporting role of contacting and communicating with the communities with regard to physical infrastructure and housing improvement plans of the agency.

h) Role of Voluntary Organisations

Voluntary organisations have been playing an important role in tackling the problems of the poor and the under-privileged. Main spheres of their services are, however, restricted to education, (including adult education), public health, sanitation, family planning and social forestry. In the field of housing, the voluntary organisations have not been able to do much because of the enormous amount of capital investment needed for such an effort.

The Subsidised Aided Self-Help Housing Scheme for the Economically Weaker Sections (EWS) being implemented in Kerala with the help of voluntary organisations, is however, one good example to show that what could be achieved by such organisations. In the above scheme, financial, technical and managerial assistance was provided by the State Government.

i) Improvement of Inner-City Slums

The slum of large cities in the developing countries present a set of shelter issues that differ in important aspects from those of squatter settlements. In many large cities, notably in Asia, inner-city slums constitute a large chunk of the population of the urban poor. In cities like Bombay, Calcutta, Colombo, Lahore, Bangkok and Madras, a substantial percentage of city's population live in such slums. As a general rule, the older the city, the more likely it is to have a large slum population. Globally, slum populations are largest in the old cities of Asia, Latin America, and Western Asia, and smallest in the young cities of Africa, especially Sub-Saharan Africa.

While 'formal' housing in the inner-city slum displays great variation, it falls broadly into two main categories derived from the original intended use of the structures. The first category consists of housing produced by the conversion and sub-division of structures originally intended for non-residential purposes or for high-income groups. This category includes the 'conventillos' of Santiago and Lima, and the 'havelis' (old mansions) and 'mohallas' of Old Delhi. The second category is made up of building types that were, from the beginning intended for occupation by low-income groups. This category house the 'chawls' (Three-to four-storey tenements) of Bombay, shop-house tenements and row tenements of South-east Asia and 'vecindades' of Mexico City.

The critical issues in planning for inner-city rehabilitations and slum improvement relate are presented and defended. Planning strategies always reflect the pressure that interest groups are able to exert, since the interests are great and the stakes usually high in the inner city, this applies to the centre great perhaps more than anywhere else. The position of the poor in the inner city is particularly vulnerable, and it is inevitable that they stand to lose most in the changes that take place, especially from changes that squeeze space that is accessible to them.

Increasing commercialisation of inner cities is, in most cases, the dominant threat to slum dwellers. Such pressures may be accompanied and exacerbated by processes of 'gentrification,' whereby areas and floor space traditionally occupied by low-income groups are gradually taken over by higher-income groups. The slum dwellers are typically unable to

resist commercialisation and gentrification, although there have been isolated instances of organised industrial workers who have been able to prevent 'take-over' of the relatively small areas in which they lived.

Rent control, in most developing countries including India, has been the most important support protecting interests of the poor in the inner-city slums. The experience of such cities as Bombay, Colombo, Delhi, Karachi, Lima and Mexico City leaves little doubt that by maintaining rents at affordable levels, rent control has helped stabilise residential neighbourhoods that would otherwise have been lost by low-income groups. Rent control has, however, become one of the most contentious issues in the inner city. It has some negative consequences. Relatively, affluent tenants find themselves in the privileged position of paying rent far below what they can reasonably afford. Moreover, rent control serves to subsidise the business and commercial establishments that manage to penetrate residential buildings. Also, stabilisation of low-income groups in the inner city is likely to be offset by a management stalemate in the situation in which government, house owners and residential all refuse to accept responsibility for the maintenance and improvement of the housing stock. The greatest objection to rent control, however, is that it greatly discourages property owners from investing in the improvement of their property, and new investors from investing in rental housing for low-income groups. This fact is supported by the progressive deterioration in the inner-city housing stock and to the sharp decline over the past three decades in the production of rental housing for low-income groups.

In Bombay, the typical rental housing solution for the poor groups are 'chawls' built by the private and public sector, which dominate the city's central area. Around 60% of Bombay's low-income population live in these 'chawls.' A Rent Control Act enacted by the British in 1939 followed by a rent freeze in 1947 established a system of protected tenancy that has ensured to this day the survival of the poor in Bombay's central areas. In 1969, the Maharashtra Government passed an Act that established the Building Repairs and Reconstruction Board. The Board was responsible for ensuring the timely repair and reconstruction of buildings occupied by tenants and not owned by co-operative societies or public bodies and used solely for residential purposes. A repair cess was levied in all buildings requiring repair effective from 1969. In the case of buildings required, the owners were required to pay a higher repair cess. In 1977, the legislation was repealed and replaced by the Maharashtra Housing Area Development Act under which the State assumed the responsibility for repair and reconstruction of building.

Most of the affected are in the four congested inner-city wards which accounting for only 26.8% total area, contain 60% of the 'cessed' buildings. In 1981, the priority list contained 2,936 buildings of which 1,136 were considered beyond economic repair. Whereas the repairs programme has proved manageable, reconstruction work has proved very difficult. The value of the Bombay experience lies in the efforts to preserve and rehabilitate the existing housing stock in central neighbourhoods, while ensuring the right of low-income groups to remain in the neighbourhoods. The experience also points to the value of directly involving the poor in rehabilitation programmes.

Colombo provides an interesting example of public sector legislative intervention, followed by subsequent upgrading programmes, aimed at stabilising the low-income population of the inner city. The ceiling on Housing and Property Law, enacted in 1973, controlled the ownership of housing property by specifying the permitted number on houses individual, family or group of persons could own. The surplus could be purchased by tenants or be transferred to the ownership of the Department of Nation Housing. The housing transferred became the responsibility of the National Housing Commissioner, and was later to be offered for purchase to tenants. In 1977, policy changes were enforced to the effect that all tenants, who paid a monthly rent of Rs. 25 (approximately \$1.50) or less, would receive their housing units free of charge from the government, as a further step towards the socialisation of housing. The value of the Sri Lankan experience is the lesson derived from the enactments of appropriate legal mechanisms which effectively stabilised the low-income groups in the inner city through the enforcement of tenurial changes.

Another interesting example of resolving the problem of inner city slum is that of land-sharing schemes in Bangkok. The schemes represent an interesting and realistic approach in countries where the urban land market is largely unregulated. The schemes tackle one of the main constraints on inner-city rehabilitation efforts-landlords opposition to the up grading of slum areas which they are seeking to recover for private redevelopment. The schemes are based on a trade-off between conflicting interest's owners versus residents-which allows the landlords to recover part of his property while permitting the residents to retain possession of the remaining part. In inner-city areas where growing demands on valuable land increase the threat of expulsion for low-income groups, land-sharing is able to break a deadlock by legitimising the right of the poor to remain where they are. The Bangkok experiments with land sharing appear to have considerable potential, not only for stabilising the position of low-income groups in the inner city. It could be applied to the government-owned land with land sharing used by the public sector as an effective tool for securing cross-subsidisation in upgrading programmes.

j) Night Shelter/Sanitation Facility to Pavement Dwellers in Urban Areas

People sleeping on footpaths or road berms present most inhuman and Pathetic aspect of housing shortage in urban centres, especially in larger towns. Driven by extreme socio-economic disparity in the villages, these hapless people are left with no other option than to migrate to cities in search of livelihood. They are mostly engaged in petty jobs. Such people are generally casual labourers, and shoe-shine boys, rickshaw pullers, hawkers, vendors and cart-drivers, etc. Having no home in or around city, they are forced to pass their nights on pavements.

Realising the agony of pavement dwellers and the shelterless, the Government of India had launched a scheme for construction of night shelters in metropolitan cities in 1990. Under this scheme, night shelters and sanitation facilities are provided to the dwellers at a per capita cost of Rs 5,000 with 20% subsidy from the Central Government and 80% from HUDCO. The scheme has since been extended to cover all urban areas, wherever the problem of pavement dwellers exists. Up to December 1993, a total number of 41 schemes had been sanctioned by HUDCO to benefit 20,000 pavement dwellers. In Delhi, the DDA has put up 17-

night shelters in different parts of the city. Each shelter is provided with jute matting, blankets, medical aid, toilet, and mini library and colour TV for entertainment of the inmates.

The guidelines of the scheme have since been revised with a view to generate more schemes during the eighth plan. The revised scheme provides for construction of 'pay and use' toilet facilities where construction of night shelter is not feasible for which a Central Subsidy at Rs 350 per user is provided.

Concluding Remarks

i) The problem of housing the urban poor is required to be tackled in a holistic manner, through multi-pronged strategies. There cannot be a lasting solution to the problem of housing for the urban poor unless housing becomes a part of the overall development strategy of the national planning in tenor and spirit. There is urgency to check the uncontrolled migration of the population from rural areas and small urban centres to large cities. It is better said than done. By and large the root of urban poverty lies in the rural poverty. In one sense urban poverty is spill-over of the rural poverty. There should be all-out efforts to improve the economic lot of the rural poor so as to arrest the process of their migration to towns and cities in search of livelihood. There is also a need to enforce the norms of land reforms in the right earnest as the structure of our villages is still largely agrarian, and the majority of rural population still depends on agriculture and agro-based industries. It is the landless labour who form the majority of the rural population. Economically, they are the most suppressed class. It is they who per force migrate to towns and cities as the villages where they are born and live, are not able to contain them due to unfavourable conditions prevailing in the countryside for their sustenance. Efficient urban land management, ways and means for encouraging community involvement in slum upgradation programmes, defining the role of private developers and voluntary organisations, and evolving realistic standards for housing urban poor are some other issues which deserve special attention to resolve the crisis of housing the urban poor.

(ii) Squatter-settlement upgrading and sites-and-services programmes are a considerable improvement on the shelter policies that prelude them. Both the programmes have displayed a tendency to serve the most able and enterprising among the urban poor. This has led to suggestions that they may have a "creaming off" as well as polarising effect among the poor. Nonetheless, the programmes are perceived as the prime instrument for improving the housing conditions of the urban poor.

CHAPTER 5

Development and Adoption of Low Cost Housing Technology

Introduction

In recent years, there has been considerable debate over the definition of low-cost housing. Without falling into any further debate on the issue, in plain terms, low-cost housing may be defined as provision of housing which caters to the minimum requirements of masses within their income capabilities, without sacrificing the quality of construction.

Most of the developing countries in Asia, Africa and Latin America share the problems of poverty, large population growth and acute housing shortage alike. The formidable challenge of providing housing to masses at affordable cost is a colossal task which most of the governments in these countries are hardly in a position to cope with. Meagre economic resources of such countries necessitate adopting appropriate building technology which could lead to economy as well as speed in construction. Prefabrication is the obvious choice to meet the ever-increasing demand of housing in these countries. However, because of the prohibitive initial capital investments, total pre-fabrication does not sound to be a pragmatic proposition in majority of the developing countries. For ensuring speed and economy in construction, the only practical solution to the problem of meeting housing requirements of masses in the present context will be go in for adoption of appropriate low-cost construction techniques involving adoption of partial prefabrication employing the use of prefabricated building components of such size and weight which could be fabricated at construction sites or in industrial production units; economical walling systems and single stack system of plumbing, etc.

Premier R&D institutions engaged in the field of building technology like the Central Building Research Institute (CBRI), Roorkee; the Structural Engineering Research Centre (SERC), Madras; the National Council for Cement and Building Materials (NCB), Ballabgarh; and the Regional Research Laboratories, etc. have played important role in the development of cost-effective construction techniques and materials. On the other hand, organisations like the National Buildings Organisation (NBO), New Delhi; the Building Materials and Technology Promotion Council (BMTPC), New Delhi; HUDCC, and the national network of Nirmati Kendras (Building Centres) have been instrumental in the promotion of the innovative construction techniques evolved by the aforesaid research institutions in the country.

Adoption of Innovative Cost-effective Construction Techniques

In order to convert the know-how' of the research labs to 'show-how' in the field, and Experimental Housing Scheme was initiated under the aegis of the NBO by the Government of India in 1962. Under the scheme, innovative construction techniques and improved

building materials, which were as product of basic and applied research, were tried out under field conditions in prototype experimental buildings for their technical and economic evaluation. Successful techniques were subsequently propagated for wider adoption. Under this scheme, construction departments sponsored projects for putting up of 40 to 50 houses incorporating new technologies, design concepts and materials etc. The NBO provided grant of 100% of 100% of the cost of new construction items. Beneficiaries of the scheme included the Central/State Government Construction Departments, Housing Boards, Technical Universities. Research Institutions, Public, Sector undertakings of Government of India and Semi-Government/Autonomous Bodies have sponsored demonstration projects under the Experimental Scheme. In all, 59 experimental projects were sponsored by as many as 42 construction departments/agencies in different geo-climatic regions of the country in which 60 innovative cost-effective construction techniques and materials were successfully tried out (Annexure I).

Adoption of Precast Elements in Partial Prefabrication

Brief details of some of the popular precast building components, namely, precast RC roofing/flooring elements, precast thin RC lintels, precast RC door and window frames, which have been successfully tried out under the NBO experimental projects and their subsequent large-scale utilisation in several housing projects undertaken by various construction departments/agencies in India, are mentioned below.

Precast Roofing Systems

Roofing accounts for up to 25% of cost of construction of a house. By using precast roofing components, 20 to 30% economy could be achieved in the cost of roof construction. Some of the popular precast roofing/flooring techniques promoted by the NBO, which have manifold advantages of saving in cost, consumption of cement and steel and increased pace of construction, are described below. All these systems make use of components which are fabricated on the ground, ready in all respect for erection in building. In-situ concreting, work is minimised in these systems so that the speed of construction is not affected. The necessity for putting up centring and shuttering as in the case of cast-in-situ reinforced concrete (RC slab completely) is eliminated under these roofing systems. The provision of centring and shuttering is a time-consuming process and elimination of the same leads to considerable saving in erection time.

Precast RC Channel Units

Present channel units (Fig. 5.1 to 5.4) are reinforced concrete structural members which possess a channel or trough shape. The sides of the unit are corrugated and grooved at the ends to provide shear key action between the adjacent units. The nominal width of the units is 300 or 600 mm and the depth is 130 mm. The length of the unit will depend upon the span

to be covered and is generally kept between 2.5 m to 4.2 m. The units have a minimum flange/web thickness of 25 mm and weight about 36 kg/m length for a width of 300 mm and about 54 kg/m length for width 600 mm.

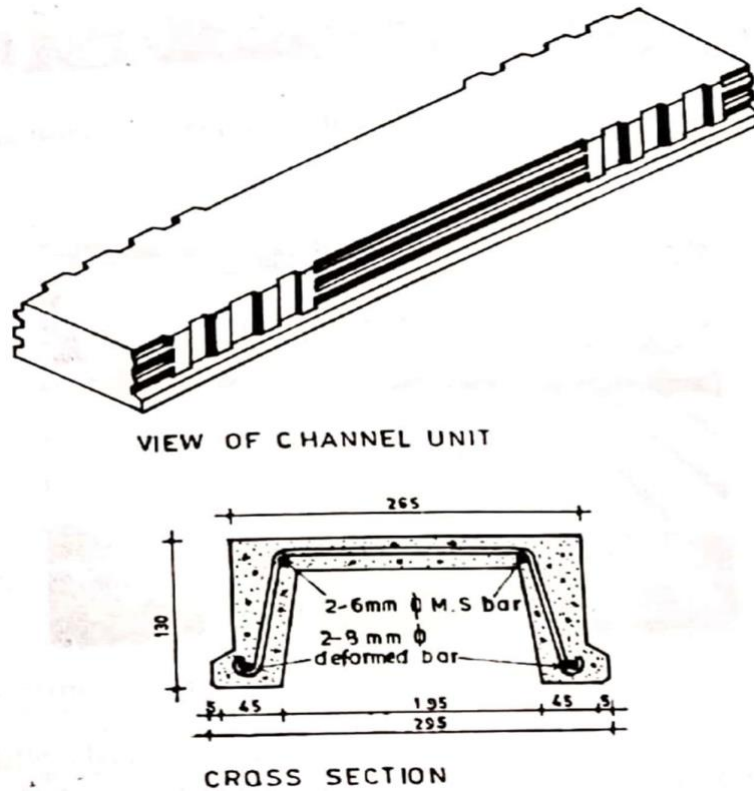


Fig. 5.1 Precast RC channel unit

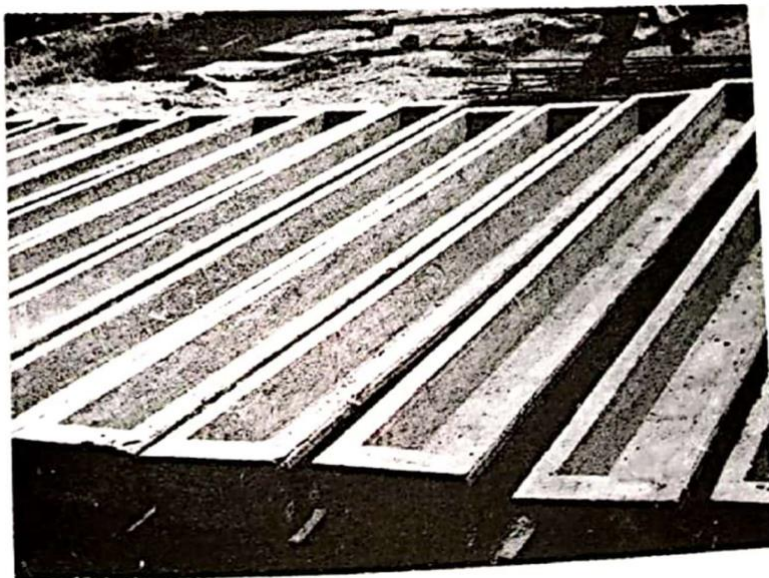


Fig. 5.2 RC channel units in casting yard

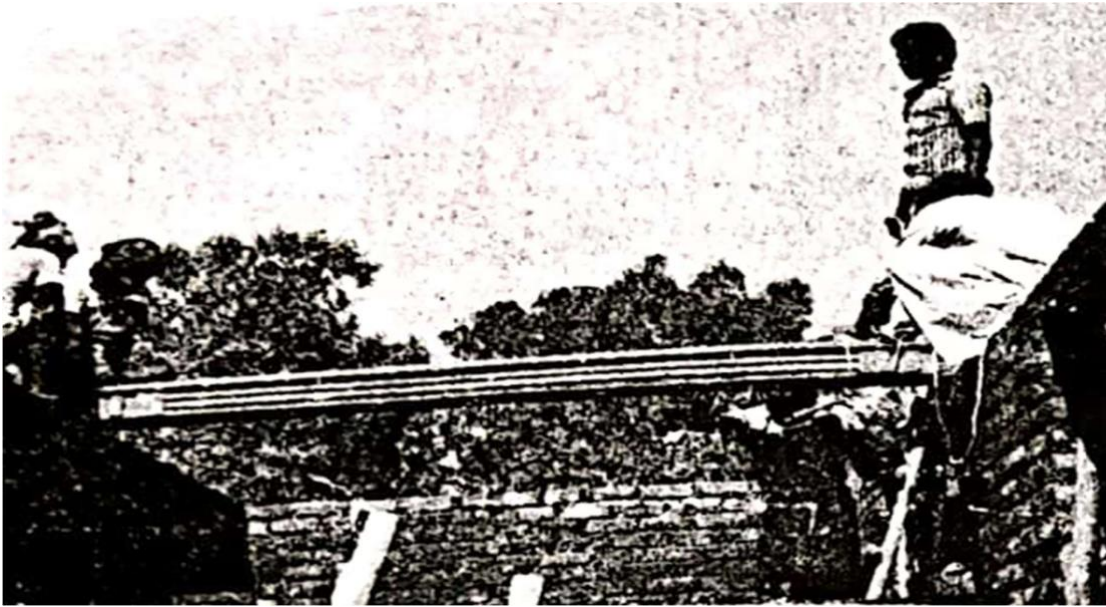


Fig. 5.3 Precast RC channel unit being placed in position

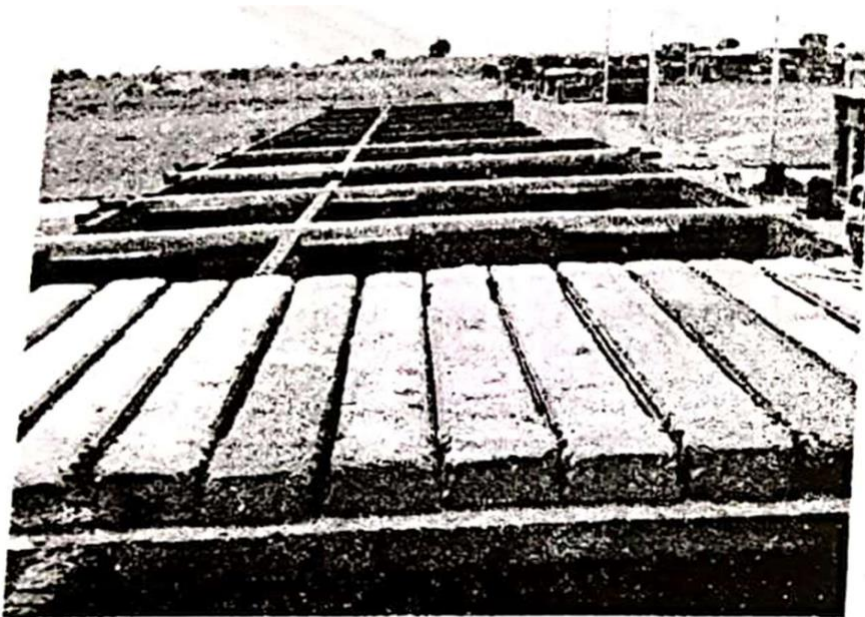


Fig. 5.4 Precast R.C. channel units placed across load-bearing walls

These units are cast in well-seasoned timber moulds with M-15 concrete with 10 mm and down aggregate. The concrete is consolidated using plate vibrator to prevent any honeycombing. The units are cured for 14 days keeping the trough filled with water and then air cured for another 14 days before placing it in position. The units are then placed side by side on the wall and joints are filled with M-15 concrete which is compacted properly. Negative reinforcement is placed in the continuous spans before filling the joints between the units. Electrical conduits, fan hooks, etc., are embedded in the cast-in-situ joint concrete. The system

provides an overall saving to the extent of 20% over the traditional RC slab. With this system of roofing, there is as well saving in construction time which is generally of the order of 20%.

Channel units have been adopted in a large number of experimental projects. Construction of 16 type-II quarters for Military Engineering Services (MES) in New Delhi, 200 low-cost houses at Neyveli (Tamil Nadu), 198 houses for the Low-Income Group (LIG) and 50 houses for the EWS at Faridabad by the Haryana Housing Board, and 20 residential units at Nagpur by the Western Coalfields Ltd. has been undertaken under the NBO Experimental Housing Scheme. Examples of large-scale utilisation of such units by the Construction Department/agencies in different regions of the country include projects involving construction of 2,000 primary schools in Uttar Pradesh (UP): 400 staff quarters in Madhya Pradesh by the State P.W.D: Kendriya Vidyalaya, New Delhi, 304 LIG and 84 EWS houses at Madurai by the Tamil Nadu. Housing Board; 500 MIG and 800 LIG houses at Sheikhpur by the Group Housing Society sponsored by Hyderabad Urban Development (HUDA), and 350 MIG (Middle Income Group) and 730 EWS houses at Velungudi and 124 LIG houses at Palayankattai by the Tamil Nadu Housing Board. Recently, 518 LIG houses have been constructed at Erode (Tamil Nadu) using channel units.

Precast RC Plank and Joist System

This system of roofing (Fig. 5.5 to 5.7) is based on the use of precast RC planks and partially precast RC joists. Planks are 60 mm thick in the centre and 30 mm thick at edges. The size of the planks will depend upon the size of the room. However, the maximum size of the plank is restricted to 1500 mm x 300 mm with 3 number of 6 mm dia. bars as main reinforcement. The planks are placed on partially precast RC joists of size 1500 mm x 150 mm. Cast-in-situ concrete laid over the joists and haunches of the plank after placing 6 mm dia. bars in both directions to make a monolithic structure. Fig. 5.5(b) illustrates connection of precast RC plank and precast brick

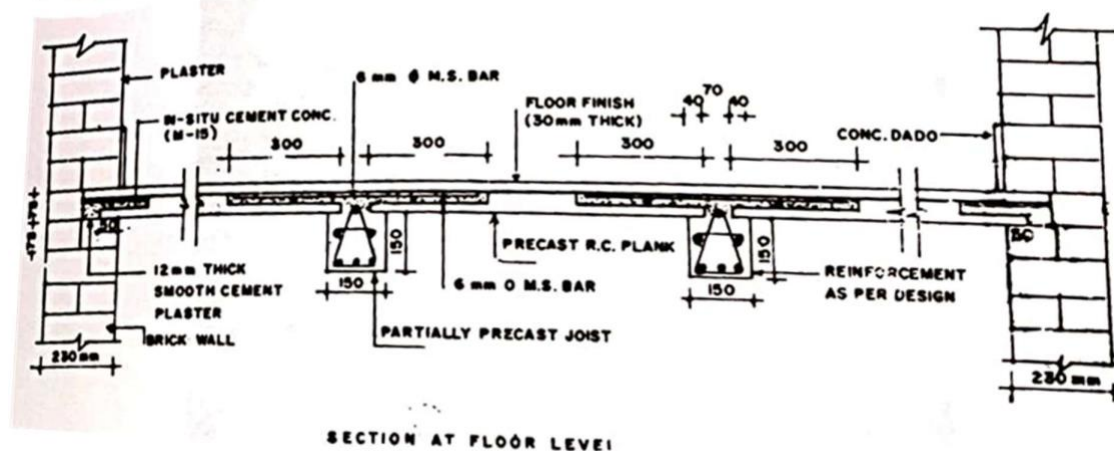


Fig. 5.5(a) Precast RC plank and joist

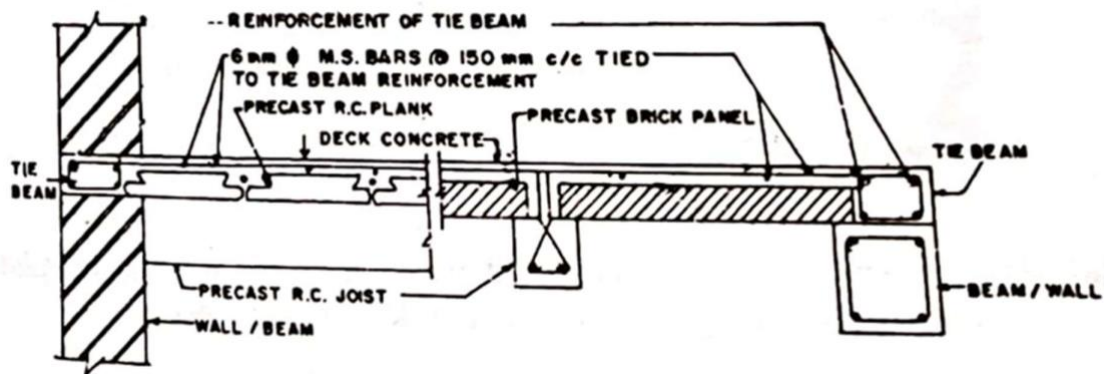


Fig. 5.5(b) Connection of precast RC plank and precast brick panel floor/roof (with deck concrete) with tie beam



Fig. 5.6 Casting of RC solid planks at the casting yard

panel floor/roof with tie beam. Provision of tie beam helps ensure structural stability in the event of earthquakes. Such beam extends to the depth of the precast components plus the thickness of structural deck concrete. The planks and joists are cast either in well-seasoned good quality timber moulds or steel moulds. Curing is done for 14 days before they are laid.

This system results in saving of the material and costs to the extent of 20 to 25% in comparison to traditional RC roof. There is also a saving in the in the construction time of the order of about 25% in laying such roof as compared to the conventional in-situ RC slab.

Under the NBO Experimental Housing Scheme, 44 LIG and 24 EWS houses have been constructed using this technique at Ludhiana by the Punjab Housing Development Board. Another 80 houses have been constructed at Lucknow by the U.P. P.W.D., 348 staff quarters

constructed at New Delhi by the All-India Institute of Medical Sciences, and 48 houses constructed at the Indian Institute Technology (IIT), New Delhi.

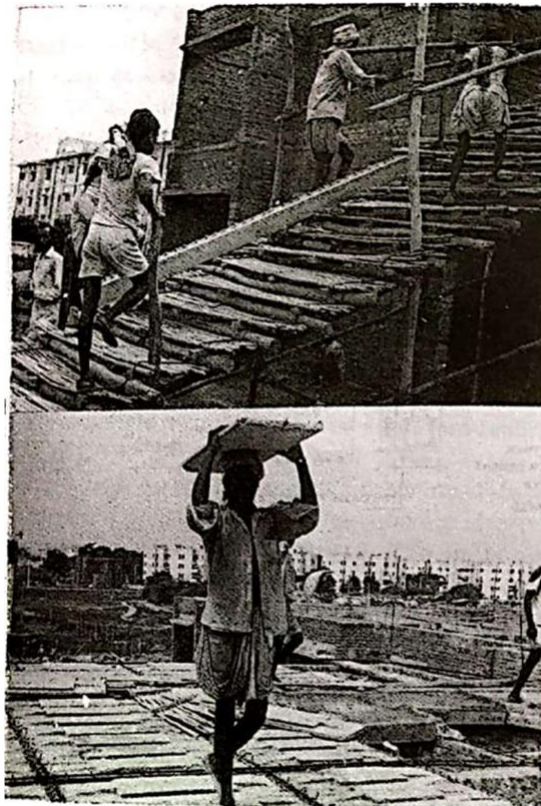


Fig. 5.7 (a) Manual carriage of RC joist, (b) Manual carriage of precast RC solid plank. on the roof comprising of RC plank and joist system

Apart from the experimental projects, this technique has been adopted for construction on mass scale by several construction agencies in the country. Utilising this system of roofing 13,000 houses have been constructed in Andhra Pradesh, 600 houses in Bombay, 850 houses in Madhya Pradesh, 100 houses Kalyani in West Bengal, and 500 houses in Ghaziabad. Recently, 14,000 houses using precast RC solid planks and joists have been constructed in different parts of Rajasthan.

Precast Brick Panel System

The system (Fig. 5.8 to 5.10) comprises of prefabricated brick-based components and partially precast RC joists. Panels are laid over the partially precast RC joists. A typical size of the brick panel is 1040 mm × 640 mm in which 16 bricks, two bars of 6 mm dia. and M-15 concrete are used. The partially precast joists are 130 mm × 100 mm in section and their reinforcement will depend upon the span. After proper curing and drying, the panels are placed over the partially precast joists over which concrete is laid in normal manner. The panels are cast in well-seasoned timber moulds over casting platform and it does not require high degree of skill. Weight of a brick panel is only 75 Kg, which facilitates its handling manually. At the time of roof assembly, partially precast joists are kept propped till the deck concrete and the concrete

over the joists have attained strength. The thickness of deck concrete over the panels is kept as 30 mm. Deck concrete is laid after placing 6 mm dia. mild steel bars, one bar on each panel in both directions. The savings in materials and cost as compared to the traditional RC slab are of the order of about 30 percent.

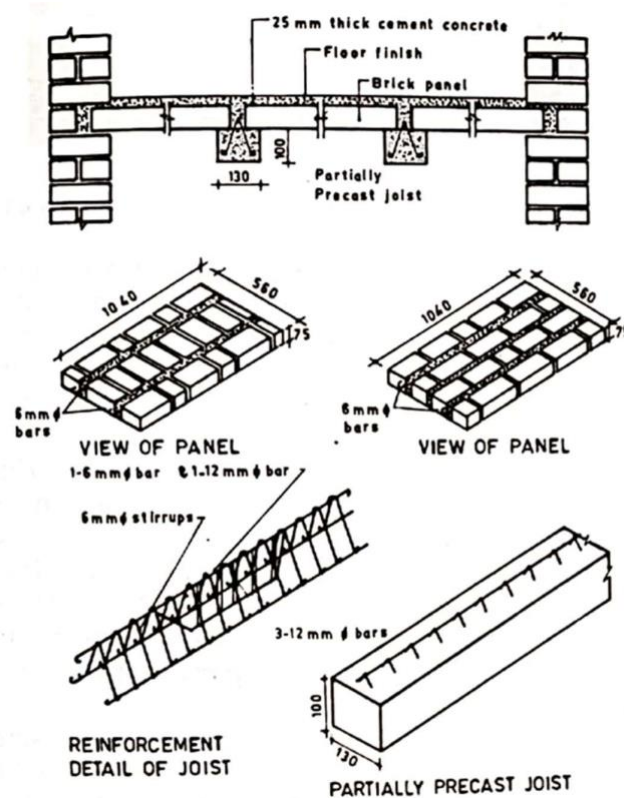


Fig. 5.8 Brick panel roofing system



Fig. 5.9 Precast reinforced brick planes at casting yard

More than 40,000 houses have been constructed utilising brick panel roofing system in India. This includes 30,000 houses in rural areas of Punjab, 200 EWS houses at NOIDA (U.P), 3500 houses at Ghaziabad, 200 houses at Lucknow, 1000 houses at Cochin, and 200 EWS at Bhopal and 600 houses for the Sri Lankan repatriates

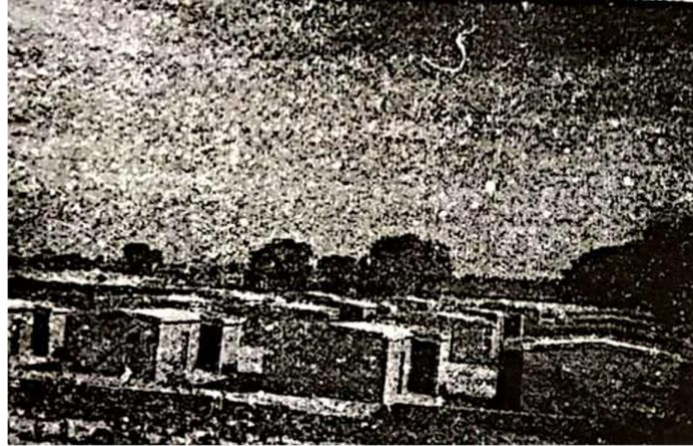


Fig. 5.10 Houses made of precast brick panel system in rural Punjab

Thin Precast RC Lintel

Normally, in building construction, 15 cm thick RC lintels are provided over door/window openings. Lintels can be as well designed as a composite member consisting of 25 mm thick RC lintel with the masonry on top. The precast thin lintel (Fig. 5.11) 75 mm thick, and 230 mm wide with 3 numbers of mm dia M.S. bars could be used for opening up to 1.8 m provided bricks used have minimum crushing strength of 10N/mm^2 and mortar is not leaner than 1: 6

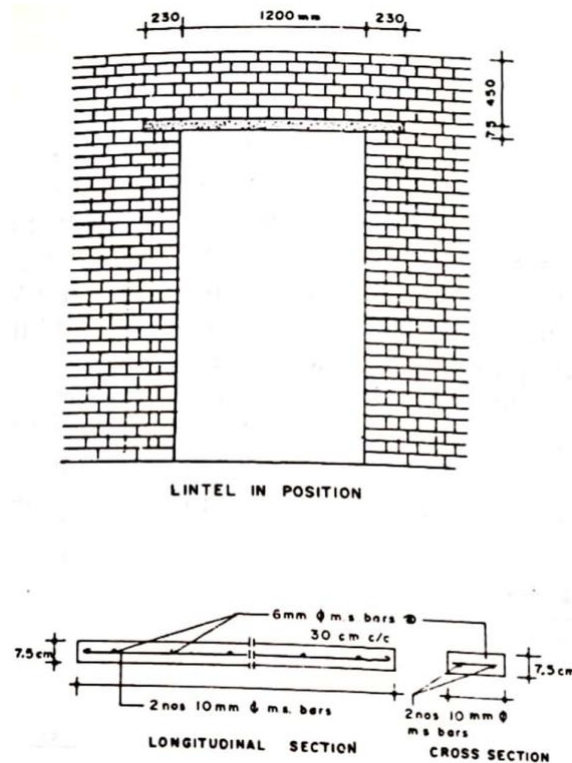


Fig. 5.11(a) Precast thin RC lintel

cement-sand and the height of masonry above the lintel is at least 45 cm. Use of precast lintels speeds up the construction of masonry walls besides eliminating shuttering and centring. About 50% saving in material and overall cost is achieved apart from saving in construction time by adoption of precast lintel.

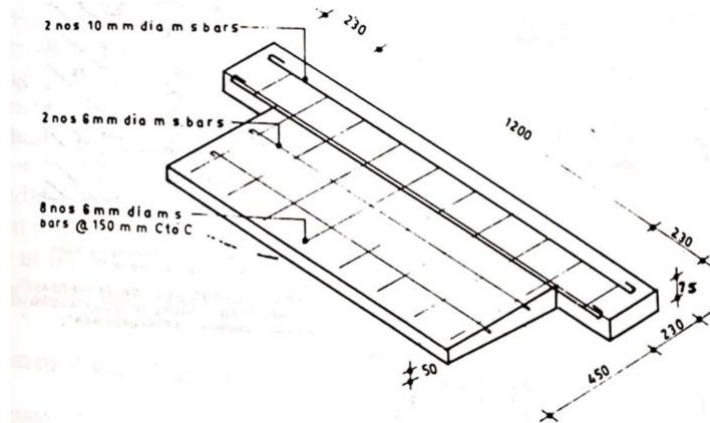


Fig. 5.11(b) Thin precast Intel with chajja

Precast thin lintels have been adopted in a number of Experimental. Housing projects. Beside this, such lintels have been used in construction of a large number of houses by the CPWD, MES, Railways, State Housing Boards and the Delhi Development Authority (DDA) and Tamil Nadu Housing Board.

Precast RC Frames for Doors and Windows

With fast depletion of the forest resources timber has now become a scarce material. This has necessitated search for substitute materials for making frames for doors and windows. Reinforced concrete frames provide an answer to the problem. Precast RC frames are alternate to traditional timer frames for doors and windows and economical compared to frames made out of traditional timbers like teak and deodar. They are more suitable than timber frames in wet areas like lavatories, bathrooms and kitchens and are resistant to termite attack and growth of fungus. The RC frames are better resistant to fire as compared to wooden frames.

Precast RC frames (Fig. 5.12) are recommended for a maximum opening of 2.25 m with cross section 60 x 100 mm or 70 x 75 mm for single shutter doors and 60 x 120 mm for double-shutter doors. Three bars of 6 mm dia. are provided as minimum reinforcement which are firmly held by means of 3 mm dia, steel bars spaced at not more than 20 cm c/c. The frames are cast in M-15 concrete.

Various methods of fixing shutters to the frames have been tried out. But the most successful and durable method appears to be use nuts running through the frame. Hold fasts, M.S. plates receiving the tower bolt, aluminium sleeves for fixing shutters are embedded in the frame during casting hinges.

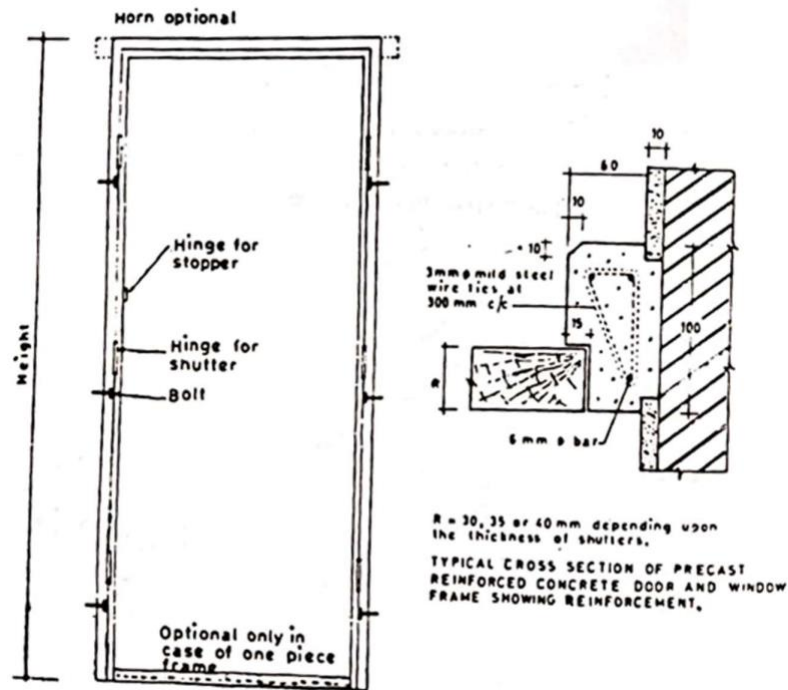


Fig. 5.12 Precast RC frames for doors/windows

Precast RC frames have been tried out in several experimental projects. Under the NBO Experimental Housing Scheme, this technique of construction of frames for doors and windows has been adopted of 10 Asst. Professors quarters at Patiala (Punjab), 992 prefabricated houses at Delhi by the Municipal Corporation of Delhi (MCD), 16 bed hostel at Roorkee (UP), 32 staff quarters for Hindustan Housing Factory (now Hindustan Prefab. Ltd.- HPL) at New Delhi, 16 type III quarters at New Delhi by the Army Headquarters, 280 low-cost houses at Neyveli by the Neyveli Lignite Corporation (NIC) and 50 and 8 EWS houses at Faridabad by the Haryana Housing Board. Subsequent to adoption of precast RC door the window frames in the experimental project sponsored by the NLC, the Corporation has since used such frames in construction of 7,000 quarters for its employees.

Adoption of Total Prefabrication for Mass Housing in India

In urban agglomerations, especially in metropolitan centres, where there exists huge backlog of housing, it has become a matter of great urgency to build large number of houses economically at fast rate to wipe out the shortage. Speed in construction could be greatly enhanced with the application of total prefabrication for mass housing programmes. By adoption of total prefabrication, completion time of housing projects could be reduced drastically, to the extent of 40 per cent.

In some advanced countries, appreciable reduction in time taken for house construction has been achieved through adoption of prefabricated method of construction- Building components complete in all respect are prefabricated in housing factories which are

subsequently transported to construction sites where only erection and assembly works are undertaken. In the erstwhile USSR, there were as many as 569 prefabrication factories producing prefabricated building components comprising of reinforced cement concrete, prestressed concrete and lightweight concrete. Standard sizes of building components based on modular co-ordination concept of planning are produced in these factories.

In India, too, prefabrication technology for mass housing has come of age. Prefabrication increasingly being adopted for large-scale housing programmes in the country. Brief description of some popular prefabrication systems adopted for mass housing programmes in India is given below.

I. Hindustan Prefab Limited

The Hindustan Prefab. Limited, New Delhi, which is a government undertaking, established in 1950, has produced a variety of prefabricated building components such as precast beams, columns, floors and roofing elements and panels etc., and has undertaken a number of mass housing projects in which the prefab. components produced by the said factory have been employed in a big way.

II. Shirke-Siporex Consortium

M/s Shirke-Siporex Consortium, Pune, manufacture precast hollow columns and beams, lightweight concrete slabs and blocks, and wall panel with the Swedish know-how which have been extensively used in mass housing programmes undertaken by them in different parts of the country. The firm has built over 1,00,000 dwelling units of all categories both in India and overseas. It has also developed anti-corrosion treatment, which has been very effective in preventing the occurrence of corrosion in M.S. reinforcement in Siporex precast concrete components used in the coastal art highly polluted areas like Bombay.

III. Large Panel Prefabrication

This is an 'on-site' prefabrication system employing precast concrete pane and hollow clay block for external walls and precast concrete and brick panels for internal walls; and precast funicular shells for roof. Being an on-site prefabrication system, no heavy equipment except a crane is required for handling of precast components. This technique of construction could be employed for construction of residential building up to 4-5 storeys. An experimental housing project sponsored under the Experimental Housing Scheme of NBO which involved construction of 144 flats in six, four-storeyed blocks undertaken by the Tamil Nadu Housing Board at Madras, has successfully employed large-panel prefabrication system under technical guidance from SERC, Madras.

Subsequent to field trial of large-panel prefabrication (Fig. 5.13 and 5.14)) at Madras, totally prefabricated concrete houses were constructed at Vashi (New Bombay) during 1972- 74 using large-sized prefabricated wall-panels, floor slabs, bath room units, staircases, etc. All the structural members and cladding components were precast and were erected using a 50- tonne travelling granty crane, especially fabricated for this purpose. The cost of construction

compared very favourably with the conventional system. Better utilisation and marginal saving were achieved in the major construction materials such as cement and steel both of which were controlled items at that point of time. The system is also currently being employed in construction of about 3,800 apartments by CIDCO (City and Industrial Development Corporation of Maharashtra Ltd.) in New Bombay.

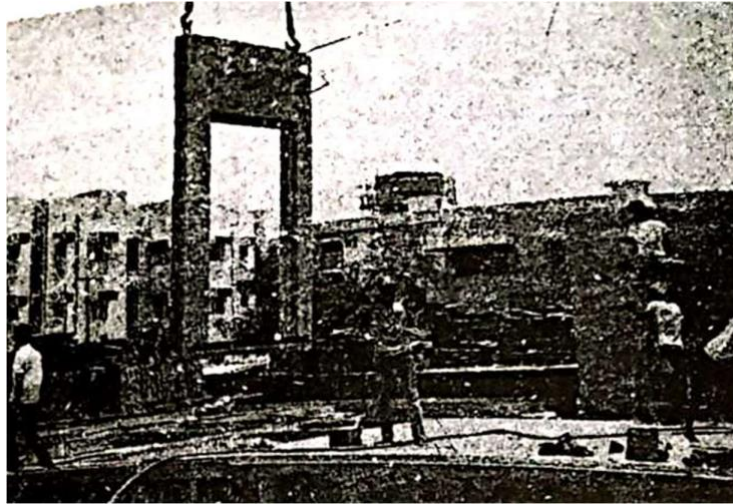


Fig. 5.13 Handling of large sized prefabricated wall panel

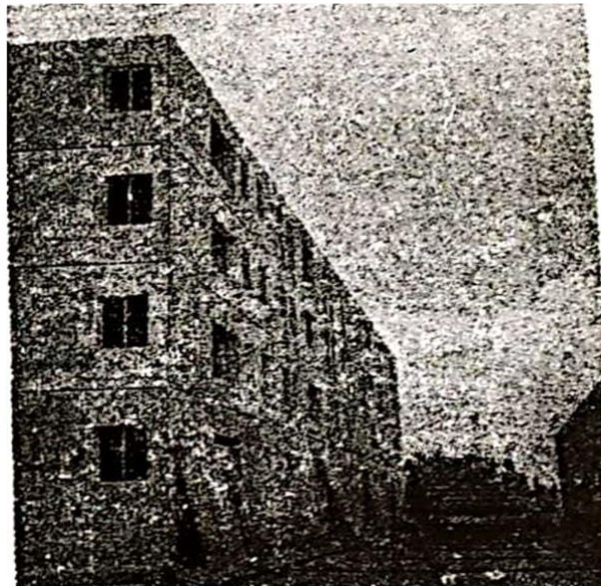


Fig. 5.14 A view of houses made of large-panel prefabrication system

Large panel precast system has also been adopted in a big way in construction of residential accommodation and office facilities for the Sardar Sarovar Narmada Nigam Ltd. in Gujarat. Experience of application of large panel precast system for mass housing has been very encouraging. It has been possible for the project authorities to construct 20 tenements covering 1,000 sq.m. per day using the aforesaid prefab construction system.

IV. Tunnel form System

In this system of prefabrication for mass housing programmes promoted recently by ECC-Construction Group of L&T (Larsen and Toubro), Bangalore, the reinforced concrete walls and floor slabs are all cast in a continuous pour of the concrete using room sized structural steel framework called Tunnel form. In a broader sense, it is an in-situ construction system like the traditional construction methodologies based on use of brick masonry for walls and RCC for suspended slabs—all expected at site. However, in this system RCC is the only material used for the single operation casting of structurally stable shell of walls and slabs for the houses. The versatility of concrete being moulded into any shape of its formwork is better exploited in this mechanised system than the other alternative choices of industrialised constructions, viz, large panel prefabrication or large precast panels for wall supporting standardised prestressed concrete flooring/roofing elements. The structure is cast in such a way that it is and includes load bearing walls, cross walls that it is totally monolithic and includes load bearing walls, cross walls and slabs. Components like facade, panels, balconies sun shades, landing staircases, cupboard, kitchen platforms, water tanks, etc, are all prefabricated and integrated into the system. Quick stripping and removal of forms are made possible by accelerated curing using hot air so as to facilitate casting of one or two flats in "One-Day cycle" Accessories like frames of doors and windows, electrical conduits, switch boards, function boxes, etc., are fixed into the forms along with the reinforcement prior to concreting. After concrete attaining the required strength, the form can be collapsed, withdrawn from the house and then shifted to the next casting. With proper maintenance a tunnel form can be used for 800 or more repetitions. Plastering is totally eliminated and concrete walls and slabs are painted directly under this system of construction.



Fig. 3.16 Placement of precast walls under tunnel form construction (after Bibl. 146—reproduced with permission from Architecture: Design, New Delhi)

The basic know-how of the technology was acquired from SPECTRA of France. Some modifications have been incorporated in the original technology to suit the local tradition and living requirements.

With thinner walls (13-15cm), the system provides 5-10% additional carpet area. Elimination of plastering on tunnel formed walls and ceilings saves on labour and materials.

The speed of construction with adoption of this system is claimed to be much faster, providing up to 125 sq.m. of construction per day from one set of formworks operated with one crane.

This technique of construction (Fig. 5.15 and 16) has been employed in the construction of 1,500 dwelling units measuring 75 sq.m. each for the employees of TISCO (Tata Iron and Steel Company Ltd.) at Jamshedpur. The entire construction was completed within a period of only 32 months. Almost 85% monolithic construction was achieved in construction of sub-residential flats by casting all walls and floor slabs of individual flats in a single pour.

Using this technology blocks up to 20 storeys have since been completed.

V. L&T Form System

This is also promoted by the ECC-Construction Group of L&T, Bangalore. Like the tunnel form system, the walls and slabs in this system also are of reinforced concrete. In this system, the walls are cast using large panel formwork in the first stage, and slabs over walls are cast in the second stage. Since the walls and slabs are cast in two separate stages accelerated curing is avoided in this system of construction.



Fig. 5.17 Construction of precast houses employing L&T form work system at Puttaparthi (Andhra Pradesh) (after Bibl. 146 -- reproduced with permission from Architecture + Design, New Delhi)

Accessories like frames of doors and windows are fixed to the formwork prior to concreting or provisions are made for door and window openings at the predetermined positions in the formwork.

The above technique (Fig. 5.17 and 18) has been adopted in construction of 240 community flats for the Shree Satya Sai Central Trust at Puttaparthi (Andhra Pradesh). The flats measuring 42 sq.m. each were built a record time of 180 days using the L&T form system.

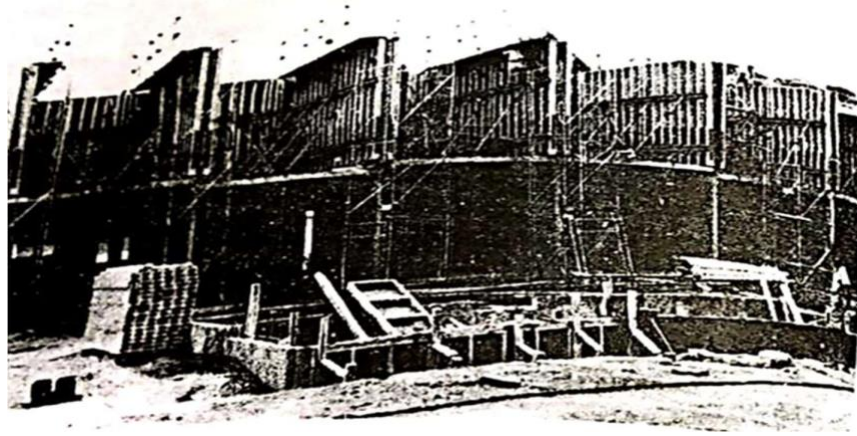


Fig. 5.18 Formwork for walls under L&T formwork system (after Bibl. 146-- produced with permission from Architecture + Design, New Delhi)

VI. Concrete Hollow Blocks for Walling and Roofing

Concrete hollow blocks (Fig. 5.19 and 20) could be ideal substitute for the conventional clay bricks whose production destroys fertile top soil to the extent of 50,000 acres every year in the country. These blocks are also ideally suited for the regions like the southern states of the country where soil of requisite grade is not available for manufacture of good quality bricks. In India, a plant for manufacturing concrete hollow blocks was setup in Madras as early as late 40s. In the USA and Canada, more than 80% of walling is currently done using concrete masonry blocks, and manufacture of concrete masonry units is an established business in these countries. In Europe, too, there has been immense increase in block masonry constructions in recent years. Such blocks have been used for load bearing constructions up to 10 storeys. Concrete masonry unit is a modular concrete product. The reasons for its popularity in the western countries are its various advantages like durability, strength, structural stability, fire resistance, thermal and acoustical characteristics, attractive appearance and, above all, economical and faster construction.



Fig. 5.19 Casting of concrete hollow blocks



Fig. 5.20 Construction of masonry wall with concrete hollow blocks

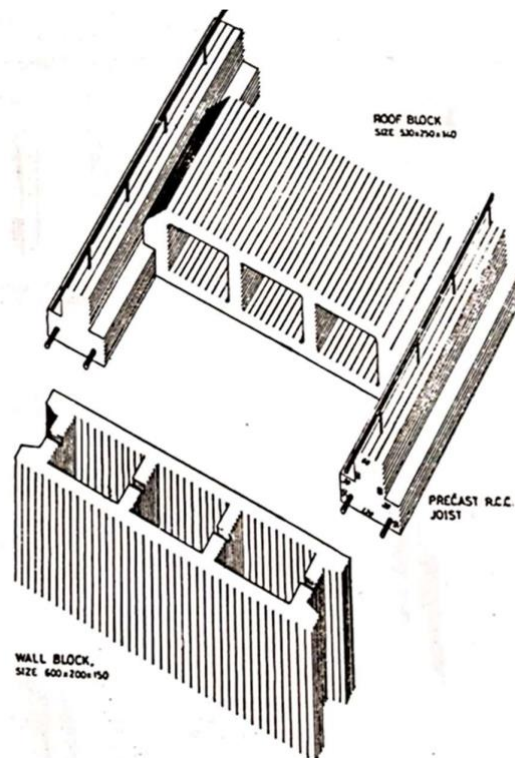


Fig. 5.21 Zip block system – precast element

There are, at present, quite a number of manufacturers of hollow block making machines in India, the Miasto Shrike and Gospel to name a few.

The Zip block system of prefabrication, promoted by M/S CVG Shelters, Madras, utilises precast concrete hollow blow blocks of 600 x 200 x 150 mm size for walling, and precast 'hourdi' type hollow blocks of 530 x 250 x 140 mm as filler material for roofing with partially precast joints for floor of roof slabs (Fig. 5.21 and 22). Under this system of construction, foundation is not prefabricated. Adoption of piles helps foundation become. integral part of the system. The Zip block system using 150 mm blocks as load bearing walls could be used for construction of houses up to three storeys. It has already been successfully tried out in the construction of several two storeyed houses in Madras.

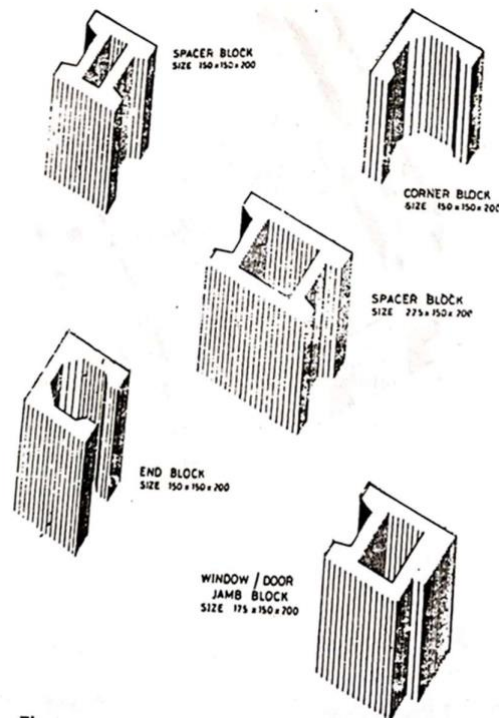


Fig.5.22 Zip block system – Special precast blocks

Plastering can be avoided by application of this system of construction. Moreover, formwork is eliminated as the zip block roofing is supported by partially precast joists, and by utilizing precast lintels and sunshades.

General Remarks on Precast Roofing/Flooring Systems

A few points which are applicable to all the precast roofing/flooring systems so far described deserve special consideration for ensuring their satisfactory performance.

- i) First, the planning of the building. It is very important that the building plan should be prepared keeping uniform span for the prefab components. By doing so, precast units of only one size will be required in size as it will reduce the number of sizes of moulds. Building planned size cross wall system is well suited for adoption of precast units.
- ii) The moulds for pre-casting may be manufactured with good quality timber or steel. Steel moulds can be used for a large number of castings and will be economical in the long run. Good timber moulds may be used for 30 to 40 casting operations.
- iii) The precast units may be designed by common methods of structural design adopting ultimate load method of design or limit state method as per the codal provisions of relevant Indian standards. Both ordinary mild steel or high strength deformed bars may be used for reinforcement.

- iv) Precast units should be adequately cured before they are used in the building. Handling methods should be given particular attention to avoid breakage and damages to the finished surface.
- v) Joints are the most critical parts in prefabrication. Their efficiency decides the success or failure of the system. Moreover, the joints of prefabricated elements tend to develop shrinkage cracks. It is, therefore, imperative to take adequate precaution to ensure watertightness of joints and structural integrity of the precast elements. As a precautionary measure, precast elements are required to be transversely connected with reinforcing bars to avoid opening of joints. To overcome the problem of differential shrinkage between the precast element and the cast-in-situ concrete in joints, use of non-shrinking cement grouts and sealants is a positive solution, and contributes to the satisfactory performance of joints.
- vi) All openings should be pre-planned and provided for in the moulds so that chipping on breaking of the precast units is completely avoided.
- vii) Fan hooks, electrical conduits, etc., should be embedded in the in-situ concrete.
- viii) The waterproofing of the roof should be done with great care as number of joists are involved in the precast roof assembly.
- ix) The Armenian earthquake (December 1988) had exposed the vulnerability of precast roofing components to the forces of earthquake. Precast concrete frame buildings suffered extensive damage during the said earthquake. Precast large panel buildings, however, gave better performance during the earthquake due to proper bracing. Different parts of such buildings were better tied together than the previously quoted buildings. As such, it becomes necessary to take adequate precautions in design and construction of precast roofing system. The revised edition of IS: 4326 provides the necessary guidelines regarding suitable strengthening measures to be adopted for design and construction of prefabricated buildings in earthquake prone areas. Moreover, occurrence of the September 1993 Maharashtra earthquake in a zone which is considered to be seismically stable, provides warning with regard to uncertainty of occurrence of earthquake. This necessitates adoption of additional. The IS: 4326-1993 (Second Revision) stipulates that all the floors and roofs to be constructed with small precast components shall be provided with tie beams. Such beam runs all-round the floor or roof and binds together all the precast components to ensure diaphragm action of the floor/roof. Tie beams shall be made of cement concrete of grade not lower than M15. They shall be provided in all longitudinal and cross walls.

Economical Walling System

Wall is one of the important structural elements in building and housing construction which, on an average, accounts for 25 to 30% of the cost of construction in civil work. Traditionally, brick masonry and stone masonry are the two most popular walling systems used in building

construction in India. Burnt clay brick is one of the predominantly used basic construction materials in the Indo-Gangetic plains while in States like Rajasthan and Gujarat, locally available stone is used as an important building material for wall construction.

As a result of research work undertaken at the CBRI, Roorkee, several innovations in design construction of load bearing brick masonry have been evolved. These innovations have been experimented with under the Experimental Housing Scheme of NBO. These include single brick thick load bearing masonry walls for four-and five-storey construction, nineteen-centimetre-thick walls (using conventional bricks) for up to two-storey construction, half brick thick walls, fly-ash gypsum brick masonry and precast stone block masonry. Adoption of these systems has not only resulted in economy but also in less consumption of bricks and mortar, utilisation of industrial waste and saving in time.

Single Brick Thick Load Bearing Wall

Till a few decades back, it was in practice to construct two-or three-storeyed buildings and houses as the pressure on urban land then was not as considerable as it is now. Design of load bearing walls was earlier made on empirical methods based on the established experience of architects/engineers. For example, wall thickness for a four-storeyed construction as per the empirical method was kept as 46 cm, 34 cm, 23 cm and 23 cm for ground, first, second and third storeys respectively. Through a rationalised design method, it has now become possible to construct single brick thick load bearing wall in multi-storeyed construction through use of better-quality bricks. In present construction practices, single brick thick load bearing walls could be adopted for construction of four-and five-storeyed residential buildings provided bricks of good quality having compressive strength of $7-10.5 \text{ N/mm}^2$ ($70-105 \text{ kg/cm}^2$) are available. Apart from saving in consumption of bricks, cement and steel, such types of load bearing brick structures provide more covered area, resulting in up to 15% economy in cost of construction.

Fig. 5.23 Five-storeyed building having single brick thick load bearing walls in all storeys



Construction of four-storeyed residential buildings having single brick thick load bearing walls have been extensively adopted by the CPWD and also by various other

construction departments such as MES, DDA, ONGC, IDPL, etc. Recently, 400 four-storeyed tenements were constructed with single brick thick load bearing walls in Erode town in Tamil Nadu.

In Calcutta, construction of a block of 50 residential units having 5 storeys with single brick thick load bearing walls on all the floors had been taken up under the NBO Experimental Housing Scheme which resulted in saving 6% the cost of construction. As a follow up of the said experimental construction, a large number of 5-storeyed residential buildings have been put up by difference bearing on of five-storey residential buildings having single brick thick load bearing walls had been undertaken by the CPWD in 1980 (Fig. 5.23.)

Nineteen-Centimetre-Thick Load Bearing Brick Masonry Walls

It has been a common practice these days to adopt 23 cm thick load bearing walls using conventional bricks for normal house construction. However, by a rational design of the structure it is possible to construct 19 cm (7.1/2") thick load bearing walls (Fig. 5.24 and 25) for up to two storeys using burnt day bricks.

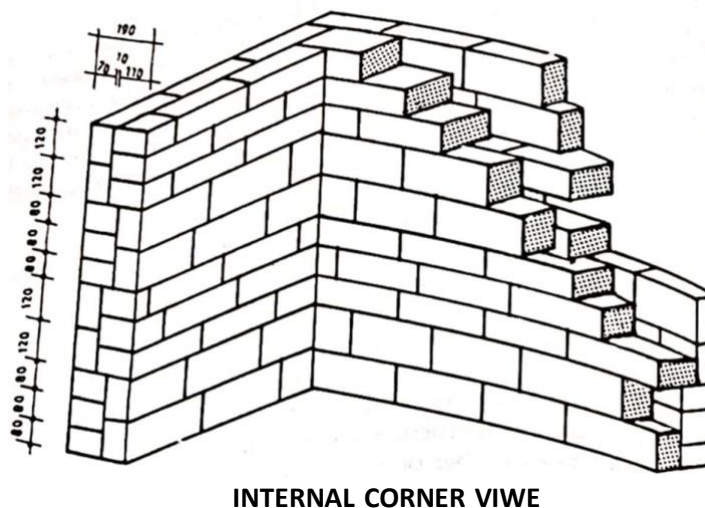


Fig. 5.24 Nineteen centimetre thick load bearing wall

In ordinary brick masonry construction, English Bond and Flemish Bond are generally adopted for getting the required wall thickness. In case of 19 cm thick walls a different type of bond is adopted and construction also differs from the traditional type. Under this technique of all construction, bricks are laid on edge as well as flat. In order to obtain a height of 23 cm, three bricks are placed on bed whereas adjacent to its two bricks are placed on edge. The next 23 cm height is obtained by placing two bricks on edge and three bricks on bed on the same side of the wall. The pattern is then repeated to obtain full height of the wall. Corners and junctions of such walls are also constructed in a different manner to break the continuity of joints.

Such walls can be designed according to the provisions of IS:1905-1980. A saving to the extent of 16% has been achieved in cost of bricks and mortars employing this technique.

However, there has not been a saving in the labour component, the reason being that two masons are employed in the construction of such walls as both its sides are to be constructed independently with bricks on edge as well as flat. As masons work on both faces of the wall a smooth face is obtained on each side as a result of which the thickness of plaster required is less. The reduced wall thickness also results in providing the same floor area in a reduced plinth area.



Fig. 5.25 A view of double storey quarters having 19 cm thick load bearing walls at Ludhiana

This technique has been tried out under the NBO Experimental Housing Scheme at Ludhiana. Subsequent to this experimentation, a large number of houses have since been constructed adopting this technique in different parts of Punjab.

Half Brick Thick Load Bearing Wall

Generally, half brick thick walls are constructed as non-load bearing walls. But with proper planning, it is possible to put up half brick thick (115 mm) walls behaving as load bearing walls. In construction of such walls, over-lapping of half brick thick walls at regular intervals act as pilasters of 23 cm x 23 cm size (Fig. 5.26 and 27). These pilasters transfer the load of the roofing systems to the foundation.

Masonry work in brick of 10 N/mm² strength in cement sand mortar 1:4 is generally adequate for constructing such walls. A ring beam of 23 m width and 7.5 cm depth with 10 mm. dia. MS bars is required to provide bearing for the roof. Saving to the extent of 15% is achieved when come pared with the traditional 23 cm thick load bearing walls.

This technique of wall construction was utilised in construction of or block of 12 single-storeyed tenements at a resettlement colony of the DDA in 1978. Precast RC channel units were used for roofing in this project. The Maharashtra Housing and Area Development

Authority (MAHADA) has adopted this technique on a large scale. The technique has also been tried for construction of 200 EWS houses at Bhopal.

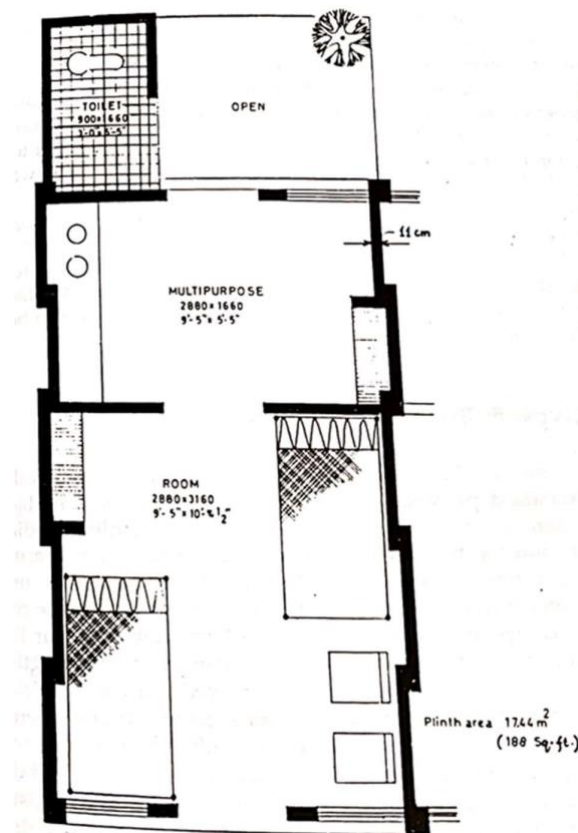


Fig. 5.26 House for the urban poor

Fly ash Gypsum Bricks for Masonry

Common burnt clay bricks can be manufactured from those soils which have got standard percentage of clay, sand, etc., within variable limits. However, there are areas especially in central and southern India where good quality soil for manufacture of burnt clay bricks is not available.

Fly ash gypsum bricks are a good substitute for burnt clay bricks for the areas in which fly ash is readily available. These bricks can be manufactured with a composition of 92% fly ash, 4.5% powdered kankar lime and 3.5% gypsum in volumetric proportion. A compressive strength of 5-6 N/mm² (50-60 kg/cm²) is expected to be achieved from these bricks. They also have good insulation properties and consequently provide better thermal comfort to the occupants in comparison with the common bricks.

The method of manufacture of these bricks is very simple and does not require elaborate arrangements. Fly ash kankar lime and gypsum are powdered and mixed in the desired proportion volumetrically in dry state. A measured quantity of water is then added and mixing is done in a pan till the mixture forms a stiff paste. The paste is poured into the steel moulds. The mould is released when the mix has hardened sufficiently. The bricks are then

cured for at least 10 days by stacking them properly and occasionally sprinkling water in order to keep them wet. The bricks are subsequently subjected to curing by air for about a fortnight. These bricks do not require to be burnt kiln.

Fly ash gypsum bricks have been used at Neyveli by the Neyveli Lignite Corporation under the Experimental Housing Scheme of NBO using fly ash from the Neyveli thermal power plant which complains high percentage of lime.

Stone Block Masonry

The precast stone block masonry (Fig. 5.28) is an improvement over the traditional stone masonry which requires excessive materials and more skilled labour and is time-consuming. The precast blocks can be easily produced at factory or at site with semi-skilled workers without any special equipment.



Fig. 5.28 A view of house with precast stone block masonry

Considering the ease in handling and other requirements the nominal length and height of the block are kept 30 cm and 15 cm, respectively, with three widths 20 cm, 15 cm and 10 cm. The actual block dimensions are short by 1 cm to accommodate mortar joint thickness. These blocks weigh from 9 to 18 kg. To get a stone texture on the outer face of wall, the blocks are cast so that the bottom face during casting forms the exposed face when laid in wall.

The blocks are cast either in individual steel moulds of fixed type or split type or gang moulds of six blocks (Fig. 5.29). These moulds are made of 4 mm thick M.S. sheet and are welded from outside so that the internal edges are sharp and free from welding flux.

A smooth level and hard surface is required for casting the blocks. The casting platform may consist of 30 mm thick 1:3:6 cement concrete finished smooth with neat cement. For production rate of 500 blocks per day a casting area of 80 sq.m. is required.

The stone masonry blocks are made of large size stone pieces bonded together with lean-cement-concrete mix of 1:5:8 (cement: sand: stone aggregate 10 mm and less). The stone

pieces used are as large as can be accommodated in the mould. These can be from 5 to 26 cm in size and obtained either from quarry or breaking the river boulders. This will result in a good flat stone-texture.

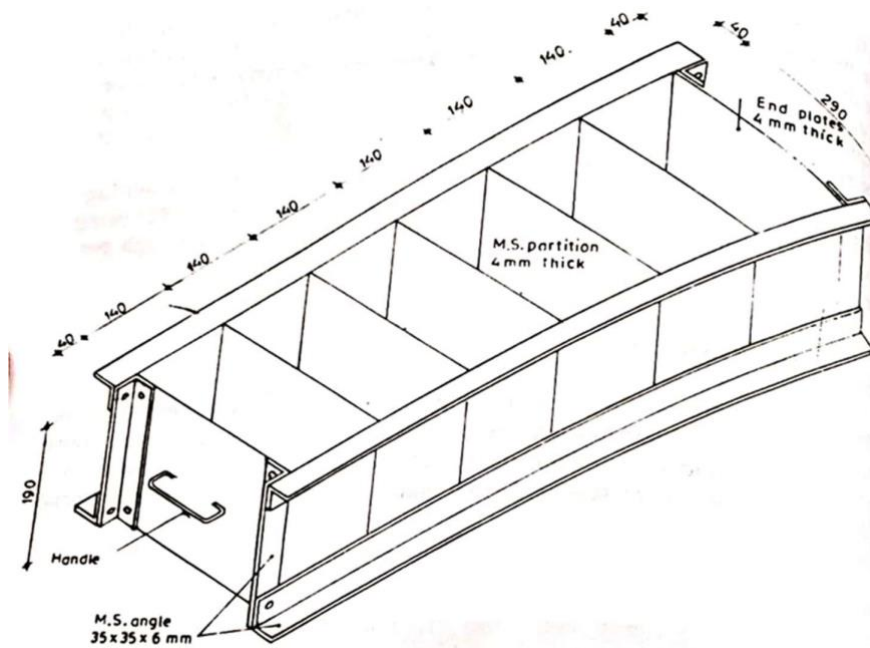


Fig. 5.29 Moulds for stone block masonry

These blocks are used both for load bearing as well as non-load bearing walls. The thickness of the wall depends upon the load coming over it. This construction system provides an economy of the order of 15 to 20% in the walling cost against the conventional method of construction, viz., random rubble masonry. Stone block masonry provides finishing of brick masonry on both sides of the masonry wall.

These blocks have been used in an NBO experimental project at Bhopal sponsored by the Bhopal Development Authority. Over 25,000 houses have been constructed with stone masonry blocks in different parts of the country. In Andhra Pradesh and West Bengal, prefab factories have come up which supply these blocks. At Dehradun (U.P.), 330 residential quarters have been constructed using such blocks. They have also been using in construction of 8 educational institutions and 1,000 residential quarters in Rajasthan where 8 million blocks were used in place of 40 million conventional bricks.

Single Stack System of Plumbing

In the conventional system of drainage for building two separate pipes are provided, one for the soil which takes discharge from the water closet, urinals, etc., and the other for the discharge from baths, sinks and wash basins, etc. Besides, a vent pipe is also fitted with both the pipes to prevent breakage of seal of the traps of different application. The single stack system of plumbing (Fig. 5.30), which replaces above system, is essentially a one pipe system in which the wastages from kitchen, both, etc., and soil from toilets is carried out of the

building in a single stack. The ventilation is also completely omitted and the service stack itself serves the purpose of a ventilation pipe. It embodies the merits of both the conventional two pipe system and the modern one pipe system. The use of this system effects saving in the quantity of pipes and fittings and also gives a cost saving of about 30% in plumbing work.

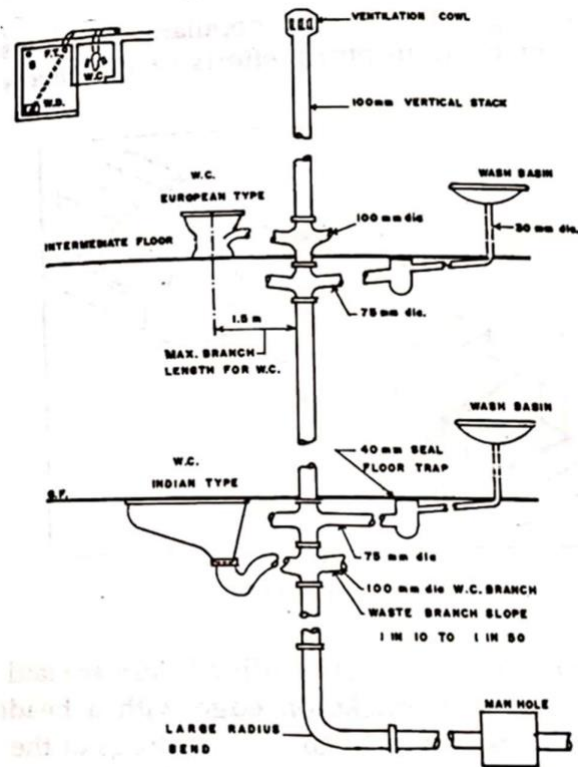


Fig. 5.30 Single stack system of plumbing

Single stack system of plumbing has been used in large number of the NBO experimental projects. Beside this, this has been also used in mass housing programmes undertaken in different parts of the country.

Role of Nir Mithi Kendra's

Establishment of the Nir Mithi Kendra's (Building Centres) across the country has played a commendable role in transfer of low-cost housing technology in the country. The First Nir Mithi Kendra was established in August 1989 in Quilon (Kerala). The Quilon Nir Mithi Kendra has played a key role in ensuring appropriate technology transfer of low-cost technology options to field needs through skill upgradation and training of artisans and production of building materials and components based on local natural or waste resources. The successful operation of the Quilon Nir Mithi Kendra has now blossomed into a national programme of the network of Building centring. A Centrally-sponsored scheme for establishing a national network of building centres was launched during 1988-89 which is being implemented through HUDCO. 366 locations have been identified so far for establishment of such centres out of which 176 are operational to this date. Description of some of the popular construction techniques successfully promoted by the Nir Mithi Kendra's are given below.

i) Rat Trap Bond

The Rat Trap Bond (Fig. 5.31) has been popularised in the southern states of the country through the inspiring efforts of the renowned architect Laurie Baker. Under this technique of bonding bricks are laid with a 75 mm cavity between two leaves of bricks on edge with a header after every stretcher with overall thickness of 23 cm. All the frogs of the bricks in such bond point towards the cavity. Use of such bond prevents transmission of dampness. It has better insulation properties than the conventional masonry wall. A saving up to 25% in the consumption of bricks is claimed through application of the rate trap bond. Such bond can be used for construction of 2 storey buildings.

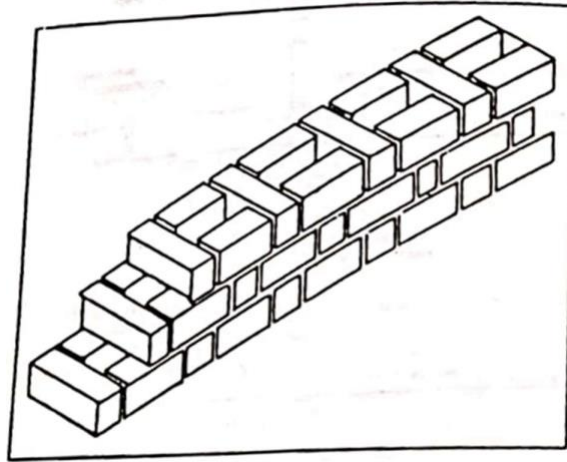


Fig. 5.31 Rat trap bond

ii) Filler Slab

The basic concept behind use of filler slab (Fig. 5.32) is to remove substantial portion of concrete below the neutral axes which does not contribute to the tensile strength of the slab. Various types of filler slab have been used, such as Mangalore pattern roofing tile, coconut shell etc. Use of filler slab leads to economy to the extent of 15% compared to the conventional RCC slab. Such slab is comparatively lighter in weight, and has increased insulation properties because of voids between the tiles.

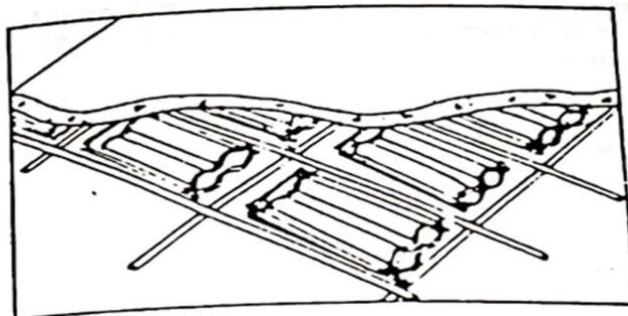


Fig. 5.32 Filler slab

iii) Brick Arches

Often costly RC lintels over door and window openings are not necessary up to 1.2 m. Ordinary brick-on-edge flat arch lintel is just sufficient (Fig 5.33). If something stronger is necessary, a hollow arrangement of brick-on-edge filled with one or two steel bars in concrete will carry sufficient amount of loads above. The latter type of brick arch leads to an economy to the extent of 50% compared to the conventional RC lintel.

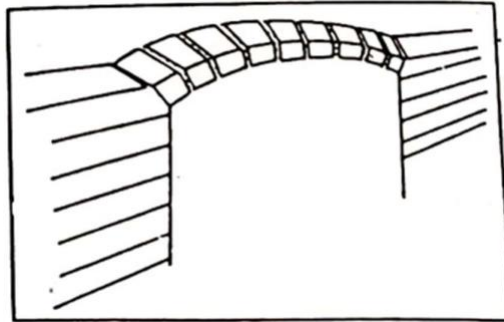


Fig. 5.33 Brick arches

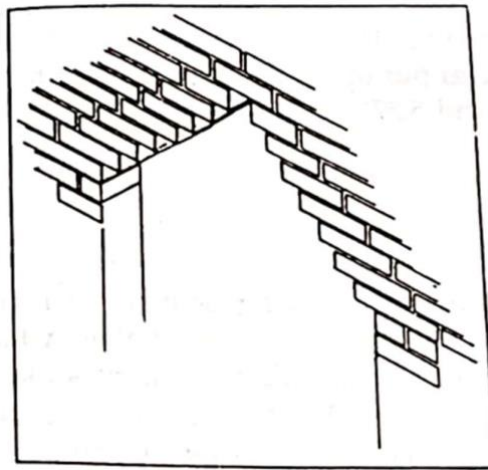


Fig. 5.34 Corbelling

The most inexpensive way of spanning an opening in a brick wall is to use the simple stepped arch or corbelled brick arch (Fig. 5.34). Each row of bricks projects about 5 cm beyond the course below until the bricks meet together in middle. No formwork or shuttering is necessary in construction of such brick-arch.

iv) Waffle Shell System (Roofing)

This system of roofing (Fig. 5.35) based on the funicular shell concept supported on a grid of cast-in-situ beams. This concept of roofing was evolved as a structural element deriving its strength through shape. It develops a state of pure compression under the action of applied loads and hence ideally suited for concrete construction. Ceiling plaster is also eliminated under this system of roofing. Apart from reduction in construction time,

application of this system of this system of roofing leads to saving in the cost of roofing to the extent of 10 to 30% compared to the cost of construction of conventional RC slab.

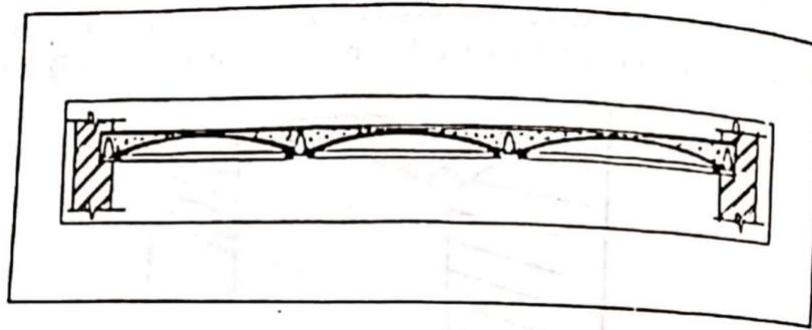


Fig. 5.35 Waffle shell system

v) Jalis

For low-cost housing, the orthodox timber windows can be replaced by 'jalis' or honeycombed wall. The jalis can be of varied patterns, and are efficient in performance. This has also been tried out in number of rural demonstrations houses put-up in different parts of the country by the NBO RHDCs. (Fig. 5.36 and 5.37)

Future Strategy

- i) In order to meet the growing demand of housing in the country, there is exigency to adopt appropriate technology which could effect economy in consumption of scarce building materials like cement and steel as well as speed in construction. In urban centres, especially in metropolitan centres, where there exists huge backlog of housing, there is urgency to build large number of houses at affordable cost to meet the requirement of housing.



Fig. 5.36 Use of jalis in place of window and ventilator in a NBO rural demonstration house

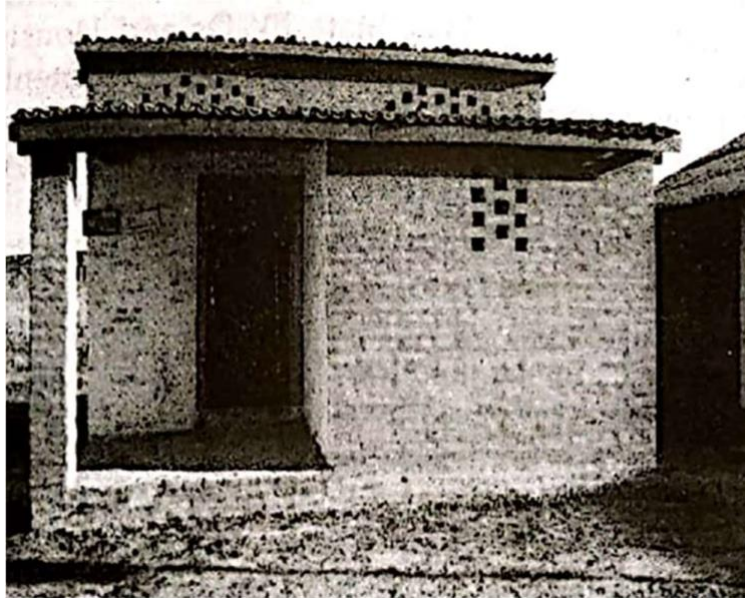


Fig. 5.37 Use of jalis in place of window and ventilator in a NBO rural demonstration house

- ii) If the present scenario on the housing front is to be improved in the right earnest, prefabrication alone appears to be the answer. Prefabrication in housing can greatly speed up construction. However, because of involvement of prohibitive initial capital investments, total prefabrication may not be the pragmatic proposition realizing the strains of present Indian economy. For ensuring speed and economy in mass construction of houses, only resort left then is to take recourse to adoption of partial prefabrication for large-scale housing programmes in which certain building components viz. roofing and walling units, lintels, etc. are precast components cast at construction sites or manufactured in industrialised production units. Size and weight of the precast building units are such as to be handled manually. Use of such precast component's effects saving in consumption of cement and steel and also speed in construction.
- iii) It is noticeable that adoption of precast roofing/flooring systems may not turn out to be an economical preposition for individual housing as it involves additional expenditure on account of necessary and the infrastructural facilities required for roofing/flooring system is by and large recommended for large-scale housing projects, especially for the categories of population belonging to Lower Income Group and Economically Weaker of the society.
- iv) There is need to lay greater emphasis on adoption of modular coordination in planning, design and construction of buildings employing use of prefabricated components. Modular co-ordination of dimensions is the process of organising the dimensions in building so as to reduce variety of sizes to which components have been made, and enable them to be used together at the building site without much modification. Internationally, a basic module of 10 cm (IM) and multi module of multiples of 30 cm (3m). have been accepted for adoption in building construction.
- v) It is observed that despite the soundness of the low-cost construction techniques, construction departments generally have some reservation with regard to

adoption of such techniques. This hesitancy on the part of user agencies has to be removed. With this objective in view, major construction departments/agencies like CPWD, State PWDs and Housing Boards, etc., should earmark 1% of their building budget for experimental construction. This will help provide requisite encouragement to engineers and user agencies to undertake constructions involving larger adoption of cost-effective construction techniques.

- vi) For wider adoption of the proven cost reducing techniques, there is need to incorporate construction items involving utilization of such techniques in the departmental schedule of rates of the various construction departments/agencies in the country. The All-India Standard Schedule of Rates and specifications prepared by the NBO at the behest of the Planning Commission in 1977 incorporates some popular precast roofing systems. Of late some construction departments like the CPWD have incorporated some of the proven innovative cost-saving construction techniques in their departmental Schedule of Rates at the initiative of the BMTPC.
- vii) Contractors undertaking experimental projects have often shown their reluctance in undertaking experimental projects involving incorporation of innovative cost-effective construction techniques since these projects, according to them are a departure from the traditional building practices. This is an important factor responsible for inhibiting the large-scale promotion of innovative cost-effective construction techniques. With a view to motivate the contractors to undertake such construction, it is imperative that some sort of fiscal incentive is given to them by way of making some bonus in the initial stages.
- viii) The dearth of skilled masons is one of the important factors inhibiting larger adoption of partial prefabrication in building construction. The masons in the existing labour market are not competent enough to undertake constructions involving use of precast building components. They require training for undertaking such construction. Establishment of more and more of Nirmiti Kendras (Building Centres) in different parts of the country would be an effective step in this direction. The existing Building Centres at Quilon, Jaipur and Nizamuddin (New Delhi) have set examples for excellence in imparting training to masons in respect of skill upgradation.
- ix) The various S.I. codes on different topics (Annexure II) related to low-cost housing (Annexure II) as brought out by the BIS could play major role in promotion of low-cost housing technology in the country.
- x) Taking recourse to modern construction management practices such as PERT (Programme Evaluation Review Technique) and CPM (Critical Path Method) could lead to avoidance of time and cost overruns as has been experienced in some of the NBO experimental projects (See Case Study).
- xi) Due attention should be given to the spatial planning in any housing project. The size and shape of the building have bearing on the cost of construction. Irregular shapes require more design efforts and will hence result in greater cost of construction.

Case Study: Adoption of Precast RC Plank and Joist System for Roof/Floor in the AIIMS Project (after Bibl. 103)

The National Buildings Organisation was approached by the All-India Institute of Medical Science (AIIMS) to suggest ways and means for undertaking a large scale housing project at lowest possible cost and within shortest possible time of completion. The project comprised of construction of 328 house for Doctors/Nurses of AIIMS, New Delhi.

The project was completed in two phases. In first phase 160 double-storeyed quarters were started in January, 1980 and were completed in six months' time through application of modern construction management practices CPM.

In the second phase 168 quarters were constructed in three storeys. All the 328 quarters are under occupation at present.

The following innovative design concepts, new construction techniques and materials were adopted in this project

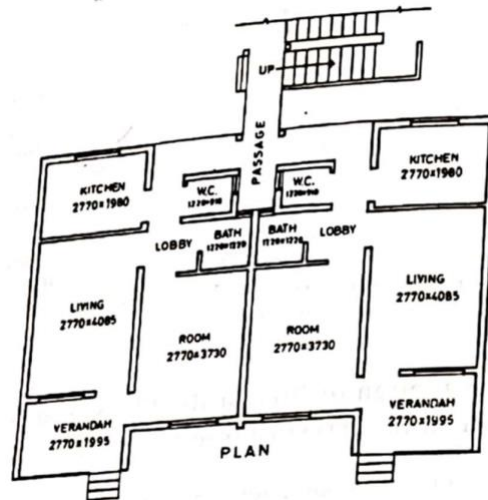


Fig. 5.38 A.I.I.M.S. housing project for doctors and nurses

- i) Modular concept in planning, designing and construction based on 3M (300mm) grid to establish dimensional co-ordination
- ii) Precast RC roof/floor system comprising of precast joists and planks to achieve saving in consumption of cement and steel and also speed in construction.
- iii) Lower ceiling height of 2.8 metres to reduce cost of construction.
- iv) Cement lime composite mortar for masonry and plastering to effect saving in consumption of cement
- v) Single stack system of plumbing to reduce cost of plumbing
- vi) PVC pipes for cold water supply to replace GI pipes which are costly.
- vii) Secondary species of timber for joinery work to replace primary species of timber which are scarce and costly
- viii) Common staircase to serve eight dwelling units in two storeyed constructions to reduce area for vertical travel, and
- ix) Ferrocement overhead tanks for water supply which are lighter and cheaper.

Fig. 5.38 shows the lay out plan. Specifications adopted in construction area indicated at Annexure III.

Assessment of Savings

The cost per unit with experimental specifications, excluding electrical services, was Rs. 20,768 (1979-80 price). The cost of corresponding conventional construction would have been Rs. 22,117 per unit. There was thus an overall saving of 1,348 in the construction of each quarter, item wise comparison of experimental and conventional construction is given at Annexure IV.

As compared to the plinth area rate of Rs. 435.80 per sq.m. of conventional type of constructions, a plinth area rate of Rs. 409.25 (1979-80) was achieved by incorporating new design concepts, cost-effective construction techniques and materials and modern construction management practice.

As a result of adoption of innovative specifications and techniques in this project, there was a saving of 36.7% in cement and 20.3% in steel. The saving of cement and steel in each item is given at Annexure V. An overall saving of 6.09% was achieved in this project.

Annexure I

Cost-effective Techniques and Alternate Building Materials Adopted in NBO Experimental Projects

A. Foundation

1. Under reamed pile foundation.
2. Hyperbolic paraboloid footings.
3. Leaner cement concrete (1:8:16) for foundation.
4. Pocket connections for foundations.

B. Superstructure Walls

5. Brick cavity walls.
6. Single thickness load bearing walls for 4 and 5 storeyed buildings.
7. Load bearing concrete panels for walls.
8. Use of modular bricks.

9. Use of improved methods of brick laying and plastering.
10. Polyethylene D.P.C.
11. Light weight aggregate concrete blocks for masonry in partition,
12. Hollow cinder blocks for masonry.
13. Damp proof course in cement mortar 1:4 with air entraining agent.
14. Large composite cement concrete panels using hollow blocks and solid bricks.
15. Flyash-lime-gypsum bricks for masonry.
16. Nineteen cm walls for superstructure masonry.
17. Stone block masonry.
18. Rat Trap Bond for Masonry walls.

C. Doors and Windows

19. Precast RC frames for doors and windows.
20. Particle board for shutters of doors/windows.
21. Steel frames for doors/windows.
22. Frameless doors.
23. Coal tar drum sheets for door shutters.
24. Red mud shutters for doors and windows.

D. Roofing/Flooring Slabs

25. Precast channel units for roofing/flooring.
26. Prestressed channel units.
27. Precast cored units for roofing/flooring.
28. Precast cellular units for roofing/flooring.
29. Precast waffle units for roofing/flooring.
30. Doubly curved tile roofing system.
31. Batten and hollow block construction for roofing.
32. Precast concrete folded plate roofing.
33. Precast hyperbolic shell units for roofing.

- 34. Precast RC T-Beam' unit for roofing.
- 35. Precast RC planks and joists for roof/flooring.
- 36. Precast 'L' pan roofing.
- 37. Precast panels with earthen 'kulhars'.
- 38. Thin RC ribbed slab for floors/roofs.

E. Flooring

- 39. Stabilised soil cement base for floors.
- 40. Magnesium Oxychloride flooring.
- 41. Floor base of 1:20:40 cement: flyash: sand.
- 42. Reduced floor thickness.

F. Structural Members (Beam, Columns, etc.)

- 43. Hollow beams.
- 44. Precast RC columns and beams.
- 45. Precast concrete staircase elements.

G. New Materials

- 46. Use of deformed bars.
- 47. Partial replacement of cement by flyash.
- 48. Hydrated lime.
- 49. Blending of Badarpur sand with Jamuna sand in mortar.
- 50. Plastic stays.
- 51. Plastic wash and hand basins and other fittings.
- 52. Secondary species of timber for doors/windows.
- 53. Reactive clay pozzolana.

H. Services (Sanitary, Water Supply)

- 54. PVC pipes for water supply.

55. Single stack system of plumbing.

56. Evaporative roof cooling system.

I. Architectural Planning and Structural Designs

57. Lower ceiling height.

58. Precast lintels.

59. Design based on ultimate load theory.

60. Design for solar energy utilisation.

Annexure II

List of BIS Standards on Various Topics Related to Low-Cost Housing

- Guidelines for Requirements of Low-Income Housing: SI:8888-1994, Part I (Urban Housing)
- Code of practice for construction of walls using precast concrete stone masonry blocks IS:14213-1994
- Specification for prefabricated brick panel and partially precast concrete joist IS:14143-1994
- Code of practice for design and construction of roof and floor with prefabricated brick panel IS:14142-1994
- Specification for precast RC channel unit for flooring and roofing IS:14201-1994
- Code of practice for construction of floor and roof using RC channel units IS: 14215-1994
- Specification for precast RC planks and joists for flooring and roofing IS:13990-1994
- Code of practice for construction of floor and roof using precast RC planks and joists IS:13994-1994

Annexure III

Brief Specification of Construction in the AIIMS Project

1. Foundations

- a) Cement concrete 1:5:10 with 40 mm nominal size brick aggregate;
- b) Lime concrete with brick balast with 10 percent lime mortar 1:1:1 (1 lime putty: 1 surkhi: 1 fine sand);
- c) Brick work in cement lime mortar 1:2:9 (1 cement: 2 lime putty: 9 sand)

2. D.P.C.

25 mm thick D.P.C with 1:2:4 cement concrete (1 cement: 2 coarse sand: 4 graded stone aggregate 10 mm nominal size) with a coat of residual petroleum bitumen of 80/100 penetration applied @ 1.7 kg/sq.m.

3. Super Structure

- a) Brick work in cement lime mortar 1:2:9 with FPS bricks;
- b) Half brick work in cement lime mortar 1:1:6 (1 cement: 1 lime putty: 6 fine sand).

4. Reinforced Concrete (RC)

RC 1:2:4 in lintels, suspended floors for bath & WC, chajjas etc, with exposed surfaces finished in cement mortar 1:3 (1 cement: 3 fine sand)

5. Roof/Floor

Precast RC planks and joist system in RC 1:2:4 (1 cement: 2 coarse sandy graded stone aggregate 10 mm nominal size).

6. Joinery

35 mm thick panelled or panelled and glazed shutters of 2nd class Deodar wood fixed to T-iron frames.

7. Flooring

25 mm thick cement concrete 1:2:4 flooring finished with a floating coat of neat cement over base course e if lean concrete and a layer of sand filling

8. Finishing

- a) Cement lime plaster 12/15 mm thick with cement lime mortar 1:1: 7 (1 cement: 1 lime putty: 7 fine sand);
- b) White-washing with lime;
- c) Painting on steel or wood work-2 or 3 coats over a coat of prime

9. Roofing

- a) Mud phuska terracing with a layer of tiles;
- b) Lime concrete terracing.

Alternative Building Materials for Low-Cost Housing

Introduction

Building materials account for 70-75% (Table 1) of the total cost of construction. Due to large-scale construction programmes in the country, the demand for conventional building materials like cement, steel, bricks and timber has outstripped their supply. Moreover, the exponential population growth and the existing housing shortage have made the situation even more alarming. There is a general shortage of conventional building materials like cement, steel, bricks and timber in different parts of the country. In 1993, there was an estimated shortage of 55,000 million bricks, 17 million tonnes of cement, 285 million tonnes of stones, 2.7 million tonnes of steel and 13 million cu.m. of timber. It is thus quite obvious that the present available stock of building materials in the country is not in a position to meet the ever-growing demand of housing. In the present situation, the country has very little option but to rely increasingly on locally available cost-effective building materials and components. The R&D efforts undertaken in the country by various research organisations like CRBI, Roorkee; SERC, Madras; National Council for Cement and Building Materials (NCB), Ballabgarh; Central Fuel Research Institute (CFRI), Dhanbad; NBO and BMTPC have led to development of various low cost, low energy consuming building materials using several industrial and agricultural wastes. There is considerable potential for exploitation of the agro- industrial wastes for their commercial production and large-scale application in construction programmes.

Table 1

Average cost break-up of building construction

Materials (73%)		Labour (27%)		Component-wise (100%)	
Cement	18%	Carpenter's wages	5%	Walls	10%
Iron & Steel	10%	Mason's wages	10%	Roofs	30%
Bricks	17%	Unskilled labour	12%	Doors & Windows	25%
Timber	13%			Foundation	10%
Sand	7%			Flooring	15%
Aggregate	8%			Finishing	10%

Substitute for Scarce Materials

Research findings of various research institutions like CBRI, NCB and CFRI etc., have led to introduction of number of alternative building materials like hydrated lime, asphaltic corrugated sheets, cellular concrete, fly ash bricks plastic pipes, secondary species of timber, pozzolana cement and overhead tanks, etc. The NBO had prepared feasibility reports for

setting up plants for manufacture of such building products. Application of these substitute materials have resulted in the saving of scarce materials like cement and steel. Some of the alternative materials for low-cost housing are discussed below.

Hydrated Lime

It is normally assumed that 70% of the cement produced is consumed in building construction out of which 40% is used for masonry mortars and plasters. Out of this, quite a large percentage of cement can be profitably replaced by lime alone or in conjunction with pozzolana.

In recent times, the use of lime in building construction has decreased due to the varying quality of material available in the market and its inherent defect such as slow-setting and laborious preparation of mortars.

An innovative process of hydration of lime has been developed which makes available hydrated lime dry form, standard quality and ready to use form as in the case of cement. It can be stored in bags for 6 months without any deterioration. NBO had set-up a demonstration-cum-production unit for demonstrating the production of standard quality hydrated lime and lime pozzolana mix in 1976. Consequently, over 50 hydrated lime plants of total capacity of half a million tonnes have been established in various parts of the country over the past one decade. These plants include 3 plants set up by public sector construction agencies, one each at Srinagar, Itarsi and Raipur. It is estimated that 1 million tonnes of cement in building construction could be substituted by the materials produced in the aforesaid units with anticipated saving of over Rs. 20 crore per year.

Lime Pozzolana Mix

The clay pozzolana, available in the market, is known as surkhi¹ which is produced by grinding brickbats and is of variable quality. A new process of production of clay pozzolana has been developed in which clays of suitable specifications are calcined to the optimum temperature and then ground to a fine powder as per the IS specifications. The pozzolana, reactive surkhi, has lime reactivity 5 to 10 times that of ordinary surkhi made powdering burnt bricks.

The lime pozzolana mix conforming to LP-20 as per IS: 4098 is made by grinding the mixture of hydrated lime, and clay pozzolana with a small percentage of gypsum/sodium metasilicate, etc. to improve their early characteristics. The investment required in production of hydrated lime/lime pozzolana mix is about Rs 200 to Rs 250 per tonne as compared to a heavy investment of Rs 1,400 per tonne required for production of cement.

Ferrocement

Ferrocement is a highly versatile form of reinforced concrete made of cement mortar and wire mesh reinforcement and possessing unique blend of strength and serviceability. The major advantages accruing from its usage include thin and light structure, the ease with which it can

be cast, its amenability to repairs in case of local damage, considerable saving in formwork and saving in cost.

It is one of the possible substitutes for timber. It can be moulded into shape by applying suitable techniques. Partitions, doors, cupboards, service core units, decorative walls, ceiling units in various designs and water tanks are some of the items which could be made using ferrocement without any difficulty.

Designs have been developed by SERC, Madras for ferrocement water tanks (Fig. 6.1) of 800 litres, 5,000 litres, 7,500 litres and 10,000 litres capacity. These tanks are economical compared to the traditional brick-walled water tanks up to the extent of 30%. They have been successfully tried out in many locations in India. It is estimated that about 5,000 ferrocement water tanks are being produced every year in Madras alone both in public and private sectors using the SERC technology.

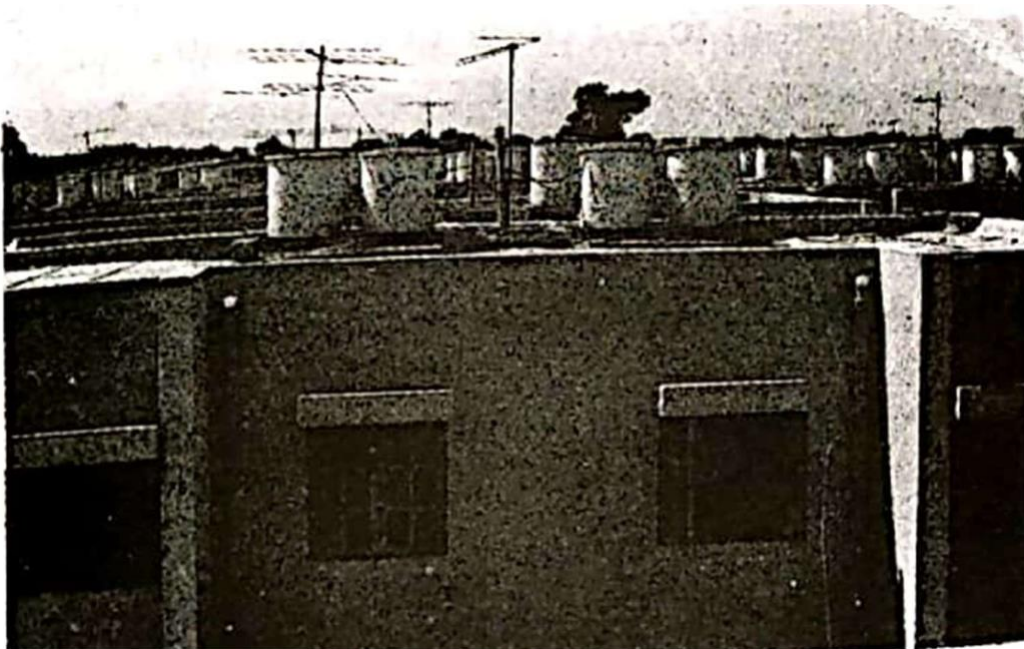


Fig. 6.1 Ferrocement water tanks installed on roof tops

Of late, Ferrocement is also being experimented for the production of cyclone and earthquake resistant core units.

Gypsum Boards

Gypsum board is ideal substitute for conventional building boards in all types of buildings. It can be used in several applications like panelling, wall-covering, partitioning and false ceiling. It can give the effect of brickwork or a smooth finishing like Plaster of Paris. Gypsum board is fire proof. It weighs one-tenth of the normal brick, the actual weight being between 6.5 to 15 kg/m² depending upon the thickness.

It is easy to use as it can be nailed, cut and screwed with ordinary tools. Such boards are available in thickness of 9.5, 12.5 and 15 mm, width of 600, 900 and 1,200 mm, and length of 1,800 to 3600 mm.

Timber Substitution

There is presently an acute shortage of good quality timber due to which timber prices have been skyrocketing. This is mainly because usually only half a dozen conventional or primary species of timber such as teak, deodar Sal, Shis hum and rosewood, etc., are preferred in building construction on their durability, decorative appearance and high strength properties. The availability of primary species of timber is rapidly dwindling due to fast depletion of the forest reserves in the country and the phenomenal rise in population over the past few decades. Other factors are slow rate replenishment of felled timber, and the increasing awareness among the people regarding the importance of maintaining ecological balance. A survey of forest reserves conducted recently reveals that about 27 million acres of the world's forests disappear every year. Half of world's original forest has already vanished. At this rate, it is estimated that global forests will completely disappear by 2020 AD. In India, in 1900, forests covered 40% of the land area. As a result of unabated deforestation during the current century, the present level of forest cover in the country has come down to the level of less than 13% of the land area.

Due to sharp rise in prices of timber, the joinery work component in construction of a new house has now a share of 20-25% in the total cost of construction. The present annual requirement of wood is estimated to be about 13 million tonnes (MT). Against this, a supply of only 6 MT per annum is available. The present demand-supply position definitely points to only towards more and more scarcity of timber in years to come.

Apart from the problems of deforestation and steep rise in the prices of wood, timber has always remained prone to attack by termites, fungi and other organisms. Most of the wood varieties used in India are susceptible to such attack. Teak, rosewood and white cedar are a few of such varieties that are durable and resistant to organism attack to an appreciable extent. However, their high cost inhibits their availability within the reach of common man.

These factors have led to the development of wood substitutes which are cheaper, curbing deforestation, environment friendly and pest resistant. Efforts in this direction have started bearing fruit. The largest construction department in the country namely the Central Public Works Department (CPWD) has already taken a bold step in this direction by banning the use of timber with effect from April 1, 1993. Some of the viable wood substitutes are described below.

i) Secondary Species of Timber

Considerable research has been undertaken in the past by the Forest Research Institute (FRI), Dehradun, regarding exploring the possibility of using lesser-known varieties of timber in building construction. The FRI has identified 117 secondary

species of timber which could be gainfully used in construction after proper seasoning and chemical treatment.

The NBO has promoted use of secondary species of timber in construction through their incorporation in the demonstration houses put up in different geo-climatic regions of the country, under its Experimental Housing Scheme. Some housing boards particularly Tamil Nadu, Andhra Pradesh, Maharashtra, West Bengal, Jammu & Kashmir and Karnataka have set-up integral plants for joinery, seasoning and treatment. It is estimated that 20% of the total timber used in building construction is now secondary timber.

ii) Plywood

Plywood utilises the secondary species of timber. Shelving's of secondary timbers and synthetic resin, are the two basic materials. It is universally available in various grades, sizes and thickness. It is used for furniture making, wall panelling, door panels, shelves, cupboards and other household items. Plywood allows easy and quick working as compared to the conventional timber.

iii) Metals

Hollow, pressed metal doors and windows have been used in India by major construction departments/agencies such as the CPWD and DDA, etc, over past one decade. These doors and windows possess the advantages of longer life, fire resistance, high fatigue strength, lighter weight, freedom from warping and swelling. Hinges, holdfasts, bolts and nuts are welded to the metal before installation. However, maintenance of such doors poses some problem. It becomes difficult for the house owners to arrange a welding set at later stage of the life span of the doors made of pressed steel.

Another improvement over mild steel doors and windows are the recently introduced galvanised steel doors and windows. Welding has been totally eliminated by assembling the joints in high precision by use of Zamac alloy and weather resistant rubbers. These doors and windows can withstand a cyclone wind velocity up to 190 km/hr.

Aluminium is yet another metal tried out for doors and windows. Aluminium door frames in conjunction with aluminium door shutters are being increasingly used in some prestigious building projects. Such doors and windows have, however, not been able to find wider application in construction of residential buildings due to their higher cost.

iv) Poly Vinyl Chloride (PVC)

Another important wood-substitute that has made inroad at least in metropolitan cities of India is the Poly Vinyl Chloride (PVC). Now PVC doors and windows are being produced in the country by the leading companies in plastic industry like Sinter Plast, the well-known 'Sintex' tank producers.

The main advantage of PVC doors and windows are their attractive finish, good insulating properties, freedom from termite or fungal attack, all-out water proofing, self-extinguishing properties in the event of fire, zero maintenance, no need for painting, and longer life. Now, computer aided design (CAD) is being used to design more stable and versatile PVC doors and windows. Concealed steel

reinforcement is being provided in PVC to produce mechanically strong doors and windows. PVC doors are coastal saline air, typical rains or sub-zero temperature. They are however, more for even unaffected by provided in

v) Precast RCC Frames

Precast RCC frames were first used in some buildings as far back as 1960. Due to the difficulties experienced in fixing of door shutters and the problems associated with their maintenance, their application could not catch up to the desired level of expectancy. Recently, firms like M/s Supriya Wonder Wood Pvt. Ltd., Bombay, have started manufacturing precast RCC frames wherein they claim to have sorted out most of the difficulties earlier precast RCC experienced with such doors.

vi) Particle Board

Particle board, a wood-based panel product made of wooden particles and resin is also being increasingly used for door shutters. It was first developed during early 40s as a substitute for timber and its primary derivatives like plywood and blockboard. Its use prohibits high wastage related to conversion of forest reserves into sawn timber or plywood. On an average 60% of a fallen tree is wasted in the process of rendering it suitable for building purposes. Use of particle board ensures 100% utilisation of the fallen tree. The particle board industry makes use of the trees belonging to the species like eucalyptus, subabool, and rubberwood, etc., planted under the social forestry programme. Such boards have uniform and plain surface and can be easily nailed. Several factories presently manufacturing particle boards in the country. These boards are available in a large number of sizes. They are, however, yet to be used on a larger scale for mass housing programmes undertaken by major construction departments in the country.

vii) Fibre Boards

The most promising wood substitute that possess all the good characteristics of natural wood is fibre board. These boards are manufactured by converting raw material that is 100% agro-wastes like bagasse, cotton stalks, rice husk, jute and corn, etc., into fibres, which are dried, formed into mats, pressed to give uniform distribution of fibres and then bound by a synthetic resin binder. The density that is achieved is very high, in the range between 240 to 1000 kg/cu.m. Such boards are claimed to possess excellent mechanical, moisture-resistant and termite-resistant properties.

Some of the major Indian companies have taken significant steps towards manufacture of good quality fibre-board. The Birla Group has set up a big plant. Nuchem Plastic, another group, has set up a big plant with the Swedish know-how at Tohana (Haryana) with an installed annual capacity of 13 million sq.m. (39,000 tonnes) of fibre boards. The product popularly known as 'Nuwood MDF' is a Medium Density Fibre (MDF) board manufactured from cotton stalks, binder (phenol formaldehyde) and paraffin wax. In this process, the agro-based material is chipped to graded sizes, made into fibres, added with resin binder to form mats and pressed to form MDF boards.

Fibre board is manufactured in varying densities-soft, medium and hard. Soft board can be used for thermal insulation and as a resilient material for floors.

Medium density boards are suitable for panelling, wall-linings, ceilings, doors and windows. Hard boards can be used for doors/windows as these are claimed to be water and abrasion resistant. Such boards have the ability to be cut into any size like natural wood. The Bureau of Indian Standards has now evolved a code IS:12406 for specifications of fibre boards.

Fibre board has been widely used as a viable timber substitute in the USA, Australia and Europe.

viii) Other Substitutes

Besides, the other viable timber substitutes for door/window shutters and frames include ferrocement, expanded polystyrene composite (EPS), polyurethane, polychoir, fibre-reinforced plastic (FRP), fibre glass, glass reinforced gypsum (GRC) composite boards, coir cement-composite panels, eucalyptus wood shutters and frames, and cement bonded particle boards. They are still in experimental stage as far as their suitability for various indoor applications in India is concerned. The ferrocement doors have been tried out on relatively larger scale, but its weight inhibits its larger acceptance among the users.

Asphaltic Corrugated Sheet

Asphaltic corrugated sheet (Fig. 6.2) is a low-cost roofing material manufactured by using wastepaper, rags and other road pickings and bitumen. This material is lightweight, water-proof, flexible and vermin proof and available at less than half the cost of cement asbestos roofing sheets. These roofing sheets have become quite popular for temporary and semi-permanent construction, rural housing and slum clearance programmes. Seven units total capacity 7 million sq.m. per year have been set up at various places in the country. Asphaltic corrugated sheets are recommended for use in regions where the temperature in shade does not exceed 44°C. These roofing sheets are better substitute for thatch roofing. The life expectancy of these sheets is about 8-10 years.

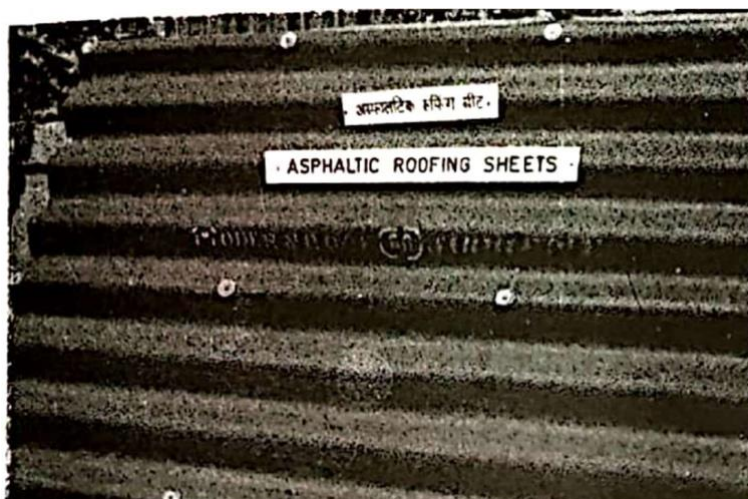


Fig. 6.2 Asphaltic roofing sheet

Cellular Concrete

Cellular concrete or aerated concrete is a light-weight material used for walling in place of bricks and for roofing as a composite construction with concrete. It is lightweight, measuring one-third the weight of the ordinary brick and has thermal insulation value. It has dimensional stability and the requisite structural properties necessary for use as a building component. The cellular concrete blocks (Fig. 6.3) can be sawn, cut, drilled and nailed. Because of lightweight, it is particularly suited to the construction of multi-storeyed buildings as it reduces the dead load of the building. Moreover, due to lightweight and high strength to weight ratio of cellular concrete products, their use leads to appreciable saving in consumption of scarce materials like cement and steel.

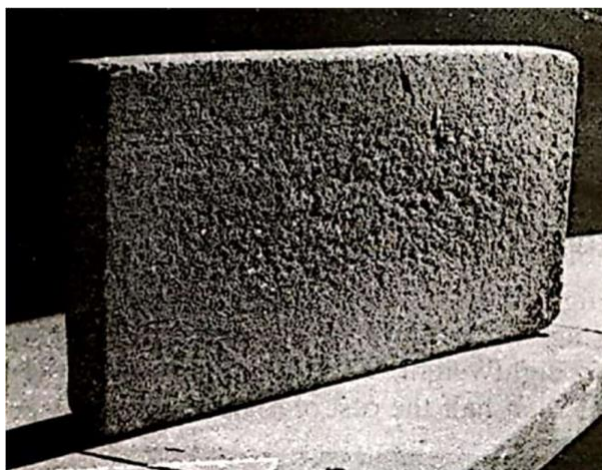


Fig. 6.3 A view of cellular concrete block

Cellular concrete is produced by autoclaving a set slurry consisting of a fine siliceous material and a binder, with the addition of foaming/air entraining agent. Based on the type of binder, it is divided into two groups namely i) cement based (Fig. 6.4) and ii) lime based (Fig. 6.5). Gas producing agents commonly used are either organic forming agents based on resin soap, glue or gas generating agents like fine aluminium powder, zinc dust, etc.

Cellular concrete based on cement-ground sand technology was first produced in the country by the Hindustan Prefab. Limited (HPL), New Delhi, during '60s. Apart from HPL, cellular concrete is manufactured by the Tamil Nadu Housing Board (TNHB) at its plant at Ennore near Madras and the Siporex (India) Ltd., Pune.

The lightweight cellular concrete blocks produced by the HPL are known as 'Vayutan.' The blocks are manufactured in the standard size of 500 × 250 × 100 mm, and are mostly used for partition walls and insulation purposes.

The lightweight cellular concrete products manufactured by the TNHB at its plant at Ennore (Madras) is known as 'Celcrete.' The plant was set up under joint Indo-Polish collaboration. It is produced with lime and sand as raw materials. The Celcrete produced by the TNHB is utilised for their housing programmes. Blocks of dimension 600 × 200 × 125 mm to 200 mm are utilised for load bearing as well as non-load bearing blocks. Experience of the TNHB reveals that through the use of cellular concrete for roofing and flooring, there is saving

of 40 to 50% in cement, 20% in steel, 75% in shuttering and 80% in water requirement at construction site with an overall economy of about 10% compared to RC roof slab.

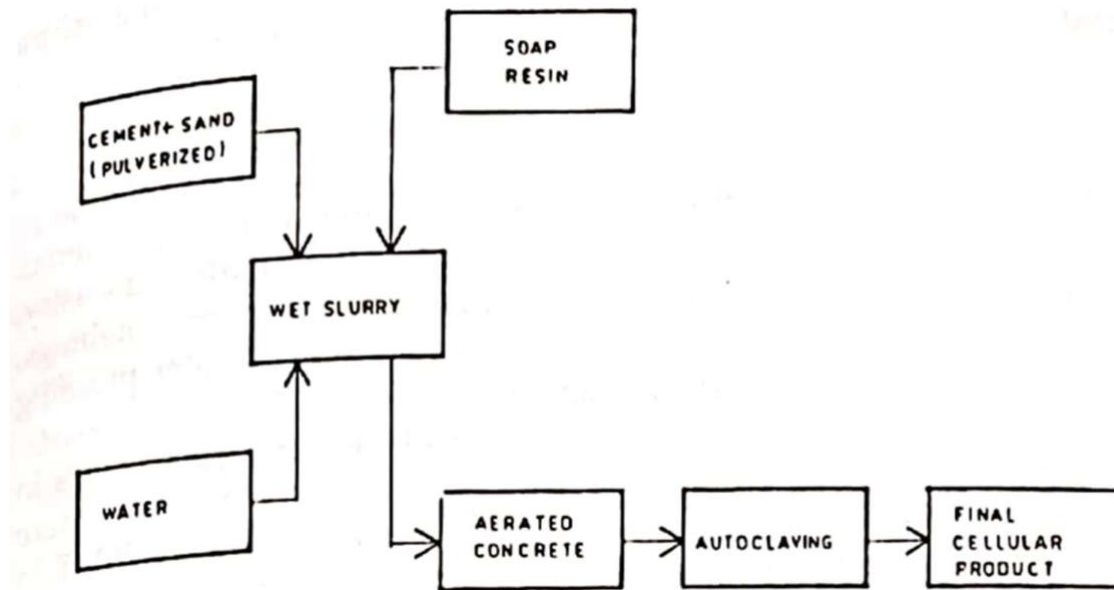


Fig. 6.4 Schematic diagram for production of cement sand based cellular concrete

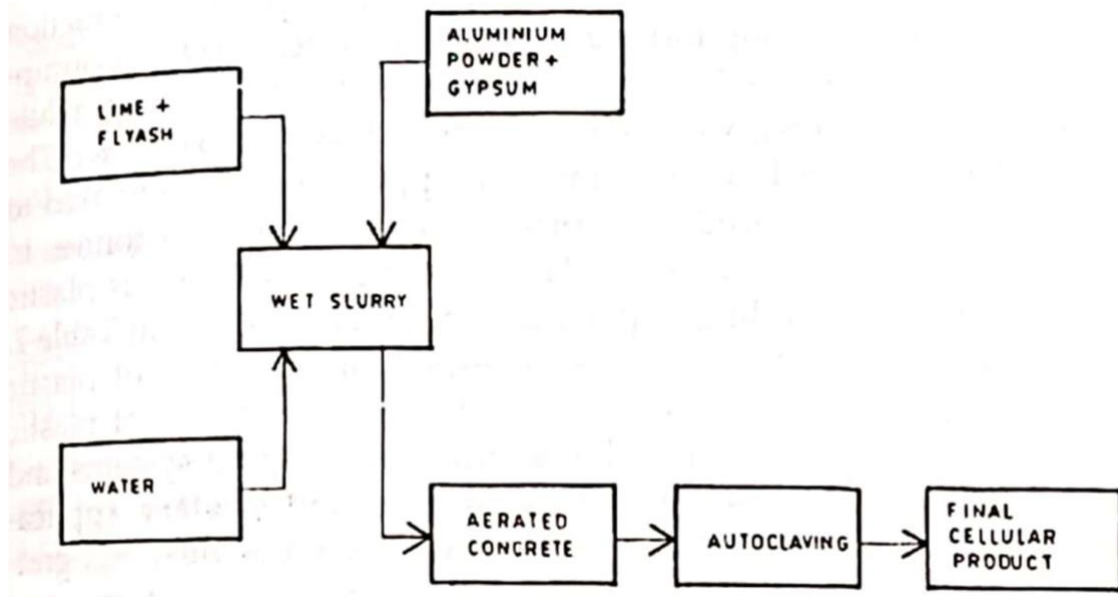


Fig. 6.5 Schematic diagram for production of lime-fly ash based cellular concrete

'Siporex' is the trade name given to the lightweight concrete produced by the Siporex (India) Ltd. This product has been widely used by several construction agencies, both in public and private sectors in Maharashtra and Gujarat.

Plastics

In recent times, plastics have proved to be a versatile building material. They have outdoor as well as indoor applications in the building industry, e.g door and profiles, water distribution pipes, interior panelling's, toilet fitting and fixtures, drain ways, water storage tanks, water proofing treatment, plumbing and sanitary fittings, etc.

Plastics in building industry were first introduced in the early '50s in West Germany in the form of door and window profiles. Since then, there has been a steep rise in the production of plastics. From a mere 3 MT in 1955, it touched the 100 MT mark in 1991. There are more than 10,000 different kinds of plastics marketed today and their performance abilities span those of every other known material from soft rubber MT in to steel.

It is estimated that on an average 25% of the total plastic production in the world is used by building industry. The current per capita consumption of plastics in the developed countries ranges from 50 to 100 Kg, while in India, it is a meagre 1.6 Kg well below the world average of 16 Kg. The present usage of plastics in India for building applications is estimated to be 0.27 million tonnes compared to approximately 3.92 million tonnes in U.S.A. and 2.76 million tonnes in Japan. The consumption of various plastic components for different building applications in India are given in Table 2.

There is, now increasing awareness in respect of utilisation of plastic as a useful building material in the country. Between 20 to 50% of plastic used in the building industry within the country is for piping systems and related products required for water supply storage and sanitary applications. A range of plastic sanitary fittings like traps, showers, flushing, gratings, basins, waste-traps, float balls and valves, flushing cisterns and overhead water storage tanks, etc., are now available. These products are resistant to corrosion, light in weight, easy in handling and installation and economical. They help in conserving cement, steel, and non-ferrous metals. Several Indian Standard specifications have been formulated for these products. The squatting pans made of Glass fibre Reinforced Plastic (GRP) are gradually replacing the traditional ceramic seats, and the same is 30% economical than the latter. Flushing cisterns made of GRP have dual control system, operating to flush water fully or partially, as required with exceptional efficiency, dependability and corrosivity. The consumption of plastics for cistern in 1994-95 was around 1200 MT.

Toilet seats and covers made of polypropylene and polystyrene are very attractive in appearance, hygienic, heat and chemical resistant, and are easy to be installed. The average weight of such toilet seats is about 1.5 kg. Their production was of the order of 900 MT during 1994-95.

PVC and high-density polyethylene (HDPE) are the two major thermoplastic materials used in the manufacture of piping system for water supply and related applications. The use of PVC pipes for water supply in India commenced in the year 1962 with the laying of imported PVC pipes near Delhi. There has since been appreciable rise in its application in the country. The demand for PVC has risen from the level of 82,000 tonnes per annum

Table 2

Application-wise consumption of plastics in building and construction sector in India (after Bibl. 133- reproduced with permission from Civil Engineering Construction Review, New Delhi)

S No	Component	Material	Consumption in MT	
			1994-95	2000 A.D.
1.	Pipes Fittings and Conduits	PVC HDPE	2,01,135 36,204	4,04,554 72,819
2.	Overhead water tanks	LDPE/HDPE	4,578	13,970
3.	Floor Tiles & Skirting	PVC	11,867	23,868
4.	Walling materials a) Wall Paper b) Partitions c) Thermoformed false ceiling & wall panels	PVC PVC HIPS	725 3,821 140	1,458 7,685 282
5.	Sanitary appliance like: a) Flushing Cistern b) Toilet Seats c) Toilet Pots d) Pumping fixture	PP/HIPS PP/HIPS PVC PVC	1,207 905 1,408 1,006	2,428 1,820 2,511 2,023
6.	Roof Sheets	GRP RMP	483 603	971 1,213
7.	Electrical Fittings	Phenolics/PP	2,715	5,461
8.	Water proofing film	LDPE	302	607
9.	Various Profiles like handrails, step nosing, expansion joints etc.	PVC	1,529	3,075
10.	Sound and heat insulation	PS & PUR	704	1,415
11.	Others	FOAM	1,006	2,023
		Total	2,70,338	5,48,183

in 1979 to 4,00,000 tonnes per annum in 1993. During the previous decade (1980-90), more than 1,60,000 km of plastic pipes were laid in the country. Rigid PVC (unpractised) pipes up to 315 mm diameter and high-density polyethylene pipes up to 400 mm diameter are produced in India along with their matching fittings conforming to relevant Indian Standards Specifications. Underground installations of PVC pipes for water mains in diameters ranging from 90 mm and above have been undertaken by the Municipal Corporation of Delhi, DDA, Tamil Nadu Housing Board, MES, PHEDs of Haryana, Punjab, Rajasthan and U.P. Jal Nigam, etc., both in their rural and urban water supply schemes.

Plastic floor tiles and coverings form the second largest application of plastics in buildings after pipes and fittings. The consumption of PVC flooring and skirting tiles in India was about 11,900 MT during 1994-95.

There are two kinds of PVC floorings-one is flexible which is normally available in sheet form, and the other is rigid which is available in tile form. Vinyl tiles consist of fillers, stabilisers and pigments bonded together with PVC binder. Asbestos-PVC tiles provide a

versatile flooring material with lasting appearance. These tiles and semi-flexible are quite brittle but once load over, they form strong and rigid surface. They are anti-skid and quiet under foot. In plastic terrazzo, an epoxy resin is used instead of Portland cement paste as a binder for marble chips.

The plastic floor tiles and coverings have been in use in India for the last three decades. They form quiet, resilient, non-combustible and dampproof wearing and decorative surface. These floorings are generally expensive but preferred by users over other floorings materials due to their visual and aesthetic value.

Low density polyethylene overhead storage tanks up to a capacity of 20,000 litres have been successfully produced and marketed in India. Plastic tanks are corrosion resistant and since there are light weight (a 10,000-litre polyethylene water tank would weigh only 500 Kg), their transportation and installation are much easier. The present consumption of plastic tanks in India is nearly 4600 MT with the annual growth rate of about 15-20%.

A suitable replacement of timber could be found in composite panels which are sandwich type using plastics foam as core and surfacing of conventional materials like aluminium, plywood, Glass fibre Reinforced Plastic (GRP), concrete, and gypsum, etc. These materials are lightweight, dimensionally stable, possess high strength to weight ratio, possess excellent sound and thermal insulation and involve low transportation and maintenance cost.

By mixing PVC with saw dust, a composite which can be easily extruded as doors and window frames and other profiles, has been developed. This has an excellent dimensional stability and can be painted, drilled and nailed like timber. Useful work has been carried out in the preparation of wood/plastic composite by high energy radiation induced polymerisation. These developments could go a long way in finding appropriate substitute of scarce primary species of timber.

Fibre reinforced concrete using plastic (polypropylene) fibres has been successfully developed in India. Known as 'PFRC-Polycerate,' this material's entry into the field of precast load-bearing manhole cover assembly (Fig. 6.6) has been successfully evaluated in the field, and adopted by several local authorities in many parts of the country.

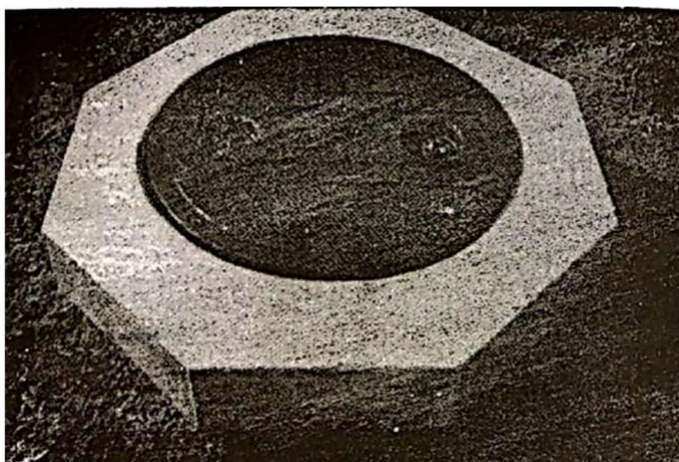


Fig. 6.6 PFRC-polycerate manhole cover (Source: Mr. O.P. Ratra, SMTPC, New Delhi)

Sand-Lime Bricks

Autoclaved calcium silicate bricks, popularly known as sand-lime brick, was invented by Van Der burgh in England as back as 1866. However, such bricks were first produced on commercial scale only in 1898. Many buildings constructed in England, Germany, the USA, the erstwhile USSR and Holland with sand-lime bricks about 80 years back are standing even today indicating their excellent performance under natural weathering conditions.

Sand-lime brick industry is well established in Germany, UK, the Netherlands, Denmark, the USA, Canada and the erstwhile USSR. Germany has more than 200 factories Producing such bricks. Many other countries for example, Switzerland, Poland, Romania, Middle East countries, Mongolia, Vietnam, Bulgaria, Finland, Chile, Kuwait, Egypt, Australia and South Africa have some production units. In India, these bricks have yet to gain popularity due to their non-availability in the market. At present, there is only one sand-lime factory near Cochin (Kerala). This semi-mechanised factory produces 30 million bricks per year using completely imported plant and machinery from Germany and was commissioned about three decades ago. Another factory near Calcutta is producing fly ash-and-lime bricks on small scale using indigenous technical know-how and machinery. Large-scale production of this type of building material has not been possible in India due to the non-availability of high-capacity press. Sand-lime brick is an alternative walling material which may overcome the shortage of ordinary clay bricks in the country. Particularly in those areas where good quality bricks are not being manufactured and sand or siliceous wastes like fly ash or ore-tailings are available, calcium silicate brick can solve the problem of good quality bricks as well the disposal of wastes. (Fig. 6.7) depicts the flow diagram for production of sand-lime bricks.

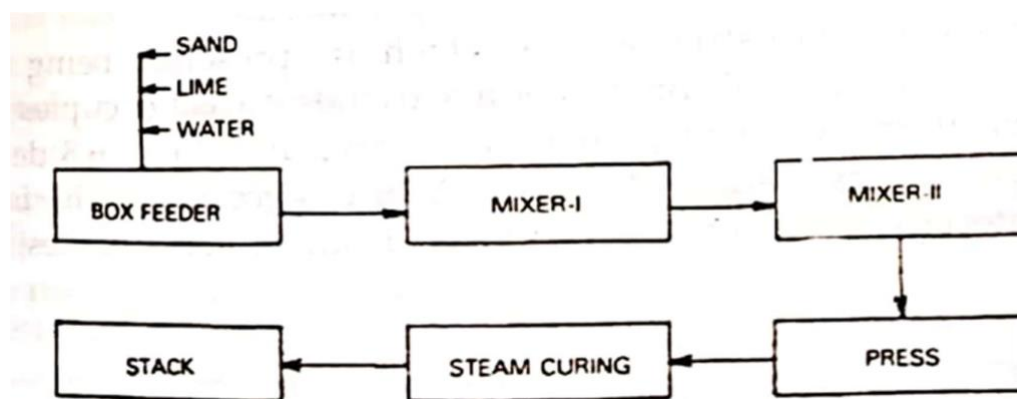


Fig. 6.7 Schematic diagram for production of sand-lime bricks

Calcium silicate brick consists of a uniform mixture of sand or a siliceous waste with a small proportion of lime, mechanically pressed and autoclaved, so that the materials are chemically banded by the action of steam under pressure. Such bricks are quite strong (compressive strength of 10 N/mm²) and can generally compare well in cost with the conventional bricks in the aforesaid areas.

Industrial Wastes

Fly ash

Fly ash is the fine residue obtained from thermal power stations using ground or powdered coal as boiler fuel. About 40 MT of fly ash (Table 3) is produced annually from as many as 70 thermal power plants in the country. This very fine industrial waste causes not only the problem of environmental pollution but also of its disposal. In India, the total installed capacity of power is around 82,000 MW (megawatts) out of which 60,000 MW is generated from thermal power plants consuming 7.5 lakh tonnes of coal daily. A thermal power plant with 1,000 MW capacity for electricity generation produces about 1.5 MT of fly ash every year, which is dumped over 100 hectares of land. Research and Development efforts undertaken in many countries in the West over past few decades have led to the development of appropriate technology which could be utilised in the production of Portland pozzolana cement, cellular concrete in combination with lime, lime-fly ash bricks, clay bonded fly ash bricks and sintered fly ash lightweight aggregate, etc. The level of utilisation of fly ash in the industrialised countries like France, Germany and the UK has been more than 70% of the total amount produced. Response of the Indian builders towards utilisation of fly ash as useful building material has, however, been lukewarm. The poor utilisation level of 5% of the Indian fly ash suggests the state-of-the-art with regard to poor utilisation of fly ash in the country.

Potentials of Fly ash Utilisation as Useful Building Material

The utilisation of fly ash as an important building material has been accepted as an engineering reality in the industrially-advanced countries for past three decades. The position with regard to its utility as useful building material in Indian context has, however, been far from satisfactory. This is so despite the fact that the Bureau of Indian Standards (BIS) has already formulated several standards for the manifold application of fly ash. Apart from BIS, several R&D organisations like CBRI, NCB, NTPC, NLC, CFRI, NRDC, NBO and BMPTC have also been instrumental in promoting application of fly ash as alternate useful building material.

Among several industrial wastes which are presently being tried out for large-scale utilisation in construction activities, fly ash occupies the most prominent position owing to its manifold application. Fig. 6.8 depicts the possible areas of application of fly ash. A brief account of the important building materials which can be produced from Indian fly ashes, is given below.

i) As Pozzolana

Most of the fly ash varieties available from the Indian thermal power stations possess the chemical and physical properties required in pozzolana. In fact, the use of fly ash as pozzolana is not only economical, but also saves considerable quantity of cement, besides improving the workability of mortar. It has been utilised as partial replacement (up to 20%) of cement in concrete, mortar and plaster in a number of demonstration buildings (Fig 6.9) constructed at Neyveli, Delhi and Madras under the Experimental Housing Scheme of NBO. Of late fly ash has

been increasingly used in production of high strength concrete in the USA, Canada and Australia.

Table 3

Annual Production of Industrial and Agricultural Wastes (1993) with their Potential

Waste	Industry	Production (MT/year)	Potential Applications
Fly ash	Thermal power stations	40.0	Portland Pozzolana Cement, bricks, light weight aggregate, lime pozzolana mixture
Blast furnace slag	Iron & Steel Industry	10.0	Portland blast furnace slag cement, super sulphated cement, production of lightweight concrete
Phospho-gypsum	Fertiliser, phosphoric and hydrofluoric acid industries	11.0	Gypsum plaster, fibrous gypsum boards and blocks, cement clinker, super sulphate cement
Lime sludge	Paper, sugar, fertiliser, acetylene and tannery industries	4.8	Masonry cement, sand-lime bricks and lime pozzolana mixture
Cinder	Thermal power stations and railways using lump coal	3.0	Lime cinder mortar, concrete buildings blocks
Mine tailings	Zinc, copper, gold, iron ores beneficiation plants	6..0	Calcium-silicate concrete, bricks and masonry cement
Red mud	Aluminium industry	4.0	Bricks, tiles, and lightweight structural blocks and roofing sheets
Water works silts	Water works	10.0	High strength bricks, cement, and lightweight bloated clay aggregate
Limekiln rejects	Limekilns quantity	3.0	Masonry
Coal washery reject	Coal washeries	3.0	Bricks, lightweight aggregate, fuel

			substitute in burning bricks
Rice husk	Rice mills	25-30	Particle boards, roofing sheets and pozzolana
Coconut husk	Coir-fibre industry	2.0	Particle boards, insulation boards, building panels and roofing sheets
Cotton stalks	Cotton plantation	10.0	Fibre boards, panels and door shutters
Bagasse	Sugar mills	5.3	Insulation boards and wall panels
Groundnut hulls	Oil mills	5.75	Particle boards, roofing sheets and chip boards
Rice straw wheat straw	Agricultural farms	Large	Roofing units, wall panels and fibre boards
Straw mill waste	Straw mills and wood-based industries	2.0	Fibre boards, particle boards and insulation boards, etc.

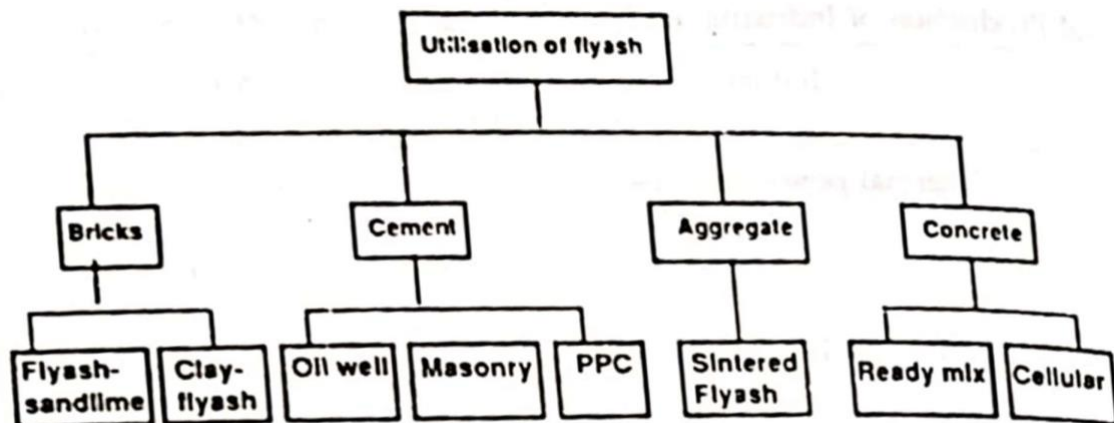


Fig. 6.8 Various utilisations of flyash

ii) Portland Pozzolana Cement

Investigations have been carried out in the country over past thirty years to assess the techno-economic feasibility of the production of fly ash-based Portland Pozzolana

cement. About a dozen plants have since been producing fly ash-based Portland Pozzolana Cement in the country.

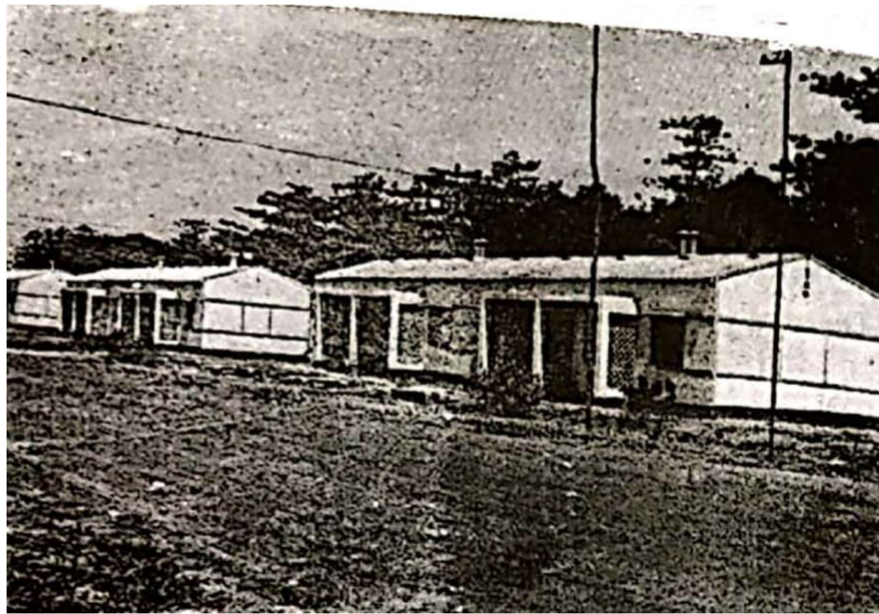


Fig. 6.9 NBO demonstration houses utilising fly ash in construction at Neyveli

The Portland-Pozzolana cement conforming to IS: 1489-1981 can be manufactured by using fly ash either by intergrading Portland cement clinker, fly ash and gypsum, or by intimately blending together Portland cement and fly ash in suitable proportions. The interginding method is convenient to adopt as it does not require any major addition to the plant and equipment already available at a cement factory.

Portland Pozzolana cement is suitable for use wherever Ordinary Portland cement (OPC) is usable under normal conditions. It produces less heat of hydration, increases water impermeability, reduces alkali-aggregate reaction and offers greater resistance to attack of aggressive waters than OPC. It is therefore, particularly suitable for marine and hydraulic structures and other mass concrete constructions.

iii) Ready Mixed Fly Ash Concrete

Portland Cement concrete in which a part of the cement has been replaced by flyash is termed as fly ash concrete. When it is prepared and supplied to consumers in a plastic, unhardened, ready-for-use state, it is called ready mixed fly ash concrete. The process of producing ready-mixed fly ash concrete consists of two operations namely (i) proportioning of fly ash concrete mix, and (ii) batching and mixing of different ingredients. The fly ash concrete is so proportioned as to attain its 28-day compressive strength equal to that of the corresponding plain concrete. The batching and mixing of different ingredients are generally done at a central batching and mixing plant. Sometimes, different ingredients are taken in a truck mixer at the batching plant and so mixed either during the journey or on arrival at the site of delivery.

The ready mixed concrete has the advantage of better-quality control, reduction in wastage and pilferage of materials, labour and supervision, which are normally associated with concrete prepared at site.

iv) *Precast Fly ash Concrete Building Units*

Fly ash concrete has been used in the production of various types of precast building units such as concrete building blocks (solid or hollow), columns beams, hollow core slabs, door and window frames, etc. It has also been tried out in the production of large number of precast flooring roofing units, such as cored units, channel units and cellular units by making partial replacement of cement. The strength and other structural properties were found to be comparable with those produced using plain cement concrete. In the NBO experimental projects, fly ash has been used in precast reinforced channel units and RC lintels with encouraging results.

v) *Clay-fly ash Bricks*

Brick industry is another field in which fly ash can be utilised as a major raw material because both clay and fly ash are of not much difference in respect of their chemical composition. Furthermore, the residual carbon content in fly ashes bring about an economy in fuel consumption during firing of bricks. In the process of brick manufacture from fly ash, about 25 to 80% of clay can be replaced by fly ash. The clay and fly ash are mixed in usual way and the bricks are moulded and fired in the usual way. The bricks thus produced are lighter as the bulk density of fly ash is about onehalf of fly ash. Clay fly ash bricks have been used in the construction of number of houses at Rama Gundam and Kotha Gundam in Andhra Pradesh. Such bricks are presently being produced on large scale by the West Bengal Power Development Corporation, Ltd. (WBPDCL) near its plant at Kolaghat. In production of Kolaghat bricks, clay and fly ash are blended in the ratio of 40: 60. Recently, CPWD has also used clay fly ashes bricks in one of its projects. About 25 million clay-fly ash bricks have already been produced in the country by the NTPC and other public sector undertakings and other private brick manufacturers.

Not all the clays and fly ashes are suitable for the production of bricks. As regards requirements of fly ash for brick manufacturer, it should have oxide composition similar to clay, the SiO₂ content should be over 40%. Al₂O₃, not less than 50% and sulphide and soluble sulphite content being insignificant.

Recently, M/s Bhanu International, Visakhapatnam, have developed a process of blending fly ash, lime and calcined gypsum or Phospho-gypsum for making concrete hollow blocks, patronised at Fal-G concrete hollow blocks. In Fal-G, the fly ash content varies between 65 to 75%, while lime sludge and calcined gypsum together constitute the remaining 25-35 per cent. The Fal-G technology enables production of bricks/blocks with simple process steps of mixing, pressure-free moulding and water curing. It is claimed that apart from the technology being environmental-friendly, the costs are comparable with the conventional products. Over a million Fal-G bricks have recently been used in the construction of 200 houses in Andhra Pradesh.

Fal-G could also be used as a substitute of OPC mortar and concrete. As a thumb rule, 1 ½ units of Fal-G cement is to be used against one unit of OPC.

The Government of India has recently finalised plan for setting up of three lime fly ash brick plants at Dadri, Delhi and Ahmedabad.

vi) Lime-Fly ash Cellular Concrete

Lightweight aerated concrete or cellular concrete can be manufactured by a process involving mixing of fly ash, quick lime and gypsum in a high-speed mixer to form a thin slurry. A small amount of foaming agent such as aluminium powder is added and mixed into the slurry liberating hydrogen gas. The aerated slurry is poured into steel moulds and allowed to set. The blocks are then removed after initial set and autoclaved at steam pressure. After autoclaving, the blocks are allowed to cool and stacked for use. These are considered excellent products for walling blocks and prefab floor slabs. The bulk density of product ranges from 500 to 1,300 kg/cu.m. Such blocks are suitable for load bearing walls in buildings up to 2-3 storeys and partition walls in multi-storeyed buildings. They can be sawn, Chiselled, planed, screwed and nailed like wood. Due to lightweight and high strength to weight ratio of cellular concrete products, their use leads to appreciable economy in consumption of cement and steel. Fly ash-based aerated lightweight concrete products are manufactured by the Ballarpur Industries Ltd. at Palwal (Haryana).

Fly ash has of late also been used in production of reinforced cellular concrete members. The problem of corrosion of reinforced cement in these units has also been sorted out. The material is suitable for flooring, roofing and walling in common dwellings and multi-storeyed buildings.

vii) Sintered Fly ash Lightweight Aggregate

Conversion of fly ash into lightweight aggregate is one of the potential ways of bulk disposal of fly ash in an economical manner. Sintered Fly ash Light weight Aggregate (SFLA) with a bulk density of 640-750 kg/cu.m. has been produced in a pilot plant at CBRI. The production of SFLA from the Indian fly ashes which was successfully carried out in the pilot plant suggests that the aggregate is suitable for use in the production of structural lightweight the concrete and precast lightweight concrete building units for use as load bearing and non-load bearing elements. It has good potential in places where fly ash is locally available and stone aggregates are costly.

SFLA is produced by (i) palletisation of the fly ash, and (ii) sintering the fly ash pellets at 1,100 - 1,200°C in a vertical shaft kiln or on a moving grate sintering strand. The technology of production of SFLA and the quality and use of the lightweight aggregate are well tried and proven in the industrialized countries like the USA, Germany, the UK and Switzerland, and may be advantageously adopted in India. This material has bright prospects. It may be worth mentioning here that SFLA has been in use in Europe since its first production under the trade name of 'Lytag' in the UK.

The West Bengal Power Development Corporation Ltd. (WBPDC) has recently entered into technical collaboration with a Japanese firm for installation of SFLA unit near its plant at Kolaghat.

viii) Fly ash-Sand-Lime Bricks

Another way to supplement the present shortfall of the traditional clay bricks could be to use fly ash for production of fly ash-sand-lime bricks commonly known as 'Calcium Silicate' bricks. The production of such bricks is a technically sound proposition and is

of particular advantage in areas where good quality brickmaking clay is not available. Cost wise, the bricks have, however, been found to be costlier than common burnt clay bricks. This type of building material is very common in many developed countries like Germany, Russia and the Netherlands. Germany has more than 200 factories producing Calcium silicate bricks whereas in India only two factories produce such building material. One is situated near Cochin producing fly ash-sand-lime bricks from completely imported know how and machinery. The second plant is located near Calcutta which produces such bricks on small scale with indigenous know-how and machinery. Calcium silicate products consume about 60-70% fly ash in their manufacturing process.

The production of fly ash-sand-lime bricks consists of four main operations namely (a) preparation of new materials; (b) mixing; (c) shaping or pressing, and (d) curing. The ultimate strength of these bricks depends upon various factors. And the notable among them being (i) nature of raw materials; (ii) moulding pressure, and (iii) steam curing in autoclaves the steam pressure and duration of curing. Initially, fly ash and sand are mixed dry and then to this is added hydrated lime in the proportion of 80:10:10 to 60:30:10 by weight, respectively. A small quantity of water is added to the mixture. The slurry so formed is next transferred to the moulds of the press where a moulding pressure of 20-30 N/mm² is applied. The green bricks are taken off the press table and loaded on trolleys for onward transmission to the autoclaves (steam curing chambers) for curing at 1.0-1.4 N/mm² of steam pressure for 4-6 hours. The cured bricks are then taken out of the autoclave and sent for use.

In April 1994, a fly ash-sand-lime brick plant with an installed capacity of 24 million bricks per annum has been set-up at Tribeni (West Bengal).

The plant is to utilise 200 tonnes/day of flyash produced at the Bandel Thermal Power Station.

ix) Other Applications

Large quantities of flyash can be gainfully used in laying of sub-base for roads, reclamation of low-lying land, filling of mines treatment of polluted waters and unsuitable solid for agriculture, and embankment. It could also be utilised for mass concreting in construction of concrete dams.

Flyash can be used as land-fill by city authorities. It has been extensively used for the reclamation of low lands at the Farakka thermal project. It can also be used for creating mounts topped with soil for growing grass in landscaping.

Constraints in Large-Scale Utilisation of Flyash

Despite much of R&D efforts undertaken in several research laboratories in the country, the actual utilisation of flyash as useful building material has not been to the extent desired. Some important factors contributing to this phenomenon are listed as below.

- i) A major impediment in the large-scale use of flyash is the non-availability of flyash in dry form. As per the practice adopted in most of the thermal power stations, the flyash collected in the Electrostatic Precipitators (ESP) is removed hydraulically. The flyash slurry is sluiced to settling ponds or dumping areas near the plant. The pozzolanicity of the freshly formed dry ash particles is seriously affected once the particles have been wetted. In some thermal power plants where flyash arresting devices such as bag filters and cyclones are non-existent, the ash spreads over a large area through the stacks.
- ii) Large variations in the quality of flyash collected from different thermal power plants with consequent adverse effects on its performance.

The nature and quality of flyash generated are greatly influenced by percentage of ash in coal, fineness of grinding and nature of combustion system. In most of the thermal power plants flyash, which does not contain unburnt carbon, is allowed to be mixed with bottom ash. Presence of carbon affects pozzolanic activity of flyash.

Degree of pozzolanic activity in flyashes determines its suitability for use as pozzolana in the production of Portland Pozzolana cement or a part replacement of cement in mortars and concrete at the construction sites. The pozzolanic activity is generally found to increase with an increase in the fineness.

- iii) Transportation and handling of flyash from thermal power plants over longer distances pose another serious constraint in its large-scale application.
- iv) Non-availability of adequate data regarding quality of flyash
- v) High initial investment in setting-up flyash based building material units
- vi) Non-incorporation of flyash based building materials in the existing Schedule of Rates and specifications of the construction departments/agencies in the country is also an impediment in its large-scale utilisation.
- vii) Widespread reluctance to use of flyash in civil engineering applications is also due to the fact that thermal power plants have not yet accorded recognition to bottom ash and flyash as two distinctly different resource materials for other industries. Once this recognition is accorded, their disposal would acquire a dignified status in the sense that bottom ash and flyash collected from different locations would not be indiscriminately mixed together, and would be suitably collected in the manner as would not destroy their characteristic features, viz, their pozzolanic characteristics and reactivity with lime.

Blast Furnace Slag

Blast Furnace slag is the only industrial waste which so far has found the most accepted application. It is waste product from steel plants. Presently, India produces more than 10 MT of blast furnace slag out of which 3.5 MT are being granulated for the manufacture of Portland blast furnace slag cement whose present production stands at around 3.15 MT. Small quantities of blast furnace slag are used to manufacture slag wool. It is also used in the road construction in steel plants.

In addition to the manufacture of blast furnace slag cement, granulated slag can be utilised in the manufacture of super sulphated cement which possesses high sulphite resistance. Air cooled slag is suitable as an aggregate in concrete. Slag fines may be used as a substitute for sand without any deleterious effects. One of the important uses of foamed slag which is produced by rapid discharge and treated water jets, finds application in the manufacture of lightweight concrete with a bulk density of 800 to 950 kg/cu.m. These foamed slag concrete blocks are used in load and non-load bearing walls. Though potential for utilising blast furnace slag for making foamed slag exists, the field trials have yet to be undertaken. Moreover, the commercial production of the blast furnace slag cement can be increased by increasing the granulation capacity of the slag at the steel plant itself.

Apart from improving the workability of fresh concrete, blast furnace slag cement has high resistance to chemical attack, and its capacity of protecting steel reinforcement makes it suitable for use in reinforced and prestressed concrete.

Red Mud

Alumina red mud or bauxite reject is one of the important inorganic waste materials obtained in large quantities from the aluminium producing plants. It is estimated that over 2 MT of red mud is produced annually in the country from the various aluminium plants. The CBRI, Roorkee, has undertaken laboratory and field trials for making building bricks out of red mud supplied from three different aluminium plants in the country. The test results have demonstrated that the physical properties of these bricks made by hand moulding or extrusion are similar to normal building bricks. In many cases a very high compressive strength was obtained due to better fluxing action produced by the red mud. The bricks made from red mud can be made and dried in the traditional manner, and fired in any of prevalent types of brick kiln.

Red mud composition gives bricks pal brown, orange or golden yellow colour depending upon the composition of raw material and firing temperature. They have thus good architectural value as facing bricks in high-rise constructions. Moreover, presence of 4-5% of alkali in the red mud provides good fluxing action resulting into good plasticity and better bond characteristics in brickwork. Production of red mud bricks on commercial scale is however, yet to commence in the country.

Red Mud Plastic (RMP)

The Red Mud Plastic, made by combining polymer (PVC) and red mud waste (from aluminium industry) is an innovative product which overcomes the negative aspects of usage of PVC namely its brittleness and short exposure life. The Red Mud waste is the sludge left after bauxite is refined to aluminium oxide. The Red Mud waste and PVC are combined in the ratio of 25% : 75% to form RMP. Possessing a better heat stability, the material is claimed to possess extra fire and thermal insulation. The inorganic oxides present in RMP add to the flame

retarding nature of the base material. RMP is also claimed to be weather proof with a long exposure life. RMP sheets have been successfully adopted in Japan, China and tropical countries.

RMP's characteristics make it a suitable material for roofing and cladding. It has made inroads into Indian market only recently. Half a dozen organisations in the country have acquired the requisite technology for its manufacture. However, only two or three companies in the country have commenced actual production of RMP corrugated roofing sheets since 1986. Its present production is around 4 MT. These sheets are light in weight (2 kg/cu.m.) and available in 1 mm thickness and standard width of little over 1 metre.

Lime Sludge

Lime sludge, which essentially contains Calcium Carbonate with varying amounts of free lime is a waste product from sugar, paper, fertiliser and Calcium Carbide industries. The annual production of lime sludge is approximately 4.8 MT. The utilities of lime sludge for the manufacture of cement and lime have been investigated for commercial exploitation. Besides, lime sludge mixed with rice husk may be used as a cementing material. Further investigations and field trials are required for proper burning of lime sludge in the commercial down draught kiln and vertical shaft kiln. Based on the CBRI technology developed for production of masonry cement from lime sludge, three entrepreneurs have been licensed to set-up lime-sludge, base industries in Bihar, Uttar Pradesh and Maharashtra.

Phospho-gypsum

Among several alternative building materials available in the country, Phospho-gypsum (calcium sulphate contaminated with phosphate) holds a great promise as a substitute building material for building construction. Phospho-gypsum is a by-product produced as a solid waste from and fertiliser and phosphoric acid plants. It is the second largest pollutant in the country after flyash. In India, the annual production of Phospho-gypsum from one dozen phosphoric acid and fertilisers plants is of the order of approximately 5 MT.

Phospho-gypsum contains impurities of phosphates, fluorides, alkalies, and organic matter which adversely affect the setting and strength development of plasters/cements produced from it. These impurities exist in Phospho-gypsum as water-soluble, lattice bound and insoluble forms. P_2O_5 , F and organic matter are generally found in the range of 0.40 to 1.5%, 0.44 to 1.5% and 0.11 to 0.60% respectively. For effective utilisation of Phospho-gypsum, it is essential to develop process for reducing these impurities or to make them innocuous to enable proper and effective utilisation of Phospho-gypsum as building material.

The CBRI, Roorkee, has evolved several methods ranging from washing and thermal to chemical treatments to reduce/ inactivate impurities. Wet-sieving of Phospho-gypsum through micron IS sieve helps in removing the major part of impurities through the rejection of coarse particles in Phospho-gypsum.

The use of Phospho-gypsum primarily depends on the extra cost involved in its purification and drying. In the Indian fertiliser industries, companies like Southern Petrochemical Industries Corporation (SPIC), Fertilisers and Chemicals, Travancore (FACT), and FDIL have developed suitable purification techniques and produced useful building materials from Phospho-gypsum. A small quantity of Phospho-gypsum is currently used as cement retarder and a larger quantity of it is used as landfill.

The most important application of Phospho-gypsum is as a raw material for cement and sulphuric acid plants. Besides this, it can be used efficiently for making light weight fibrous gypsum plaster boards and blocks, panels for wall partitioning, slotted tiles for wall cladding, decorative boards for artificial ceilings and for making super sulphated cement. Recently, a durable water-resistant gypsum binder has been developed from calcined phospho-gypsum and other additives for use in masonry work, blocks, fibre reinforced boards and plastering for external and internal applications.

The Union Ministries of Chemicals and Petrochemical, Environment and Forests, and Urban Affairs and Employment have joined hands to promote the use of Phospho-gypsum as a building material in a big way. The first such project is being set up at Visakhapatnam with the German knowhow. Initially, the Phospho-gypsum-based products are to find their application in official and residential complexes.

Apart from reducing environmental pollution, the gainful application of Phospho-gypsum as a useful building material as envisaged in the aforesaid project will go a long way in instilling confidence among the entrepreneurs and users in respect of larger application of Phospho-gypsum based products in building construction.

Cements produced from Phospho-gypsum have delayed setting and slow rate of strength at early ages, but strengths at later ages (28 days and beyond) are comparable with those of Ordinary Portland Cement (OPC).

Agricultural Wastes

In the rural areas 370 MT of agro-waste is available of which 50 MT comprise cotton stalks, jute stalks, maize stalks and other such crop wastes. Another 50 MT comprise rice husk, bagasse, raw-dust groundnut shells and such small wastes. These could be gainfully utilised as viable building materials.

i) Rice Husk

About 25-30 MT of rice husk is produced annually in the country. The highly reactive pozzolana produced from rice husk and clay may be used for making masonry mortar and plaster of different grades. Rice husk is obtained as a waste material from rice mill.

Investigations have been undertaken with a view to utilising them as a pozzolanic material. Technical know-how has been developed and pilot plant trials carried out at various research institutions in the country like RRL, Jorhat; NCB, Ballabgarh, and IIT, Kharagpur. The CBRI, Roorkee, has developed a method of production of rice husk-based building materials which can be used as a substitute for

cement in many ways. Besides, it can also be used for making building blocks and boards with suitable binders. Plants of 2-5 tonnes per day capacity have been set-up in the small-scale sector in the rice-producing States in the country for the production of rice husk pozzolana lime binder.

Rice husk being predominantly siliceous in nature, has been utilised as an additive with highly plastic soils in reducing the losses during drying of bricks. Experiments undertaken on red and mixed black red soils over laying Andhra Pradesh, Orissa 1 and Tamil Nadu show that good bricks from these soils which are apparently highly sensitive to drying can be produced by the addition of rice-husk ash up to 30%, and firing the bricks in the temperature range of $1000 \pm 20^\circ\text{C}$.

The National Building Research Station, Karachi (Pakistan) has successfully tried out rice-husk ash (RHA) in combination with lime in production of hollow load bearing blocks, mortar and plaster for construction of low-cost houses undertaken by the Institute. Thirty per cent of Portland cement was substituted by RHA in construction of precast lintels and roof beams.

ii) Coir Waste

The production of coconut waste in India is estimated at 2 MT. About 50% of the coconut is used for making coir/mats/cushions, etc. The remaining raw material is mostly used as fuel. The CBRI, Roorkee, has developed a process for making corrugated roofing sheets utilising coir, waste or woodwool waste and cement. These sheets are light and tough, and possess good bonding strength having good thermal insulation properties. They are about 50% cheaper in comparison to asbestos cement sheets. So far, two entrepreneurs have been issued licences for production of such sheets in India.

Bagasse

Bagasse is the fibrous residue from sugarcane processing. The estimated annual production of bagasse in the country is around 5.30 MT. Due to its fibrous components, bagasse can be utilised for production of insulation boards and building wall panels with the use of a suitable organic binder. Field trials have been carried out for making insulation boards from this waste material, and indigenous technology is available for production of bagasse-based building products in the country. However, the economy of manufacturing insulation board from bagasse on a commercial scale is yet to be worked out. There is great potential for utilisation of bagasse in manufacture of insulation board in view of the large-scale availability of the main raw material which is one of the prerequisites for establishment of the waste-based industry.

Other Agricultural Wastes

Furthermore, about 10 MT of cotton stalks, 4 MT of jute stalks, 5.75 MT of groundnut hulls, 2 MT of sawdust (saw mill waste) and 1 million of coconut pith are available in the country for their exploitation as viable building materials. The R&D efforts have been undertaken in the country for the development of building materials, especially for production of roofing units and walling boards by utilising these wastes. However, no indigenous technology has been

developed in the country for commercial production of building components based on these agro-wastes.

Future Strategy for Promotion of Alternative Building Materials

Industrial and agricultural wastes like flyash, Phospho-gypsum, blast furnace slag, red mud, rice husk and coir waste, etc., hold great promise as building materials. Several governmental efforts have been undertaken in recent years the country to popularise use of such materials. Some of the measures initiated in this regard are listed as below:

- i) The National Housing Policy (NHP) announced by the Government of India in May 1992 has emphasised on economising the use of scarce building materials and promoting use of cost-effective and environmentally appropriate technology and indigenous resources. It also calls for providing an impetus to larger use of materials based on industrial and agricultural wastes by public and private construction agencies. The housing policy also lays stress on incorporation of low- cost specifications laid down by the state/central construction agencies. The CPWD has already inducted flyash in their schedule of specifications. Moreover, the NHP advocates earmarking 10% of the total construction by public construction agencies over the use of cost-effective technologies and materials.
- ii) With a view to providing fiscal incentive to the entrepreneurs willing to undertake manufacturer of building products based on industrial wastes like flyash and Phospho-gypsum, the Government of India has been announcing fiscal incentives in the past three union budgets. Excise duty has been exempted on materials and components using these industrial wastes as a raw material.
- iii) In the Union Budget proposal for 1992-93, further concessions were announced in terms of custom duty reduction on important plant and machinery required to be imported for manufacture of flyash and Phospho-gypsum based building materials and components. Excise duty was also waived for production of bricks/tiles with 25% or more of red mud (a waste product of aluminium industry) light concrete blocks/components. The excise duty on pre-fabricated building components is also proposed to be reduced to the extent of 5-15% encouraging further use of flyash and Phospho-gypsum etc., to effect economy in consumption of cement.
- iv) Flyash should be accorded the status of a national resource material for the building industry. Presently, clay brick industry is destroying the top fertile soil to the extent of 50,000 acres every year. Gainful utilisation of flash in production of bricks/blocks could serve the dual purpose of conserving valuable agricultural land and augmenting brick production of the country. Over past few years, there is a renewed awareness with regain to larger application of flyash as useful building material in construction activities. Increased production of clay flyash and lime- flyash bricks/block over recent years bears testimony to the above fact. A Working Group on Flyash Utilisation constituted by the Government of India has strongly recommended commercial utilisation of flyash through manufacture of building bricks. Under the National Policy on flyash utilisation, six flyash brick units are to

be set-up in the vicinity of Rajghat and Badarpur thermal power stations in Delhi U.T. The Government of India has recently also finalised a plan for setting up three flyash brick units at Dadri, Delhi and Ahmedabad under the Indo-Dutch co-operation programme.

- v) With a view to promoting use of flyash based products in the country, no new licences for burnt clay kilns should be given within a radius of 50 km. of thermal power stations, unless brick manufacturers start using flyash to replace natural soil, which is fast depleting. A switch-over to flyash-based brick will help save at least 30 to 35% of the useful top soil.
- vi) The HUDCO and NHB are promoting new building materials production units based on agricultural and industrial wastes by participating in the equity of new units. The HUDCO is also extending term-loan support to entrepreneurs for setting-up industry for production of building components based on agro-industrial wastes.
- vii) While technologies for making clay bonded sand-lime and Phospho-gypsum based building products have been developed at the CSIR laboratories, the BIS has either developed or is in the process of developing national standards on such bricks and other applications of flyash and Phospho-gypsum in construction works.
- viii) The Building Material and Technology and Promotion Council (BMTPC), under the Ministry of Urban Development (MUD), has undertaken identification of potential technologies which could help larger utilisation of industrial wastes like flyash, phospho-gypsum in construction industry. The Council has prepared technical profiles in respect of clay flyash burnt bricks, flyash-sand-lime bricks, cellular concrete components, cement fibre roofing sheet, alumina-red mud bricks and Phospho-gypsum based buildings components for their larger exploitation in building activities. The Council also incorporates proposals for foreign assistance in areas of building technology. It has also taken initiative in respect of encouraging larger entrepreneurs' interface with financial institutions.

Low-Cost Infrastructure Services

Introduction

Housing is far more than shelter. It is both a basic community services and facility so essentially required for a wholesome family and community life. The quality of life in human settlements depends on environment, available facilities and resources. The basic needs for human settlements are ample clean air for breathing, potable water for drinking and efficient system of solid, sullage and waste disposal for hygienic and sanitary surroundings. Provision of infrastructure services catering to above basic needs of human settlements form an integral part of any programme related to housing development both in urban and rural areas.

Present Status

The present position with regard to available infrastructure services in both cities and villages in most of the Third World countries is far from satisfactory. Large population growth and excessive influx of the poor migrants from villages to cities have to a large extent, degenerated civic life in major cities. The pressure of population and acute housing shortage in cities have overstrained the existing infrastructure services in cities, especially in metros.

According to the World Health Organisation (WHO), over 1.3 billion people in the developing world gained access to safe water for the first time in the preceding decade during the International Drinking Water Supply and Sanitation Decade (IDWSSD) launched by the United Nations with the primary objective of providing impetus to the programmes of water and sanitation worldwide. Some 750 million gained access to improved sanitation facilities during the raid period. Yet in 1990, over a billion rural and urban residents were still without safe water supply, and nearly two billion lived without proper sanitation. According to the ESCAP report on 'State of Urbanisation in Asia and Pacific', only 42% to adequate sanitation. In slum areas, these percentages plummet to 20% for water, and 11% for sanitation. Surveys carried out by the National Institute of Urban Affairs (NIUA) indicate that one-fourth of the urban residents in India do not have direct access to safe drinking water, and over half lack access to safe sanitation. According to the WHO Commission on Health and Environment, only 210 of India's 3,768 towns and cities have partial sewage and treatment facilities with only eight having the privilege of full facilities. Moreover, about 20% of the urban population is still without provision of water supply.

The position in villages is still worse. In 1988, drinking water supply through tap was available to only 15% of the households. Inadequacy of services in the countryside is further evidenced by the fact that according to the 44th Round of NSS (1988-89), around 89% of the rural households did not have sanitary latrines.

Technological Options

Over past three decades many cost-effective technological options have been evolved in several developing countries in respect of supply of potable water, provision of sanitary latrines in homes, drainage of waste water, sewerage system, collection and disposal of garbage and improvement of environment in housing and human settlements. Brief description of the available cost-effective technological options which could lead to environmental upgradation in housing and human settlements is given below.

Low-Cost Sanitation

The modern technological solutions offered for improving sanitation and environment for built-up areas and new settlements in metropolitan cities, small and medium towns, and villages are varied in nature and magnitude. A study published in 1980 by the World Bank identified over 20 systems for sanitation, varying from simple on-site latrines to water-borne sewerage. Out of 20 only three systems have been found appropriate for adoption on mass scale in India. Water-borne sewerage represents a high level of user convenience but it is extremely costly, and demands large quantities of trouble-free operation. Sanitation, more than any other infrastructure services, offers prospects for reducing costs through the use of alternatives to conventional sewerage.

As a result of the sustained research undertaken in the field of low-cost sanitation, within and outside the country, different types of low-cost sanitary latrines have been developed which can be adopted depending upon their appropriateness to local situation. Criteria for selection of a particular type of sanitary latrine for a specific community is guided by certain factors such as soil conditions, cost, technical assistance, housing condition, and cleaning material, and at times the requirement for manure. Pour-flush latrines and twin leachpit (Fig. 7.1) systems have proved to be among the most successful of the low-cost sanitation systems. Unlike other wet on-site systems of sanitation, such as aqua privies and septic tanks, they are inexpensive. They are widely used and promoted in the South Asian countries including India, and efforts to introduce them in parts of urban Africa have had encouraging results.

Sulabh International, a non-governmental organisation (NGO) engaged in the cause of liberation and rehabilitation of scavengers in the country, has constructed over 7,50,000 Sulabh Shauchalayas (pour-flush tanks) in 760 towns all over the country. Under a low-cost sanitation programme launched by the Central Government, under the aegis of HUDCO in 1989, some 2,34,066 dry latrines are being converted into pour-flush water seal latrines thus leading to liberation of 16,431 scavengers. Under 'Bhangi Kashta Mukti Yojana,' launched in Gujarat in 1968, more than 1,50,000 service latrines have so far been converted into pour-flush water seal toilets with leach pits for treatment of human excreta. Such toilets have worked well over a period of 16 years, and have stood the test of time.

Septic tanks, despite their high costs, have been widely adopted in high-income areas in developing countries, and the use of a single tank to serve many households has been found to reduce considerably the cost of this form of sanitation.

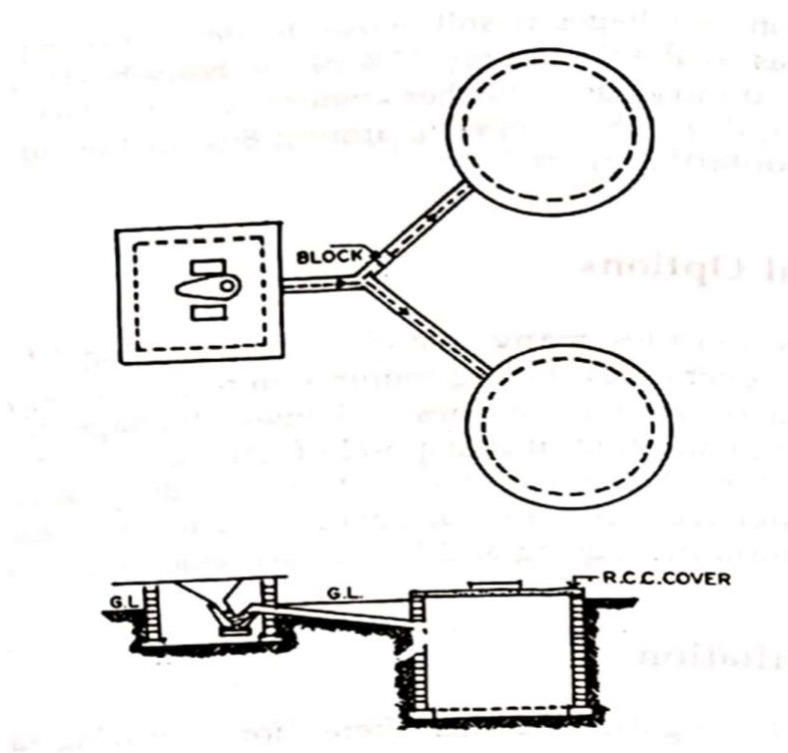


Fig 7.1 Sulabh Shauchalaya (Double-pit pour flush latrine)

Community toilets have been tried in many developing countries. Public motivation and education programmes are often necessary to ensure the success of such toilets. The Sulabh International has put up more than 3,000 community toilet-cum-bath complexes in 625 towns spread all over the country. The 'pay-and-use' concept of community toilets promoted by 'Sulabh International' has been recommended by the World Bank for adoption in 19 countries in South-East Asia, Africa and Latin America. 'Sulabh (Fig. 7.2) is an ordinary pour-flush latrine to look at, but for specially designed slopy pan which requires no more than two litres of water to flush excreta against 12.5 litres required in case of a normal flush toilet. The two covered pits receive the excreta in rotation so that only one is used at a time. This twin-pit combination is claimed to last for at least 100 years.

Among the other notable developments in the field of low-cost sanitation is the 'Shallow-sewer' system for urban slums, which was initially developed in Brazil for its application in a high-density squatter settlement. The system consists of sewers laid at a shallow depth (30 cm) cover and away from imposed vehicular loading) with small inspection chambers used to replace manholes. Such a system has been successfully operated in many squatter settlements in Brazil and Pakistan. The Orangi (Karachi) Project involving community participation has successfully made use of the 'Shallow-sewer' system under its slum upgradation programme. Orangi is Karachi's largest squatter settlement with a population of nearly 1 million. Almost 75% of the households in Orangi have sanitary latrines connected to

an underground disposal system a low-cost design costing less than US \$50 per household, and built by the residents with supervision from a community development group. The above project has served as a model for other slum upgradation programmes undertaken by several municipalities in Pakistan.

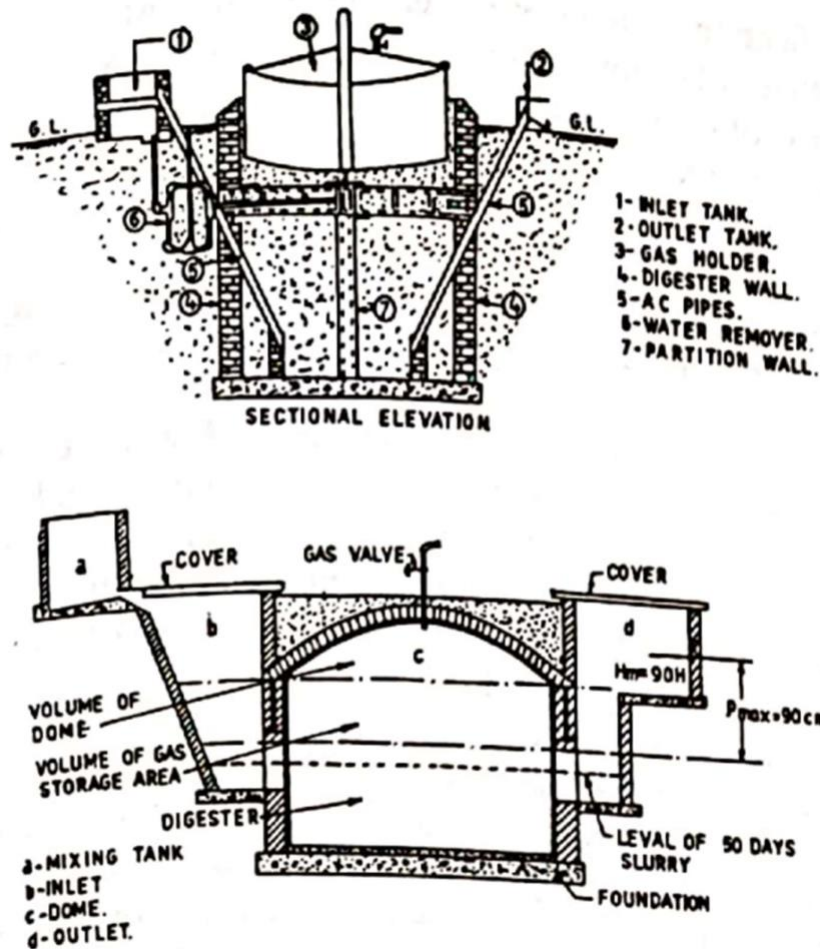


Fig. 7.2 Common design of biogas plants

In Delhi, over 100 mobile toilet vans (MTVs) have been installed in several slum pockets to improve sanitation in such areas. Each of these vans is equipped with some 10 WCs, each WC, in turn, being capable to handle about 100 persons.

Domestic Waste Disposal

Proper collection and disposal of domestic waste and garbage are essential for preventing the environmental degradation in any human settlement. Many refuse-collection systems are in vogue in different parts of the world. The level of refuse-collection also varies considerably. Door-to-door collection facilities are often provided in high income areas, while communal storage facilities are the norm in low-income settlements.

The characteristics of a particular refuse determine the appropriateness of collection vehicles and the system for its treatment and disposal. A variety of containers are used for storing domestic refuse, varying from plastic and paper bags through bamboo or straw baskets, to metal containers, cardboard boxes, and the like in different parts of world.

Masonry storage facilities are a popular method of collection of domestic wastes in developing countries. Large capacity (5-10 cu.m.) roll-off containers capable of being loaded directly on to the refuse vehicle have been successfully used as an alternative masonry storage facility, and a range of such vehicles and containers has been introduced in many developing countries.

Open dumping accompanied by incineration is the most prevalent form of disposal in developing countries. Of the various processes available for waste treatment and disposal, only sanitary landfill and, in some cases, composting are economically feasible in most developing countries. The high organic content of developing country refuse (which reach up to two-thirds of total weight) makes composting a feasible option. In Bangalore and Delhi vermicompost is produced out of organic wastes through their decomposition by earthworms. The compost thus produced is a rich manure for better agricultural yield.

In India, pellets made out of organic wastes have been tried out as an alternative source of fuel. Plants producing such pellets have been set-up in Madras, Bombay, Bangalore and Baroda.

Water Supply

Provision of safe and adequate drinking water is not only essential for survival but also for healthy living. Pressure of overurbanisation in our urban centres especially the metros has led to the situation of extreme inadequacy on water-supply front. This is borne out by the fact that against the per capita water requirement of 225 litres daily, the actual per capita supply of potable water to residents of Madras is only 78 litres daily. In Delhi and Bombay, level of water supply is of the order of only 172 litres and 136 litres respectively. In the overall scenario, in Delhi, against the requirement of 2838 mld (million litres daily), there was supply of only 2498 mld in 1994, leading to a shortfall of 340 mld. Bombay had a supply of 2538 mld against a demand of 3360 mld. Ahmedabad gets only 432 mld against a demand of 535 mld.

Population growth of cities has already led to exhaustion of available water resources. Over-exploitation of surface and ground water resources have forced the authorities to look for distant sources. In Bombay, water is supplied from various sources located 29 km to 120 km away from the city. The acute water shortage in the city of Madras has affected its growth and development. A special train named 'Water Express' brings water for the citizens of Madras.

Discharge of untreated industrial and domestic waste water into water bodies has deteriorated the quality of water into water bodies has deteriorated the quality of water resources. The city of Delhi has already exhausted the water of river Yamuna and converted it into a 'perennial drain'. During 48 km journey of Yamuna through Delhi, 1198 million litres

of water is withdrawn from the river, and as much as 1489 million litres of sewage gets dumped into the river every day.

In keeping with the spatial expansion of cities, the water supply network has also been expanding. In some of cities, especially the old city areas, the pipes are very old and get corroded and develop cracks ever a period of the time. Consequently, the city sewage enters the water supply system contaminating the water flowing through it. Every year there are a number of incidents of cholera, gastroenteritis and other water borne diseases. The city authorities that are already lagging behind in meeting the demand of potable water, find it difficult to replace the old pipes. Recently, Municipal Corporation of Delhi (MCD) set aside Rs 6 crore for replacing pipes in the walled city of Sadar-Paharganj areas.

Accelerated Urban Water Supply Programme (AUWSP)

With a view to improving the position with regard to supply of potable water in the less privileged smaller towns, the Government of India introduced Accelerated Urban Water Supply Programme (AUWSP) during the Eighth Five-Year Plan (1992-97) for towns having population of less than 20,000 (1991 Census). The cost of implementation is to be shared between the Central and State in the ratio of 50:50. There are approximately 2,251 towns in this category and the provision of funds required on pro-rata basis is estimated at Rs. 2,000 crores approximately. As against this, the provision of Central assistance earmarked for this scheme during the Eighth Plan is Rs. 50 crores.

Rural Water Supply

As regards rural water supply position, only 41 per cent of the rural population on global scale have convenient access to safe drinking water supplies. In most of the developing countries in Asia, the percentage of rural population covered with provision of safe water supplies is a meagre 26. Open wells, ponds, streams and rivers still continue to be the traditional sources for water supply in majority of the Indian villages.

According to the IMRB study in India, the traditional open dug wells continue to be the primary source of water for all purposes including drinking water. The handpump comes next. The dug well-handpump-tap is generally the order of preference. Nearly 10 per cent of the households collect drinking water from exposed sources such as ponds, lakes and canals.

Accelerated Rural Water Supply Programme (ARWSP)

With a view to meeting the requirement of water supply in the villages without no safe sources, the Central Government introduced the Accelerated Rural Water Supply Programme (ARWSP) in 1972-73 to assist States and Union Territories with 100% grant-in-aid to implement

schemes in such villages. These programmes continued till 1973-74, and when in 1974-75, the Rural Water Supply Programme was introduced under the Minimum Needs Programme (MNP), the ARWSP was discontinued. In 1977-78, when the progress of supply of safe drinking water to identified problem villages was not as per expectations, the ARWSP was reintroduced to augment efforts under MNP.

In order to ensure maximum inflow of scientific and technical input into the rural water supply sector and thus to deal with quality problems of drinking water, the National Drinking water Mission (NDWM) was launched in 1986. The government has since renamed the National Drinking Water Mission as the Rajiv Gandhi National Drinking Water Mission (RGNDWM).

As a result of top priority being accorded to the rural drinking water supply programme which meets one of the basic needs of the rural people, the outlay in the eighth plan has been kept at Rs. 5,100 crores under the Central Sector and Rs 4,954.52 crore in the State Sector MNP. The concerted governmental efforts have led to provision of at least one safe source (of water) in majority of the villages (total 5,83,003) in the country.

Energy

There is an increasing pressure on the conventional sources of energy due to steep rise of population growth in developing countries. Moreover, there is also the question of affordability of the available traditional sources of energy for the poor in rural as well as urban agglomerates. Recent technological advancements have provided some solution in this regard. There have been number of innovations in the field of non-conventional sources of energy which could be gainfully utilised by both the rural poor and the urban poor as well. Some of the viable sources of energy suitable for adoption by households in villages and towns are described below.

1) Biogas Plant

The normal method of disposing of animal dung is to dry it into cakes for burning it, as fuel which not only leads to insanitary conditions in the house and the neighbourhood but also deprives agricultural land of an excellent inorganic manure. Animal dung contains ingredients which could be exploited for fuel and manurial values. In recent years, over 2 million biogas plants have been installed in a large number of villages across the country. The gas produced out of such plants is used as fuel for cooking food in homes and also for the purpose of street-lighting (Fig. 7.2). As fiscal incentive, the

Union Ministry of Non-Conventional Energy Sources (MNES) makes available a Central subsidy to the tune of 25-40% of the cost for installation of family-size plants.

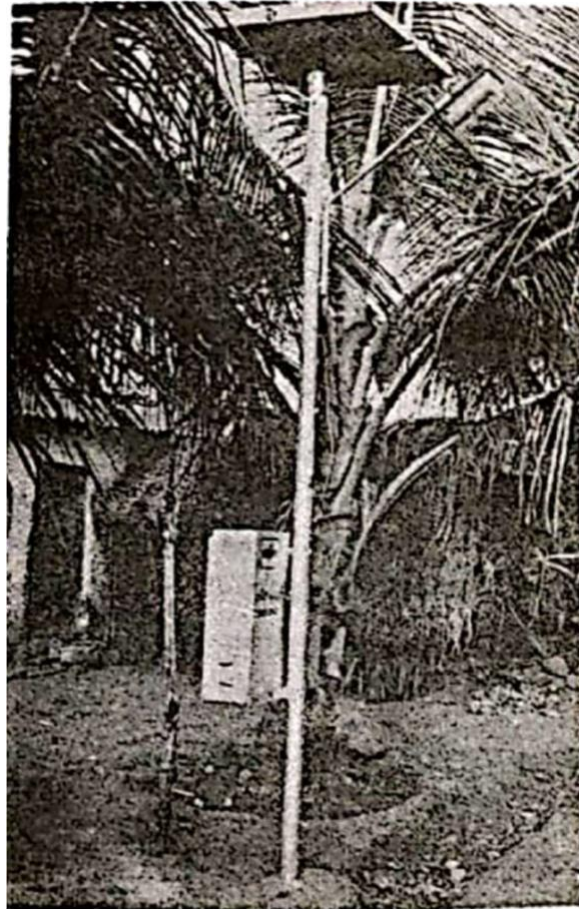


Fig. 7.3 “Stand-alone” street light for rural area (Source: BHEL, New Delhi)

The Sulabh International has also experimented with human excreta. The gas produced in biogas plants containing human excreta has been utilised for street light of a major thoroughfare in Patna. The organisation was since set-up 61 night-soil based biogas plants in different parts of the country.

Since the launch of the National Project on Biogas Development in 1981-82, over 2.1 million biogas plants have been set-up across the country. Uttar Pradesh, Maharashtra and Andhra Pradesh are promoting mainly night soil-based biogas plants.

2) Solar Energy

Solar energy keeps life's clock tickling on earth. And in today's times with the energy crunch, eyes are increasingly turning to the giant power house in the sky the Sun. An awesome source of power, solar energy equivalent to almost 75,000 trillion* (1 trillion = 10¹²) KWH hits the earth every day. And a mere 0.1% of this staggering figure is sufficient to meet the world energy needs.

In recent years there has been increasing application of solar energy for domestic and community needs. The solar energy can be gainfully used for varied applications such as street lighting in villages and hamlets where transmission costs of electricity from the grid are high, rural domestic and community lighting, water pumping in problem villages-far away from the power grid and installation of solar power plants for whole villages. Solar power is widely used in the remote villages of Madhya Pradesh, Karnataka and Tamil Nadu. By the end of March 1993, nearly 60,000 solar photovoltaic energy systems had been deployed in India's remote areas and the countryside, all designed and made indigenously.

There is a countrywide programme for demonstration and utilisation of solar photovoltaic systems, especially in rural areas. The most common systems being promoted are 'Stand-alone' street lights (Fig. 7.4) developed by the Bharat Heavy Electricals Ltd. (BHEL). They stand alone without any need of costly overhead or underground transmission lines. This also caters to domestic lighting in villages which are formally covered by rural electrification programme, solar lanterns, solar-powered TV sets and lighting of public buildings like community centres, primary health centres and places where adult literacy classes are held usually at night.

According to the annual report (1992-93) of the Ministry of Non-Conventional Energy Sources (MNES), the target for village street lighting systems which was kept as 5000 was exceeded by 900 up to December 1992, while that for villages power plants of 100 kilowatts was nearly met by setting up plants at 87-kilowatt capacity.

Solar photovoltaic lanterns are gradually becoming a more popular source of domestic lighting. According to an estimate, there are at present (1995) 700 solar lantern users in Maharashtra. The current price of a solar lantern is around Rs 950, which considering that there are no recurring expenses or maintenance charges, and the life of lantern, if properly handled, is five years at a minimum, is not unreasonable. Even the poorest household incurs more expenditure on lighting during the same period.

Beside these solar heaters and cooker are also being used for domestic application. Application of solar heaters result in saving of conventional sources of energy such as wood, kerosene oil and electricity. The Solar cookers are pollution free. Under a government-sponsored programme initiated in 1982 to address the rural energy problems and commissioned by the MNES, the solar cookers are being promoted in different parts of the country. Till date as many as 3 lakh solar cookers are reported to be in use in different parts of the country.

Smokeless Chulha (Hearth)

Cow dung, charcoal and firewood which are most commonly used as fuel in Indian villages produce a lot of smoke which is very injurious to health. To avoid this nuisance, at the behest of the NBO exhaustive investigations on efficacy of the existing chulhas were undertaken by the Planning Re-search and Action Institute, Lucknow in 1964. The study had identified some performance-efficient chulhas which could work without causing any smoke hazard. Such

chulhas being pollution free, women find them much convenient for cooking. The NBO has promoted such chulhas in the country through their incorporation in cluster demonstration rural houses put up in 81 villages lying in different geo-climatic regions of the country. Each improved chulha saves 700 kg. of fuelwood a year. In recent years, the MNES has been actively implementing the scheme for installing smokeless chulhas in the villages. 19.5 million rural households in the country have already been covered under this scheme. Another 5 million additional households are proposed to be covered during the Eighth Five Year Plan.

CHAPTER 8

Rural Housing

Introduction

Housing scenario in the rural India is more appalling than the urban agglomerations. As per NBO estimates, there was a shortage of 20.4 million dwelling units in 1991 in the country. The quality of housing observed in the rural areas is an indicator of the abject poverty, tremendous socio-economic disparity and extreme backwardness prevalent in the countryside. Inadequacy of infrastructure facilities in villages is very obvious from the fact that protected drinking water supply through tap was available only to 15% of the households in villages, and about 89% households in areas did not have sanitary latrines in 1988. Slush, dirt and unhygienic environment are generally synonymous with the human settlements in rural areas.

Traditional Practice of Rural House Construction

A majority of rural houses in the country are constructed with mud wall and thatch roof (Fig. 8.1). In places where stone is locally available, as in Rajasthan, Gujarat and hilly areas like Himachal Pradesh, houses are constructed with stone wall and roof comprising of stone slab and slate etc. Timber and bamboo are used in north-eastern States of Assam, Meghalaya,

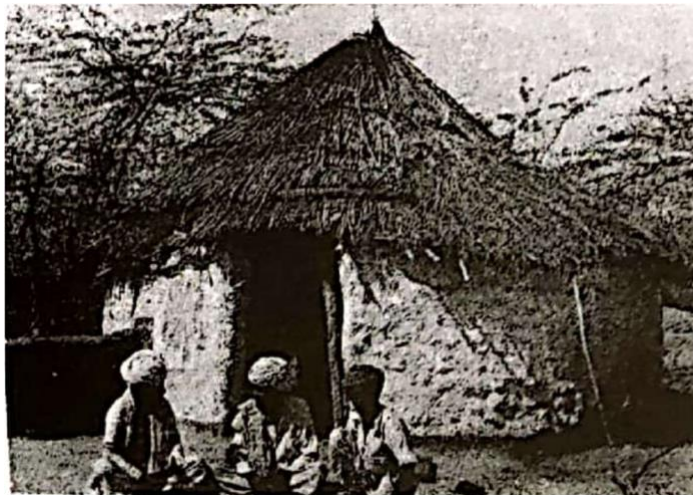


Fig. 8.1 Traditional rural housing in a Rajasthan village

Tripura, Nagaland, Mizoram, Arunachal Pradesh and Manipur. Bricks and tiles are used as building materials in the rural areas of Uttar Pradesh, Madhya Pradesh, West Bengal, Bihar,

Punjab, Karnataka, Kerala, and Tamil Nadu, etc. In construction of traditional rural houses age-old conventional methods are normally employed.

Appropriate Rural Housing Technology

In the present context, the appropriate technology for meeting the requirements of rural housing in the country will be the one which emphasises on judicious blending of the technological advancements in the field of building technology, living habits and financial capabilities of the rural people with due importance being accorded to optimisation of the locally available building materials.

The NBO Regional Housing Development Centres (RHDCs), the Central Ministry of Rural Areas and Employment, HUDCO, various state Rura Housing Co-operatives, and Government of Kerala etc. have been instrumental in promotion of cost-effective and appropriate rural housing technology in the country. Improved designs of village houses were evolved by the RHDCs to make them more hygienic and liveable. The centres hand put up clusters of demonstration low-cost rural houses in selected villages across the country. Over 81 such clusters (Fig. 8.2 and 8.3) comprising 10-20 houses along with measures for environmental improvement have been put up in different geo-climate regions of the country with the objective of educating the rural people about the innovations in low-cost housing technology. This has been done with emphasis on optimisation of locally available building materials, which could help bring down the cost of construction as well as make rural houses more durable and liveable. The list of the cost-effective construction techniques and indigenous materials, which have been extensively tried out under the cluster demonstration rural housing projects, is provided at Annexure I. Items of environmental improvement provided in the aforesaid projects included sanitary latrines, paved streets, smokeless chulhas, bathing platforms, supply of potable water and provision of streets and open space etc.



Fig. 8.2 NBO cluster demonstration rural houses in a Gujarat village constructed in stone masonry

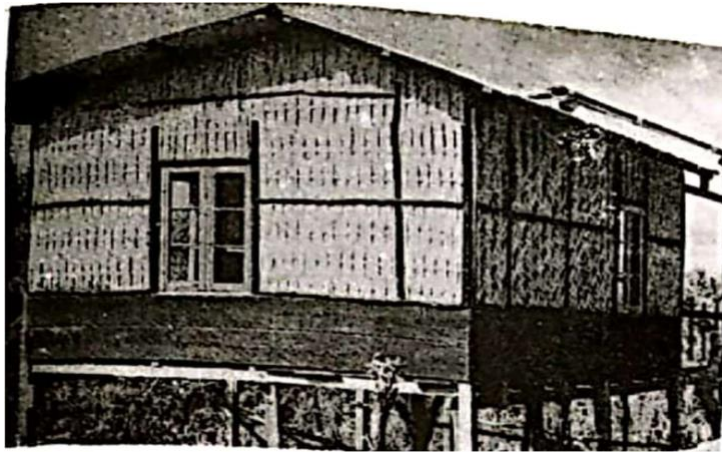


Fig. 8.3 NBO demonstration house in a Mizoram village depicting larger use of timber

Mud Housing Technology

Mud is a principal building material in many developing countries. The local soils have been used for the construction of such houses. As per the estimate, 8 out of 10 rural houses in the country are either kutcha or semi-pucca and are built with mud walls and thatch roof employing self-help. Virtually, in every continent of the world—from the dry deserts of the Middle East to the rain forests of South America, from impoverished Asia to the affluent Europe, examples of mud buildings are abounded, and apart from their regional variations, mud is the predominant building material all over the world, especially in the countryside. Versatility of earth as a building material is illustrated Fig. 8.4.

In Asia and Africa, as well as in parts of Europe and Latin America, where most people still live in villages, there is an age-old relationship with the earth. In Nigeria's hot and humid climate, the mud-bamboo and mat house are the only viable choice, its latticed walls open to the breeze and deep overhangs keep the interior cool and dry. Similarly, in the hot arid Dra Valley in Morocco, where earth is the only material available, mud house huddle closely together to keep out the desert dust and heat. The arch and dome, which have now become established symbols of Islamic architecture, have their origin in the mud brick structures of the Middle East.

There are at least 20 different techniques in use the world over. Though they vary enormously, six of them are commonly used.

a) Mud Lump (Cobwall):

This is the simplest method of construction of mud walls. In this technique, local soil which is neither too clayey nor too sandy is mixed with requisite amount of water to form lumps of good consistency. Lumps are placed one over another manually. Each layer is placed on alternate days to make allowance for sufficient time for drying. Wall so constructed is then plastered with a coat of mud-mixed with some organic material such as cow dung. Generally, height of such wall is low, not exceeding 2.4 m and thickness about 30 cm.

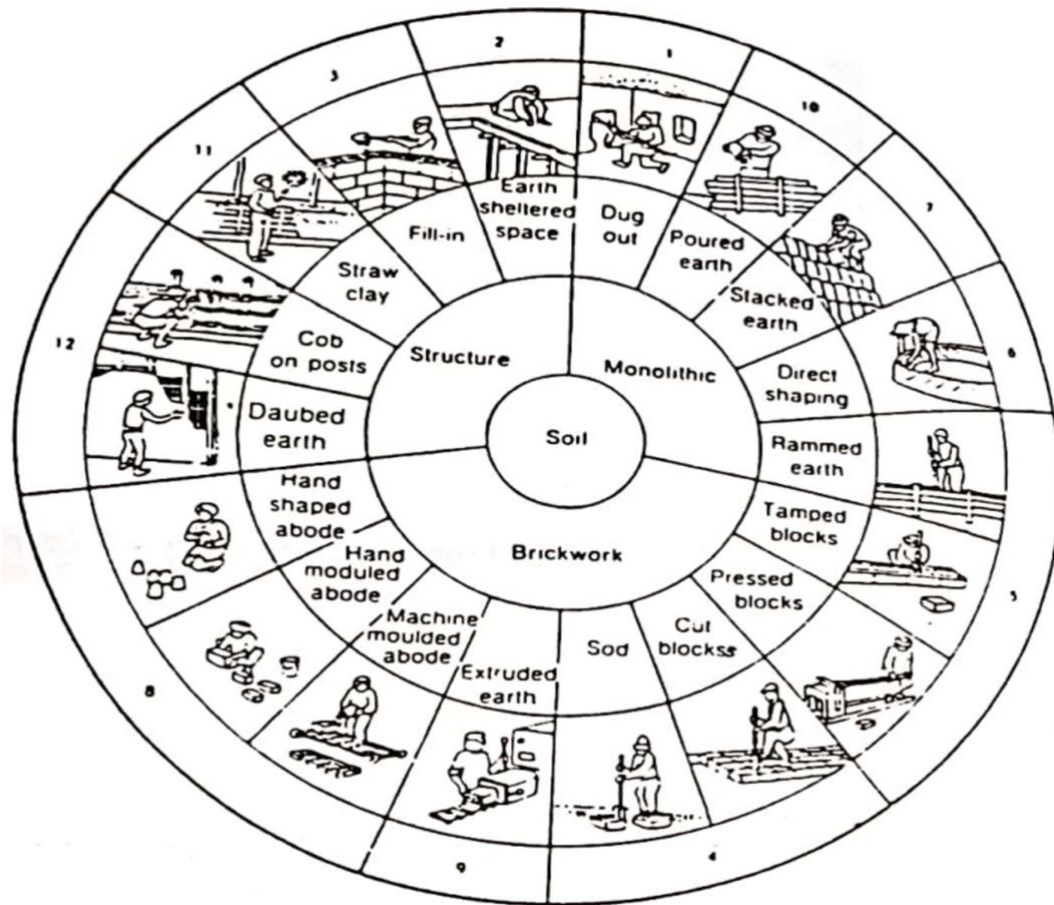


Fig. 8.4 Use of earth as building material (after Bibl. 141 reproduced with permission from SKAT, Switzerland)

The roof for such construction normally comprises of bamboo or wooden trusses placed at regular intervals on the mud wall which support the wooden purlins and rafters. Thatch or country tiles is used for covering the roof. The roof overhang of such houses is generally kept between 30 to 45 cm. Such houses are built at very low cost through self-help, though they are not very durable and strong.

b) Wattle and Daub

This is one of the simplest and cheapest forms of wall construction. For many people in Africa (e.g., Swaziland) and Asia it is the only way they can afford to build their houses. Wattle and daub basically refer to a framework of vertical poles, tied together with horizontal branches and plastered inside and outside with mud.

c) Sun Dried Mud Brick (Adobe)

Under this method, walls are prepared from sun dried bricks. For preparation of sun-dried bricks, any soil which is found to contain 80% clay and 20% sand, or 40% clay and 60% sand may be used. Water is added to and soil in the requisite quantity and then the prepared mud is moulded the shape of bricks/blocks of suitable size with the help of wooden mould. The green dried mud bricks are used for construction of walls. The bricks are laid in courses in mud mortar. The wall is raised in stages, about 1 metre high every day. The mud wall so prepared is well formed and strong.

Sun dried mud bricks are known as 'adobe' in Mexico and the USA. Such bricks are very popular in south India, Egypt, Peru, New Mexico, Southern Africa and West Tunisia. The State of New Mexico in South West- ern USA has developed a regulatory code for mud buildings which covers mainly adobe (sun dried mud bricks).

Manually prepared sun-dried mud bricks are now becoming common in rural India. Though they are normally larger than kiln-fired bricks, they do not conform to any standard size. Such bricks usually make stronger walls than mud.

d) Rammed Earth (Pise de terre)

Rammed earth (pise-de-terre) (Fig. 8.5) is a technique where mud is Rammed manually between two moveable timber shuttering's (vertical frames) which are erected where the finished wall will be. The preferred type of soil for such construction is 'sandy loam.' The soil prepared for ramming is just moist enough to form a compact ball when it is pressed between cupped hands.

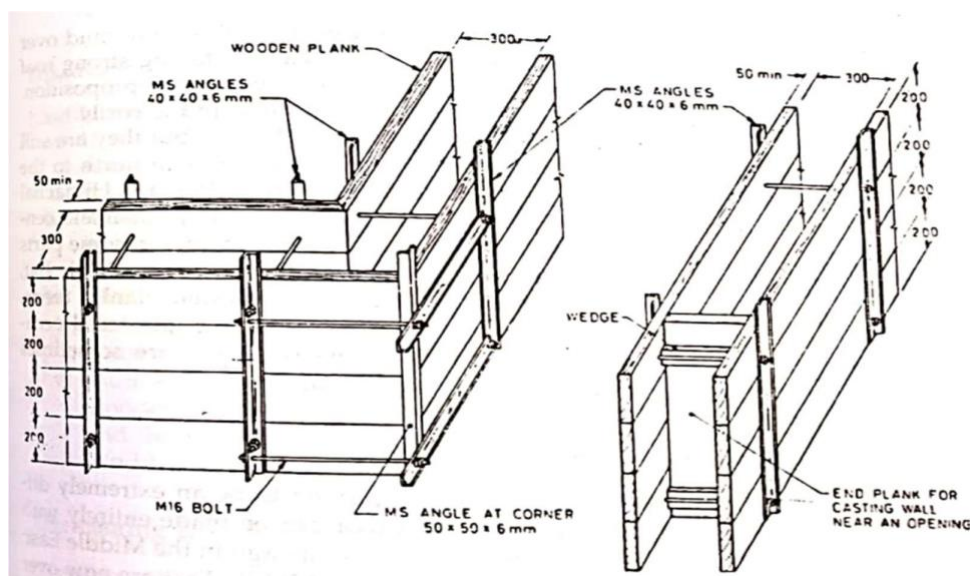


Fig. 8.5 Formwork for rammed earth construction

The walls are built in courses. The height of the course is determined by the size of form work, which is generally not more than 80 cm. for 30 cm. thick wall. After one layer has been rammed, the shuttering is raised, a second layer of mud rammed onto a first layer, and so on. Rammed earth technique of wall construction has been used for centuries in Venezuela and Colombia.

In India, rammed earth was first tried out in 1948 in the construction of 4,000 houses at Karnal (Haryana). These houses have been in use for more than four decades now. The Bureau of Indian standards has also brought out a code of practice (IS: 2110-1980) for the constructions rammed cement soil walls for rural houses.

e) Compacted Soil Block

The technique of production of compacted soil block is relatively young in Comparison with others. Compacting the soil in a small form produces a block, which by virtue of the increased density, is both stronger and more water resistant than an ordinary mud brick. Such blocks can be produced by compressing earth into a solid block in machine called the 'Cinva Ram'.

Mud Roofs

Mud roofs are made in several parts of the world by plastering mud over a mat of bamboo sticks supported by wooden beams. Making strong roof an entirely with mud has proved to be an extremely difficult proposition. Flat mud roof involves heavy supporting timber work and is costly.

In India, mud roofs are not as common as mud walls, but they are still widely found in a belt ranging from Jammu and Kashmir in north to the Deccan plateau in south, an area covering Jammu, Punjab, Himachal Pradesh, some parts in Uttar Pradesh, Madhya Pradesh, Rajasthan and central parts of Maharashtra. In south, mud roofs are found only in some parts of Karnataka and Andhra Pradesh. The Indian mud roofs are normally flat, with mud used to cover a supporting platform of wooden planks, reeds, bamboo matting or stone slabs. Mud is beaten down and plastered occasionally with an emulsion of cow dung. Layers of leaves are sometimes added to prevent the mud from dropping through.

Mud Brick Vault and Dome

Making strong roof entirely with mud has proved to be an extremely difficult task. The only technique by which a roof can be made entirely with sun dried mud bricks was invented over 6,000 years ago in the Middle East: the vault. Some ancient vaulted roofs in the arid Middle East are now over a thousand-years old and are still intact.

The mud brick vault-and dome building system was evolved centuries ago in West Asia. It was a response to necessity, for roof-spanning materials such as timber and reed became more and more scarce as populations grew in the hot, semi-arid regions. It is also the unique response of the ancient West Asian architects to the fact that sun-dried mud brick has strength in compression but not in bending or tension.

Domed or vaulted roofs are extremely used in Iran and Turkey. In Europe, vaulting involves laying masonry over a wooden form work which is later removed when the vault became dry. But in the West Asian construction, no supporting formwork is required for fired or sun-dried brick. If sun-dried bricks are used with mud mortar, the bricks fuse together on drying, since they are of the same material. This gives the vault additional strength.

Characteristics of Mud

Advantages

- 1) It is a truly natural, cheap and local material-available in most parts of the world.
- 2) It provides excellent heat insulation, so the internal part of a mud building is cooler in summer and hotter in winter than a building made with steel and concrete. Due to this property, mud houses are most suited in tropical and sub-tropical climatic conditions.
- 3) It is strong in compression and so makes good walls. It can also be made strong in sheer and tension through additives and reinforcement.

Disadvantages

Mud has some serious disadvantages which are as follows:

- 1) It is eroded easily by water, which makes its use difficult in areas with high rainfall or possibilities of flooding.
- 2) It has low strength, which means mud roofs are difficult to make.
- 3) It is susceptible to mechanical damage. Rodents can easily make hole in mud walls and under the floor, or thieves can dig their way into the house.
- 4) Mud does not grip wood properly, so gaps often develop around wooden doors and windows in mud walls. Consequently, mud houses often have few openings and are badly ventilated.
- 5) Mud soaks up water and becomes very heavy. As a result, the wooden beams supporting a mud roof begin to sag, the mud cracks and the roof starts leaking.
- 6) Mud houses behave very poorly in the event of earthquakes.

Overcoming Disadvantages of Mud

Centuries of trial and error have generally produced designs appropriate for local conditions. There is, however, always scope for improvement. A Cornish proverb has it that "All a mud house needs is a good head and a good pair of boots," meaning thereby that all a raw-earth

building needs to survive is a 'pucca' (stone or concrete) foundation, and overhanging roofs to save walls from water.

Most of the disadvantages associated with mud houses can be overcome by suitable improvements in design and technology, such as soil stabilisation, appropriate architecture, and improved structural techniques. As the majority of the people in the developing countries will, out of necessity continue to live in mud houses, ways of improving on the traditionally built mud houses have been evolved out of sustained research in the direction

The Central Building Research Institute (CBRI), Roorkee, has developed non-erodable mud plaster for making mud houses more durable. The Water-repellent treatment for mud walls consists of mixing cut-back bitumen with ordinary mud plaster prepared with bhusa (paddy/wheat chipping) and cow dung. The water-repellent mud plaster is prepared by mixing up to 5% (by weight) cut-back with ordinary mud plaster comprising of mud and bhusa (paddy/wheat chipping) in the proportion of 70 Kg bhusa to one cubic metre of soil. The cut-back, an emulsion of bitumen (90/100 grade), is prepared by hot mixing 50 Kg bitumen, 10 litres of kerosene oil and 0.5 Kg of wax. For every cum. of soil 70 Kg of cut-back is added. The mix of soil, bhusa and cutback is thoroughly mixed by turning over and kneading to get a homogeneous mix. The mortar so prepared is applied in the usual way. When the plaster has dried, a coat of gobri-leaping (cow dung mixture) 1-part gobri and 1 part soil and cutback in the same proportion as for plaster is applied over the dried surface of plaster. The plaster thus rendered can withstand the rains for 3 to 5 years without major maintenance.

Fire Retardant Treatment for Thatch Roof

Thatch is the most common roof for mud houses. It is, however, vulnerable to fire hazards. In order to make the thatch roof fire retardant, the top and bottom surfaces of the thatch roof are plastered with 20 to 25 mm thick layer and 10 mm thick layer of bitumen stabilised mud plaster, respectively. Over this plaster, two coats of gobri leaping are applied to improve the finish. Furthermore, this is left to dry to receive spray of bitumen solution. Bitumen solution is prepared by mixing hot molten bitumen and kerosene oil in the proportion of 1:2. One coat of this emulsion is either sprayed or applied by brush on the top surface of the plastered thatch. The second coat follows after three to four hours. Two coats of lime are then applied over the bitumen-sprayed surface.

Soil Stabilisation

Researchers have identified around 130 different stabilising agents including cement, lime and bitumen. Before deciding the correct type of stabiliser to be added to the soil, thorough testing of the soil is essential.

The best type of soil for stabilisation with cement is one, which has a proper grading of particles of different sizes, from the sand range to fine clay. Very clayey soil, particularly black cotton soil, is best stabilised with lime, and bitumen is suitable for sandy soils. In some

cases, a combination of stabilisers should be used. Highly plastic soils, and soils containing organic matter are not suitable for stabilisation.

The following percentages of additives are found most suited to achieve the desired strength to manufacture stabilised bricks/block, resistance to weathering and to effect economy. The exact quantity of stabiliser to be added depends on the type of soil. As a rough guide, the following could be adopted under normal situations:

Sandy soils: 5% cement stabilisation

Clayey soils: 5 to 6% hydrated lime and 2% cement Normal

red soils with up to 25% clay: 2 ½ % hydrated lime.

Lime combined with fly ash makes a cement almost as good as Portland cement. This is used as a stabiliser for both sandy and clayey soils.

Of all the stabilisers, cement has been the most extensively used. Soils which are treated with lime are not as strong nor as water resistant. Soil stabilised with cement or lime can also be compacted in a Cinva Ram. In India, apart from the Cinva Ram, some of the popular soil-block making machines in vogue are 'Ellson block master,' 'Balram' and 'Astram.' On average, about five hundred 23 x 19 x 8 cm. pressed-soil blocks can be made with the 'Astram' machine per day, and about a thousand 23 x 11 x 7.5 cm. blocks with the 'Balram'.

The Stabilised bricks are usually sun-dried, and laid in ordinary mud mortar, although stabilised mud mortar would be better. They are protected with a covering of non-erodable mud plaster from the outside, and with ordinary mud plaster from the inside.

The stabilisation of soil with the addition of Portland cement, lime, bituminous binder, industrial wastes like fly ash and bagasse, etc., can improve the bearing strength of bricks and render it increasingly water and moisture repellent. Of the various additives, cement bricks (with 5% cement) though better in strength (about 4 N/mm²) * are poor in water and moisture resistance, whereas bitumen-stabilised bricks (with 1% bitumen), though poor in strength (about 3 N/mm²), are good in water and moisture resistance. Stabilised bricks can be laid in ordinary mud mortar or stabilised mud mortar; the latter one, of course, is preferred from strength and durability considerations. Experiments have been carried out at the CBRI, Roorkee to reduce the amount of bitumen from 1.0 to 0.5% through the use of industrial wastes like fly ash and bagasse as additives. The use of rice husk cement has also been tried for the stabilisation of soil with promising results.

Application of Science and Technology for Rural Areas (ASTRA) at the Indian Institute of Science (IISc), Bangalore, and the NBO RHDC, Bangalore, have played an important role in the promotion of soil cement blocks as useful building component for house construction in Karnataka. As a result of promotional work undertaken by the ASTRA and the NBO RHDC, five stabilised mud housing programmes involving construction of about 7,000 houses have been undertaken in different parts of the State.

The major advantage of the stabilised soil block vis-a-vis the burnt brick is the significant saving in energy (about 70%). Such blocks are cheaper by 20 to 40% compared to

burnt bricks. A study of the performance of the soil blocks used in the construction of houses undertaken by the Karnataka Housing Board has revealed that the compressive strength obtained from such blocks, ranges from 3-7 N/mm². Up to two-storeyed houses, 5% cement or 2 ½ % cement and 24% hydrated lime stabilised soil blocks possessing a minimum compressive strength of 3.5 N/mm² could be used. Soil stabilised blocks are gradually being accepted as alternative to burnt bricks in Karnataka. Large number of private builders in rural and urban areas in the southern states of the country, especially in Karnataka are using such blocks in house construction.

Soil Stabilisation by Traditional Water Proofers

Various traditional materials ranging from plant juices to cow dung are mixed with mud for rendering mud walls. Such treatment renders mud walls more water repellent.

In northern Ghana, an extract of boiled banana (or plantain) stems is mixed with lateritic soils. In Upper Volta, a plant extract locally known as 'am' is used as varnish. In Northern Nigeria, 'laso' (an extract from the vine 'vitis pallida,' locally known as 'dafara') and 'makuba' (made from fruit pod of the lowest bean tree) are used for waterproofing of mud walls. Cow dung mixed with clay has been widely used in Ghana and India. In Sudan, 'jaloos' (mud) houses are treated with 'Zibla,' a local water proof material made from cow or horse dung. Straw has been mixed with mud since Biblical times, especially in West Asia. In Ethiopia, straw (preferably 'chid' the straw of millet) is used in mixing 'chika' or soil paste.

Rural Housing Programmes

In a country like India where about 75% of the population lives in 5.83 lakh villages, a rural housing programme is necessarily an aided self-help programme. Such aided self-help is the technical systemisation of the well-known traditional principles which have a bearing on co-operative efforts. Over the past two decades, number of rural housing programmes have been undertaken in the Government and co-operative sectors. Some of the prominent rural housing programmes undertaken in the country are described below.

1. In order to ameliorate the living conditions of the landless rural poor, the Government of India had introduced a scheme envisaging provision of house-sites to the landless agricultural families in rural areas free of cost in 1971 in the Central Sector as a component of the Fourth Five-Year Plan (1969-74). Under the said scheme, the State Governments and the Govt. of Union Territories provided house-sites measuring minimum of 82 sq.m. (100 sq.yds.) free of cost to 14.5 million landless rural families who did not own house sites nor built-up houses. The beneficiaries were expected to build modest houses through self-help. The scheme was transferred to the state sector with effect from April 1, 1974, and was also included in the Minimum Needs Programme (MNP). The scheme had been continued in the Fifth Five-Year Plan (1974- 79) wherein its scope was extended to other landless workers who were Bonafede

artisans such as potters, carpenters, fishermen, blacksmiths, etc., who had no house-sites of other land of their own. The scheme was accorded high priority through its inclusion in the 20 Point Programme announced by the Prime Minister in July 1975. The scope of the said scheme was further broadened in the revised New 20-Point Programme announced in 1980, according to which assistance was to be provided to some 3.6 million households during the Sixth Five-Year Plan (1980-85). So far 194.23 lakh rural landless workers have been provided house-sites, and 60.25 lakh families have been construction assistance on allotted house-sites.

2. Rural Housing Co-operatives can play an important role in the integrated rural development by mobilising saving and channelising community action not only for the construction of houses but also for the management of community facilities. This is being taken advantage of in some developing countries notably India, Pakistan, Malaysia, Thailand and Sri Lanka.

In India, with a view to encourage housing in rural areas, the Maharashtra Co-operative Housing Finance society provides long-term credit on priority basis. Similarly, in Andhra Pradesh, the Scheduled Castes and Scheduled Tribes Co-operative Housing Federation has provided a large number of houses to the rural poor through utilisation of loans received operative Housing Society, the Karnataka State Co- operative Housing Finance Society have played a significant role in providing affordable houses to the rural people.

3. In 1973, the Government of Kerala had launched a scheme popularly known as 'One Lakh Housing Programme.' Under this programme, 100 houses were to be constructed in each of the existing 1,000 Panchayats in the State at that time. The houses were to be provided to one lakh landless poor families whose poor affordability prevented them from having their own houses. The construction of houses under the scheme was undertaken by the Panchayats. The State Government mobilised funds for undertaking such a massive programme for rural housing through collections, donations and grants. The final figure in respect of houses constructed was, however, 60,000 which was still a creditable achievement by any standard. The major lacuna in this scheme was non-participation of beneficiaries in construction of the houses and they kept complaining about the quality of construction of the houses and they pressurised the Government for grants to repair the houses year after year.

In 1982, the Government of Kerala introduced a new scheme called the Subsidised Aided Self-Help Housing Scheme for the Economically Weaker Sections known as 'SASH' with the involvement of voluntary organisations. The main thrust in 'SASH' was to involve beneficiaries fully in construction of houses to overcome the lacunae experienced under the One lakh Housing Schemes. The programme, though conceived in 1982, was implemented in full swing during 1984-85. The entire loan assistance for this programme was provided by HUDCO. Over 800 voluntary organisations have since participated in the implementation of the aforesaid housing scheme in Kerala.

4. Over the past one decade, the Ministry of Rural Areas and Employment (MRAE) under the Government of India, has undertaken number of schemes catering to the housing

needs of the rural poor. The Indira Awas Yojana (IAY), launched during 1985-86 is fully subsidised by the Central Government. This is a scheme of the construction of low-cost houses for the poorest of the poor belonging to the SCs/STs and freed bonded labour in the rural areas. The overall target during the Seventh plan (1985-90) was for construction of one million houses. The scheme is now also extended to non-SC/ST families, with ceiling of expenditure on such families being 4% of the total allocation. The IAY has since become part of the Jawahar Rozgar Yojana (JRY). The latter was initiated in 1989 by merging the National Rural Employment Programme (NREP) and the Rural Landless Employment Guarantee Programme (RLEGP). Over 18.43 lakh houses have been constructed with a financial involvement of Rs. 2,197.49 crore since its inception till November, 1994.

5. The HUDCO is actively engaged in evolving designs and financing of low-cost rural housing programmes in the country. It started financing of rural housing schemes in 1977. It has since financed 1,496 projects envisaging construction of more than 27.3 lakh housing with a loan commitment of Rs. 1043.32 crore.
6. The Government of India has recently approved a scheme to supplement the efforts of the States and Union Territories to give an impetus to housing for the weaker sections and people in the rural areas during the Eighth plan-period (1992-97). This is to involve allocation for capital subsidies for rural housing as well as allocation of resources for internal subsidy and setting-up 300 rural centres to popularise low-cost building materials.

Under the new scheme, the States and Union Territories will receive matching allocation for rural housing schemes meant for the weaker sections. It covers sites and services; shelter upgradation and construction of new houses. Funds are to be provided to States on equal proportion to the expenditure incurred by them on housing programmes for the poor under the MNP. This support will be available to the States in a flexible manner to help them continue with their existing programmes which are suffering for want of adequate resources.

The allocation for rural housing during the Eighth plan is Rs 350 crores. A sum of Rs 12,000 is to be given to those below the poverty line for this purpose. The subsidy will be shared equally by the Centre and the States. For borrowers, who have to pay loans over a period of 15 years, there will be an interest subsidy up to 4.5 per cent. Housing subsidies are presently provided through the Indira Awas Yojana for Scheduled Castes and Scheduled Tribes, allotment of house sites-cum-construction assistance under the MNP. The HUDCO supported loan schemes for economically weaker sections, whereas the LIC/GIC funded loan schemes for the State Governments.

Typical Specifications for Low Rural Housing

For the guidance of prospective builders and construction agencies entrusted with the task of providing houses to masses at affordable cost, the typical specifications for low-cost

housing to be adopted are provided at Annexure II. Selection of a particular specification is generally influenced by the builder's choice for the specific alternative specification and the availability of the locally available building materials.

Annexure I

Local Materials and Construction Techniques promoted by the NBO RIHDCs for Construction of Low-Cost Houses in Rural Areas

Foundation

1. Mud concrete.
2. Lime Stabilised mud concrete.
3. Lime Concrete.
4. Cement Concrete (1:8:16).

Walling

1. Sundried brick masonry with mud/water proof mud plaster.
2. Brick masonry mortar.
3. Bitumen-soil stabilised bricks for masonry.
4. Stone masonry in mud mortar.
5. Laterite stone masonry.
6. Lime mortars and composite mortars for masonry.
7. Clay block masonry in mud mortar.

Roofing

1. AC/CGI Sheet Roofing over local wood rafters and purlins.
2. Asphaltic sheet roofing.
3. Country tile roofing over precast RC battens. 15
4. Country tile roofing over wooden battens (Secondary-Species).
5. Mangalore tile roofing/country tile roofing over split bamboos/secondary species of timber.
6. Stone Slab Roofing over wooden joists.
7. Brick tile roofing over wooden battens.
8. Thatch roofs treated with fire retardant.
9. Brick Panel roofing over RC/Wooden joists.
10. Jack arch-terrace (Madras Terrace).

Floorings

1. Mud flooring.
2. Mud Gobri flooring.
3. Stabilised Mud flooring.

4. Moulded Blocks from the block making machine.
5. Stone Slabs locally available such as cuddapah slabs.
6. Cement flooring over Jhama Brick Bats.

Finishing

1. Lime Surkhi Cement Pointing and plasters.
2. Water Proof Mud Plaster.
3. Secondary Species of Timber.

Flooring

1. Stabilised soil cement base for floors.
2. Magnesium Oxychloride flooring.
3. Floor base of 1:20:40 cement: flyash: sand.
4. Reduced floor thickness.

Structural Members (Beams, columns, etc.)

1. Precast RC columns and beams.
2. Precast concrete staircase elements.

New Materials

1. Use of deformed bars.
2. Partial replacement of cement by flyash.
3. Use of hydrated lime.
4. Plastic stays.
5. Plastic wash and hand basins and other fittings.
6. Secondary species of timber for door/windows.
7. Reactive clay pozzolana.

Services (Sanitary, water supply)

1. PVC pipes for water supply.
2. Single stack system of plumbing.

Architectural Planning and Structural Designs

1. Lower ceiling height.
2. Precast lintels.
3. Jalis for ventilation.

4. Design for solar energy utilisation

Annexure II

Typical Alternative 100 Specifications for Low-Cost Housing in Rural Areas

1. Foundation

Boulders, gravel, kankar, local burnt bricks, bricks, stones or laterite in lime or mud mortar.

Stabilised soil using cement, lime or bitumen.

Rammed earth (Please see note 1 below).

2. Damp Proof Course

2 courses of burnt clay bricks soaked in 2% soap solution laid in cement mortar containing 1% soap solution.

2 courses of burnt bricks dipped in tar laid in tar-sand mortar consisting of 15 kg. of tar in 100 kg. of sand.

3. Walls

Burnt clay bricks, stone, laterite, laid in lime or mud mortar/unburnt clay bricks or blocks; stabilised mud bricks or blocks with cement, lime or bitumen as stabiliser and laid in stabilised mud mortar containing 2% lime.

Pise construction i.e., in-situ rammed earth at optimum moisture content.

Timber or bamboo post with panels of split bamboos, reeds or palmyrah matting with or without plaster. (Please see note 2 below).

4. Structural floor/roof

Flat roofs

Local wood joists and planks, bamboos, reeds covered with a layer of 8 to 12 cms. thick compacted soil.

Stone slabs.

Sloping roofs

Trusses of local timber, ballies and purlins of local timber, ballies and bamboos covered with thatch, reeds, palmyrah leaves, country tiles, slates, Mangalore tiles (Thatch and palmyrah leaves to be made fire-proof).

Asbestos cement sheets, GI sheets.

Asphaltic roofing sheets (Please see note 3 below).

5. Flooring

Well compacted soil rendered dust free by frequent gobri leapings. Stabilised soil layer with cement, lime or both as the stabiliser.

Burnt clay bricks.

6. Doors and Windows

Local timber, air seasoned/bamboo and iron fittings

7. Finishing

Internal: Ordinary mud plaster

External: Non-erodible mud plaster

Notes

1. Rammed earth wall is suitable only when sand content is 60 to 70% and rainfall is scanty.
2. This specification is for areas prone to earthquake and where heavy rainfall occurs.
3. Asphaltic roofing sheets are to be used only if the shade temperature is not likely to exceed 40°C.

Housing in Disaster Prone Areas

Introduction

Natural disasters like floods, earthquakes, cyclones, landslides, floods, avalanches, volcanic eruptions and tsunamis, etc., cause great loss of human life and property. Human settlements in India, too, are often subject to the vagaries of natural disaster. Earthquakes, cyclones and floods are more frequent natural calamities hit the Indian subcontinent. In this chapter nature and extent of damage caused to structures by such calamities, and the precautionary measures to be adopted in the event of their occurrence to make the structures, especially the non-engineered buildings want more resistant to the devastating effects of natural catastrophes, have been discussed at length. In view of the large-scale damage of the non-engineered buildings and houses belonging to the low strength masonry (LSM) category, in the earthquakes of recent past occurring in North Bihar (October 1988), Uttarkashi (October 1991) and Killari, Maharashtra (September 1993), a writeup on repair and restoration of earthquake damaged non-engineered buildings is also included in this chapter.

Earthquake

Earthquake is nature's one of the most catastrophic natural disasters known to the mankind since time immemorial. On an average, nearly 10,000 quakes stronger than magnitude 4 on Richter scale are observed every year across the globe. In the past, several devastating earthquakes in various parts of the world have caused immense loss in terms of human lives and property. According to the Secretary General's report in the 43rd UN General Assembly, more than 50% of deaths due to natural calamities in the world during the period 1900-87 has been due to earthquakes alone (Table 1).

About two-thirds of India are known to lie in seismic zones of moderate to severe intensity. Conventionally, Himalayan-Naga Lushai region, Indo-Gangetic plains, Western India and Kutch and Kathiawar regions are geologically unstable parts of the country, and some devastating earthquakes of the world have occurred in these regions. The 1993 earthquake in the Marathwada region (Latur and Osmanabad districts) of Maharashtra, which is known to be a stable zone from the point of view of tectonic movements within the earth, has however, posed a serious threat to the earlier demarcation of the seismic zones within the country. The Peninsular India where the above earthquake occurred had hitherto been branded as a seismically stable zone. Besides, a heavy casualty in terms of lost lives (about 10,000) and property in the Deccan plateau has necessitated the urgency to review the seismic map of India as indicated in IS: 1893-1984

Table 1

Deaths in natural hazards (1900-87)

(after Bibl. 7-reproduced with permission from the Indian Concrete Journal, Bombay)

Natural hazard	Deaths, percent	Region of the world	Deaths, percent
Earthquakes	50.9	Asia, south and west pacific	85.5
Floods	29.7	Europe	7.0
Cyclonic storms	16.8	America	7.0
Volcanic eruptions	1.9	Africa	0.5
Tsunamis	0.5		
Landslides	0.1		

Earthquake Forecast

Despite tremendous advancements in the field of seismology, it is not yet possible for seismologists the world over to make any prediction regarding timing and placing of any future earthquake, unlike the cyclones. In recent years, there has been greater reliance on the Very Long Baseline Interferometry (VLBI) equipment for a possible solution in this direction. There has been yet another advancement in the field of earthquake forecast. Some French scientists have recently reported the viability of the Synthetic Aperture Radar (SAR) technique being used for earthquake prediction with fair degree of precision. They claim to have experimented the above technique for recording changes in ground motion produced by the 1992 earthquake in Landers, California. Their findings, as reported in the prestigious science journal, 'Nature' (Vol. 364, dt. 8.7.1993) provides an account of first ever actual mapping of earthquake done with the help of European Remote Sensing ('ERS-1') satellite.

North. India faces grave threats of a devastating earthquake of magnitude eight or nine on Richter scale comparable to the Alaskan earthquake of 1964 which was one of the greatest ever measured, arising out of the possible release of the accumulating stresses along the plate margins of the Indian and Asian plates over past 300-700 years in the Central Himalayan region. The building up of stresses along the plate margins in the region is on account of persistent-thrust Indian plate is exerting on the Asian plate along the plate boundary. The Indian plate is believed to be moving at a speed of 2 cm a year, the same rate at which the two sides of the notorious San Andreas fault (crack) that runs for more than 1,000 km through California (USA), are sliding past each other. The Indian earthquake prediction is based on the recent studies carried out by the groups of American and Indian scientists, of the past Indian earthquakes and analysis of plate tectonics of the region backed up by satellite observation. The scientists have employed Global Positioning System (GPS) for undertaking study of the movement of tectonic plates in the Central Himalayan region. The GPS, a network of satellites

and ground stations, was used to accurately detect movement of a set of points on the Indian and Asian tectonic plates between 1991 and 1994. In recent past, the Indian Remote Sensing satellite IRS-1B had observed ruptures on ground and formation of horst, a block of the earth's surface bounded by faults near Killari which is reckoned as the probable reason for the powerful earthquake that rocked Killari and its surrounding villages on September 30, 1993 leaving about 10,000 dead. In light of the above, the recent forecast of a catastrophic earthquake in the Central Himalayan region cannot be taken lightly.

Lack of sensitive earthquake monitoring systems such as 10 broadband digital stations which are capable of recording even minutes tremors to a great degree of accuracy, and poor history of earthquake recording (compared to China's earliest recorded earthquakes dating back to 1500 BC, the earliest date of the Indian earthquakes, goes back by 350 years only) have been inhibitory factors in proper monitoring and realistic forecast of Indian earthquakes in the past.

Validity of Plate Tectonics

The occurrence of earthquakes in zones designated as 'stable' from seismic considerations questions the validity of the well-established theory of Plate Tectonics according to which the potential earthquake belts globally are the zones along the margin of lithosphere plates which constitute earth's crust. The most well-known of these are the Circum-Pacific belt, around the perimeter of the Pacific Ocean and Alpine belt running across Central Asia to the Mediterranean Sea. India, particularly its Himalayan mountainous region, forms a part of the Alpine belt which extends from Indonesia to Spain, through Myanmar, India, Middle East and Southern Europe. The Circum-Pacific belt has been the home of about two-thirds of world's largest earthquakes. In not-too-distant past (December 1989) there has been report of occurrence of earthquake in Eastern Australia, a zone which is well away from the edges of tectonic plates, as has also recently happened in seismically stable zone of Marathwada (Maharashtra) in September 1993. More recently, the devastating earthquake (of magnitude 7.2 on Richter scale) which rocked the Kansai (Kobe-Osaka) region of Japan claiming more than 5,400 human lives and destroying around 1,00,000 buildings (albeit many of them were pre-1980 construction, i.e. before stricter seismic regulations for design and construction of earthquake resistant houses were enforced in Japan) had struck in a region which was hitherto reckoned as one of the most quake-safe regions in Japan. The Kobe earthquake occurred not along the major inter-plate faults (cracks) where continent-size plates grind against one another but on intra-plate faults that spiderweb a single giant plate. These instances speak of uncertainties of occurrence of earthquakes. To be more precise, no place on the earth is free from the possibility of earthquake and necessitate review of the existing seismic map on global scale.

Damages to Houses

Of all the man-made structures, the collapse of buildings during an earthquake has been the largest contributor to the loss of lives and injuries to the people. Study of some catastrophic

earthquakes which have occurred over the past two decades indicates that it is the non-engineered buildings constructed spontaneously and informally in traditional manner without resorting to any or little intervention by qualified architects and engineers, which bear the maximum brunt of this natural disaster (Table 2). This fact is supported by the collapse of a large number of unreinforced masonry houses in the past earthquakes of Guatemala (1976), Armenia (December 1988), Kangra (April 1986), North Bihar (August 1988), Uttarkashi (October 1991) and Killari, Maharashtra (September 1993) and Algeria(August 1994). In the Guatemala earthquake, more than 25,000 people had perished due to collapse of about 2,50,000 adobe houses. The Armenian earthquake had well exposed the vulnerability of precast constructions too. Massive death toll (about 25,000) in the said earthquake was attributed to collapse of hundreds of precast concrete and unreinforced masonry buildings. On the contrary, the experience of Loma Prieta earthquake (October 1989) suggests that the properly-designed earthquake resistant structures very well stood the devastating effects of earthquake leading to much less casualty in terms of loss of human lives (only 300 compared to Armenia's 25,000) and damage to buildings. Ironically, the earthquake intensity on Richter Scale in the two earthquakes were nearly similar. Likewise, the casualty figure in case of the Killari earthquake measuring 6.4 m on Richter scale had been around 10,000 while earthquake of almost similar intensity (magnitude 6.6) in the Northridge, earthquake (January, 1994) in California claimed only 57 lives.

Table 2

Building types for housing in India (1981 census)

(after Bibl. 7-reproduced with permission from The Indian Concrete Journal, Bombay)

Category (vide MSK)	Type of wall material	Rural, percent	Urban, percent	Total, percent	Destruction Vulnerability In earthquake
A	Clay mud	37.1	4.0	41.1	High
	Unburnt bricks, adobe	6.2	1.4	7.6	High
	Stone(in mud)	8.1	2.0	10.1	High
B	Fired Bricks	13.4	13.0	26.4	Medium
C	Wood	0.7	0.4	1.1	Low
	Cement concrete	0.3	1.4	1.7	Low
Others	Metal sheet	0.1	0.4	0.5	Low
	Other (grass, reeds etc.)	10.0	1.7	11.7	Low

The safety of non-engineered buildings from the devastating effects of earthquakes warrant serious consideration in view of the fact that more than 90% of the population in zones prone to earthquake in the world live and work in such buildings and most losses of lives during earthquakes have occurred due to their collapse. It is unfortunate that risk to life in such belts is increasing instead of decreasing on account of rising population, abject poverty of the people, scarcity of modern building materials, viz., cement and steel, lack and awareness and skill. The increasing magnitude of seismic risk in earthquake prone areas,

especially in the developing countries, could be gauged from the fact that number of dwelling units and other related small-scale constructions might double over the next 20-25 years due to the population explosion in most of the developing countries including India. The large-scale loss of lives and property in the recent Maharashtra earthquake of relatively moderate intensity (measuring 6.4 on the Richter Scale) is also attributed to poverty, ignorance large population and non-engineered constructions in the quake hit areas of Latur and Osmanabad districts

Traditional Housing in Disaster Prone Areas

In Jammu and Kashmir, which lies in a zone of severe seismic intensity, the age-old 'dhaji diwari' technique is adopted to minimise the destructive effects of earthquakes. This method of construction uses a timber frame construction with diagonal bracing in the body of mud walls, which makes the structure more rigid and helps give better performance in the event of earthquakes. In Assam, also in a severe-intensity earthquake zone, the system of 'ikra' walling (Fig. 9.1) is adopted which comprises of timber-frame construction with panels of reed or split bamboo with mud plaster on either side. Mud houses with thatched or bamboo matting walls and thatched roof on account of their light weight and flexibility generally withstood the earthquake forces without any damage during the 1988 North Bihar earthquake.

The colossal destruction of houses in the moderate earthquakes of Killari (September 1993) and Uttarkashi (October 1991) was due to poor

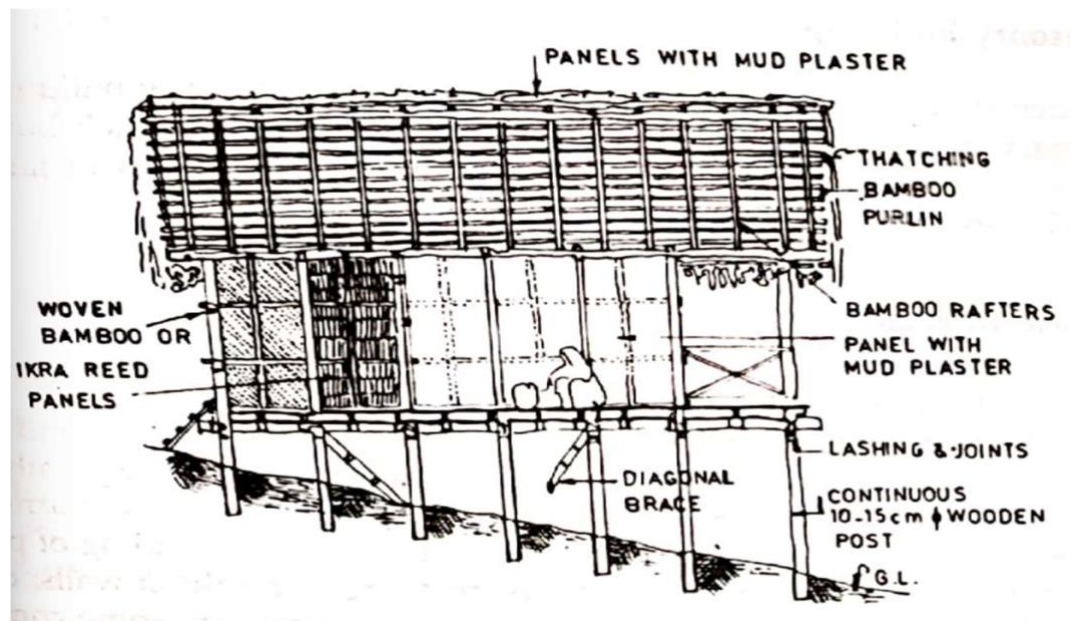


Fig. 9.1 Section of traditional Assamese house



Fig. 9.2 A view of completely damaged stone masonry house with slate tiles as roofing material

quality and weakness of the stone masonry construction (Fig. 9.2) using, in most cases, non-plastic soil mortar. The tragedy of the present times is that the traditional earthquake resistant constructions like 'dhaji-diwari' in Jammu and Kashmir or Himachal Pradesh and the 'ikra'-wall construction in most seismic region of the north-eastern states has gone into disuse in favour of the more damage-prone stone and brick constructions.

Types of Damages and Failures of Non-Engineered Buildings

Masonry Buildings

Induction of tensile and shear stresses in walls of masonry buildings is the primary cause of different types of damage suffered by such building in the event of earthquakes. The typical damages and modes of failure are briefly described below:

Typical Damages

Non-Structural Damage

The non-structural damage is the one due to which the strength and stability of the building is not affected. Such damage occurs very frequently even under moderate intensities of

earthquakes. Cracking and overturning of masonry parapets; large cantilever cornices and balconies; failing of plaster from walls and ceiling; cracking and overturning of partition walls; cracking and falling of ceilings; and cracking of glass panes are some common examples of non-structural damage.

Damage and Failure of Bearing Walls

1. Failure due to racking shear is characterised by diagonal cracks which could be due to diagonal compression or diagonal tension.
2. Crack patterns in load bearing walls due to bending and shear are illustrated in Fig. 9.3.

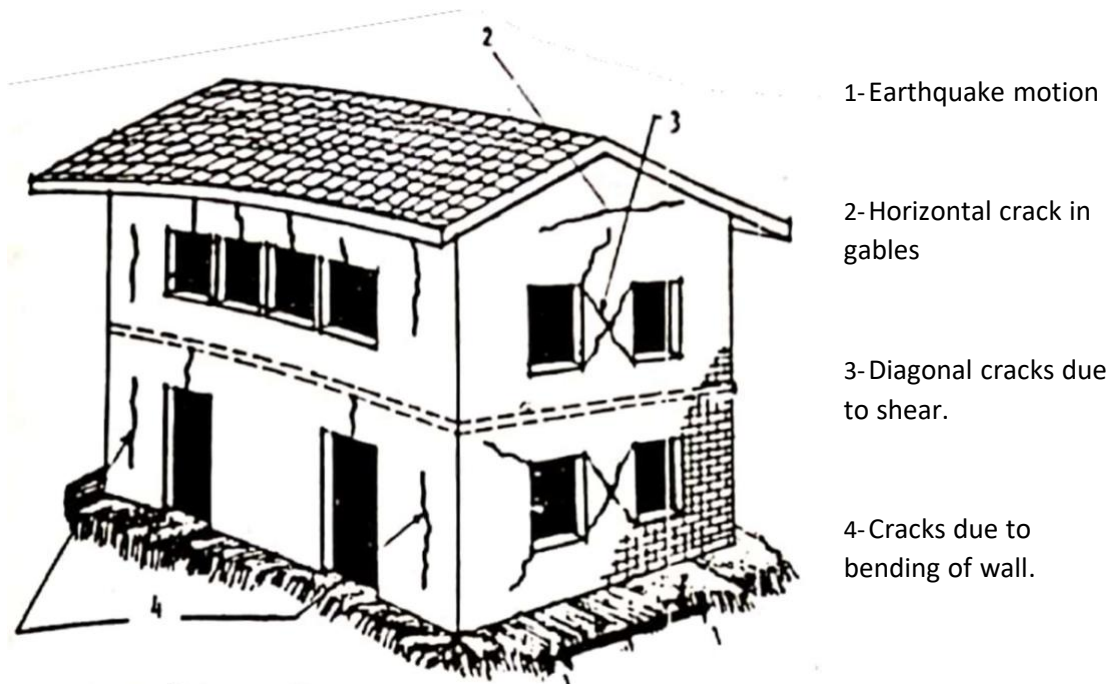


Fig. 9.3 Cracking in bearing wall building due to bending and shear (after Bibl. 152)

3. Unreinforced gable end masonry walls are very unstable and strutting action of purlin's imposes additional force to cause their failure. Horizontal bending tension cracks are caused in the gables.
4. Damage in unsymmetrical building occurs due to torsion and warping in an earthquake. This mode failure causes excessive cracking due to shear in all walls. Larger damage occurs near the corner of the building.
5. Arches across the openings in walls are often badly cracked since the arches tend to lose their end thrust under in-plane shaking of walls.

Modes of Failure

Failure of Ground

1. *Inadequate depth of foundation:* Shallow foundations deteriorate as a result of weathering and consequently become weak for earthquake resistance.
2. *Differential settlement of foundation:* During severe ground shaking, liquefaction of loose water-saturated sands and differential compaction of weak loose soils occurs which leads to excessive cracking and tilting of building which may even collapse completely.
3. *Sliding of slopes:* Earthquakes cause sliding failures in man-made as well as natural hill slopes and any buildings resting on such slopes have a danger of complete disastrous disintegration.

Failure of Roofs and Floors

Improperly tied roofing material is dislodged due to inertia forces acting on the roof. This mode of failure is typical of sloping roofs, particularly when slates, clay tiles, etc., are used as roofing material. Brittle material when asbestos cement may be broken if the trusses and sheeting purlins are not properly braced together.

Causes of Damages in Masonry Buildings

The following are the main weaknesses in the materials and unreinforced the masonry constructions and other reasons for the extensive damage of such buildings (Fig. 9.4).

- Heavy weight and very stiff buildings, attracting large seismic inertia forces
- Very low tensile strength, particularly with poor mortars
- Low shear strength, particularly with poor mortars
- Brittle behaviour in tension as well as compression
- Weak connection between wall and wall
- Stress concentration at corners of windows and doors
- Overall unsymmetry in plan and elevation of building
- Unsymmetry due to imbalance in the sizes and positions of openings of openings in the walls
- Defects in construction such as use of substandard materials, unfilled joints between bricks, not-plumb walls, improper bonding between walls at right angles etc.

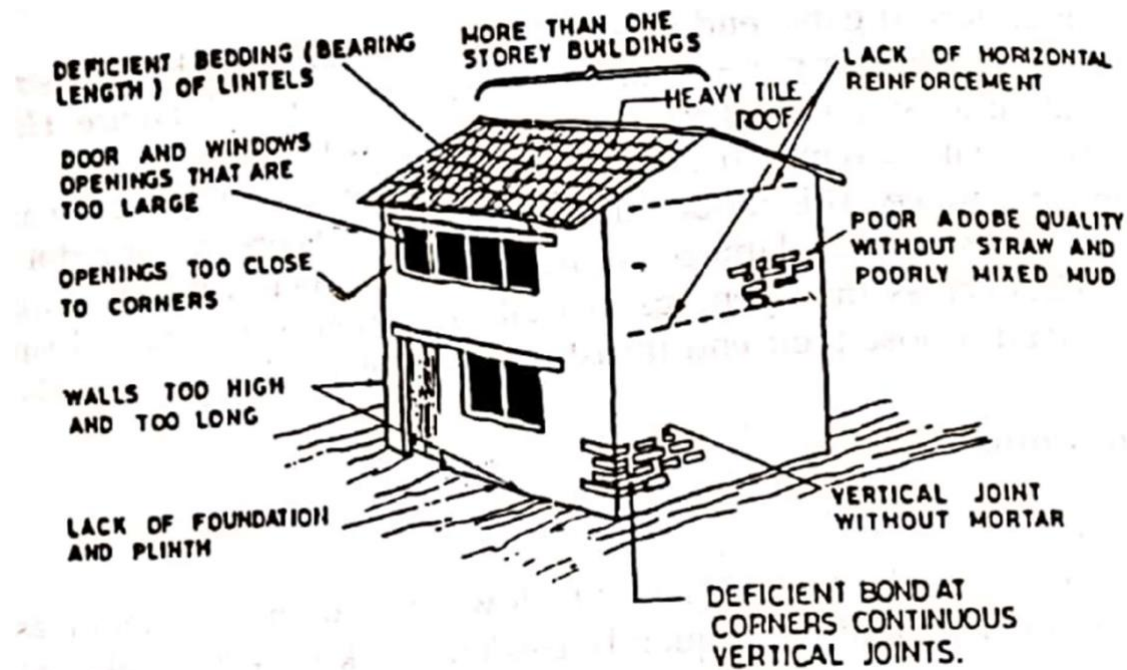


Fig. 9.4 Graphic summary of causes of failure (after Bibl. 152)

Repair and Restoration of Earthquake Damaged Non-Engineered Buildings

Concepts of Repair and Restoration

The methods of repair and restoration depend largely on factors like structural scheme, materials used in construction of building, feasible technology for repair and restoration and the availability of funds. Some methods like Splints and bandage' wire mesh with gunite, 'epoxy injection' etc., have been tried out for repair and strengthening of earthquake damaged building in some countries.

The underlying concepts in repair and restoration of earthquake-damaged buildings are described as below.

Repairs

The main purpose of carrying out repairs of damaged building is to bring back the architectural shape of the building dealing with non-structural components so that all services could start working and the functional utility of the building is restored quickly. Repair does not improve the structural strength of the building, and can be very deceptive for meeting the strength requirements of the future earthquake. The actions involved in repair include the following:

- a) Patching up of defects such as cracks and fall of plaster
- b) Repairing doors, windows, replacement of glass panes
- c) Checking and repairing electric wiring
- d) Checking and repairing gas pipes, water pipes and plumbing services
- e) Re-building non-structural walls, smoke chimneys, boundary walls, etc.

- f) Replastering of walls as required
- g) Rearranging disturbed roofing tiles
- h) Relaying cracked flooring at ground level
- i) Redecoration-white washing, painting, etc.

Restoration

This activity envisages restoration of the original structural strength of the building by appropriate treatment of the damaged structural components of the building such as load bearing walls, beams, columns, etc. It may involve cutting portions the elements and rebuilding them or simply adding more structural material so that the original strength is more or less restored. The process may involve inserting temporary supports, underpinning, etc. In most instances, the relative cost of retrofitting to reconstruction cost determines the decision regarding undertaking any restoration work. As a thumb rule, if the cost of repair, restoration and strengthening is less than about 50% of the reconstruction cost, retrofitting is the preferred choice.

Some of the techniques of restoration are stated as below:

- a) Removal of portions of cracked masonry walls and piers and rebuilding them in richer mortar. Use of non-shrinking mortar will be preferred.
- b) Addition of reinforcing mesh on both faces of the cracked wall, holding it to the wall through spikes or bolts and then covering it suitably. Several alternatives have been used.
- c) Injecting epoxy like material, which is strong in tension, into the cracks in walls columns, beams etc.

Repair Materials

The most common materials for damage repair work of various types are cement and steel. In many situations non-shrinking cement or an admixture like aluminium powder in the ordinary Portland cement will be admissible. Steel may be required in many forms, like bolts, rods, angles, channels, expanded metal and welded wire fabric. Wood and bamboo are the most common materials for providing temporary supports and scaffolding etc., and will be required in the form of rounds, sleepers, planks, etc. Besides, special materials and techniques are available for best results in the repair and strengthening operations. They are described below:

Epoxy Resins

Epoxy Resins are excellent binding agents with high tensile strength. There are chemical preparations the compositions of which can be changed as per requirements. The epoxy components are mixed just prior to application. The product is of low viscosity and can be injected in small cracks too.

The higher viscosity epoxy resin can be used for surface coating or filling larger cracks or holes. The epoxy mixture strength is dependent up on the temperature of curing (lower strength for higher temp.) and method of application.

Epoxy Mortar

For larger void spaces, it is possible to combine the epoxy resins of either low viscosity or higher viscosity, with sand aggregate to form epoxy mortar. Epoxy mortar mixture has higher compressive strength, higher tensile strength and a lower modulus of elasticity than Portland cement concrete.

Thus, the mortar is not a stiff material for replacing reinforce concrete. It is also reported that epoxy is a combustible material.

Therefore, it is not used alone. The sand aggregate mixed to form the epoxy mortar provides a heat sink for heat generated and provides increased modulus of elasticity as well.

Gypsum Cement Mortar

It has got rather limited use for structural application. It has the lowest strength at failure among these three materials.

Quick-setting Cement Mortar

This material is patented and was originally developed to be used as a repair material for reinforced concrete floors adjacent to steel blast furnaces. It is a non-hydrated magnesium phosphate cement with two components, a liquid and a dry, which can be mixed in a manner similar to Portland cement concrete.

Guniting

Guniting, also called shotcrete, is pneumatically applied on concrete. Two different methods of operation are employed. In the first method the wet concrete mix, usually Ready Mix, is pumped through a hose to a nozzle at the point of application. Compressed air is injected near the nozzle forming a spray that is directed against the surface. Concrete applied by this method is termed 'Shotcrete' by the equipment manufacturers and is called wet mix. In the second method, the concrete may also be Ready Mix, except that it is delivered dry on the job. It is then placed in a guniting machine from which it is delivered through a hose by compressed air to a nozzle at forming a spray. Concrete placed by this method is termed 'guniting' by the equipment manufacturers and is called dry mix.

Guniting is widely used for strengthening or repairing concrete and masonry structures. Experience has shown that guniting bonds well with brick walls in the repair of masonry buildings. In fact, guniting bonds to almost any surface-hard, soft, wet or dry, and even to polished steel. Guniting is shotcrete because of its better bonding capability on account of higher velocity of concrete emerging out of the nozzle

Restoration Techniques

While considering restoration work, it is important to realise that even fine cracks in load bearing members, which are unreinforced, like masonry and plain concrete reduce their resistance very largely. Therefore, all cracks must be located and marked carefully and the

critical ones fully repaired either by injecting strong cement or chemical grout or by providing external bandage. The techniques are described below along with other restoration measures.

Small Cracks

if the cracks are reasonably small (opening width - 0.075 cm), the technique to restore the original tensile strength of the cracked element is by pressure injection of epoxy. The procedure is as follows (Fig. 9.5 and 9.6).

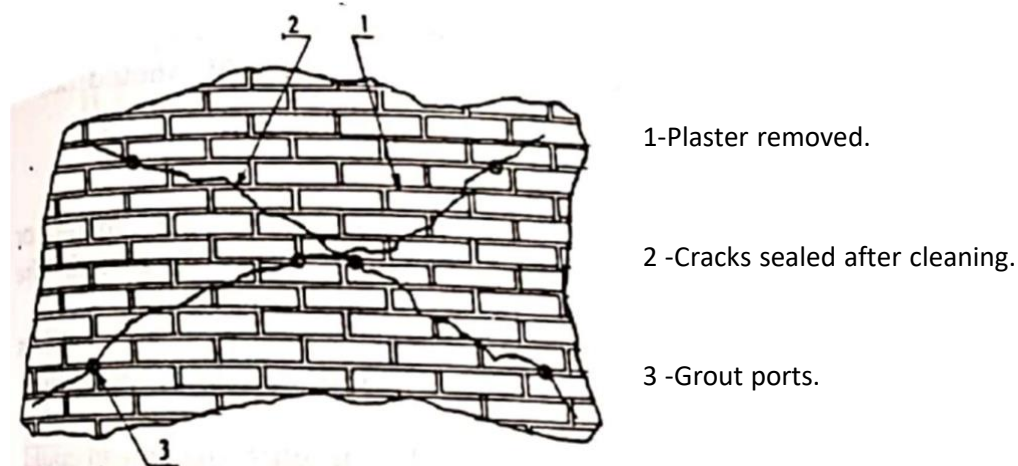


Fig. 9.5 Grout or epoxy injection in cracks (after Bibl. 152)

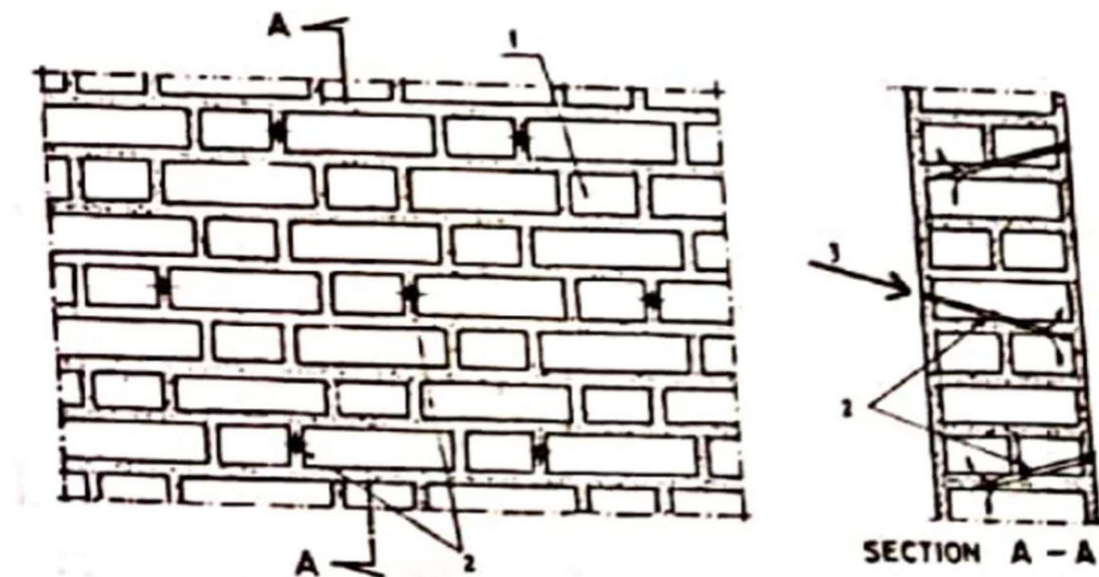


Fig. 9.6 Grout or epoxy injection in existing weak walls (after Bibl. 152)

The external surfaces are cleaned of non-structural materials, and plastic injection ports are placed along the surface of the cracks on both sides of the member and are secured in place with an epoxy sealant. The center-to-center spacing of these ports may be approximately equal to the thickness of the element. After the sealant has cured, a low

viscosity epoxy resin is injected into one port at a time beginning at the lowest part of the crack in case it is vertical or at one end of the crack in case it is horizontal.

The resin is injected till it is seen flowing from the opposite sides of the member at the corresponding port or from the next higher port on the same side of member. The injection port should be closed at this stage and injection equipment moved to the next port and so on.

The smaller the crack, higher is the pressure or more closely spaced should be the crack, higher is the pressure or more closely spaced should be the ports so as to obtain complete penetration of the epoxy material throughout the depth and width of member. Larger cracks will permit larger port spacing, depending upon width of the member. This technique is appropriate for all types of structural elements-beams, columns, walls and floor units in masonry as well as concrete structures. Two items should however, be taken care of in such type of repair.

- a) In the case of loss of bond between reinforcing bar and concrete, if the concrete adjacent to the bar has been pulverised to a very fine powder, this powder will dam the epoxy from saturating the region. So, it should be cleaned properly by air or water pressure prior to injection of epoxy.
- b) It is earlier stated that cracks smaller than about 0.75 mm may be difficult to pressure inject. So, cracks smaller than this should not be repaired by this method.

Large Cracks and Crushed Concrete

For cracks wider than about 6 mm or for regions in which the concrete or masonry has crushed, a treatment other than injection is suggested. The following procedure may be adopted.

- a) The loose material is removed and replaced with any of the materials mentioned earlier, i.e., expansive cement mortar, quick setting cement or typosum cement mortar.
- b) In the case of damage to walls and floor diaphragms, steel mesh could be provided on the outside of the surface and nailed or bolted to the wall. Then it may be covered with plaster or micro-concrete.

Fractured, Excessively Yielded and Buckled Reinforcement

In the case of severely damaged reinforced concrete member, it is possible that the reinforcement would have buckled, or elongated or excessive yielding may have occurred. This element can be repaired by replacing the old portion of steel with new steel using butt welding or lap welding.

Splicing by overlapping will be risky if repair has to be made without removal of the existing steel, the best approach would depend upon the space available in the original member. Additional stirrup ties are to be added in the damaged portion before concreting so as to confine the concrete and enclose the longitudinal bars to prevent their buckling in future.

In some cases, it may be necessary to anchor additional steel into existing concrete. A common technique for providing the anchorage uses the following procedure:

A hole large then the bar is drilled. The hole is filled with epoxy, expanding cement or other high strength grouting material. The bar is pushed into place and held there until the grout has set.

Strengthening RC Members

The strengthening of the RC members is a task that should be carried out in consultation with a structural engineer. Here a few suggestions are briefly described to illustrate the ways in which the strengthening could be undertaken.

- 1) The RC columns can best be strengthened by jacketing. This is done by providing additional cage of longitudinal and lateral tie reinforcement around the columns and casting a concrete ring (Fig. 9. 7 and 8). The RC beams and shear walls could be also strengthened in similar way.

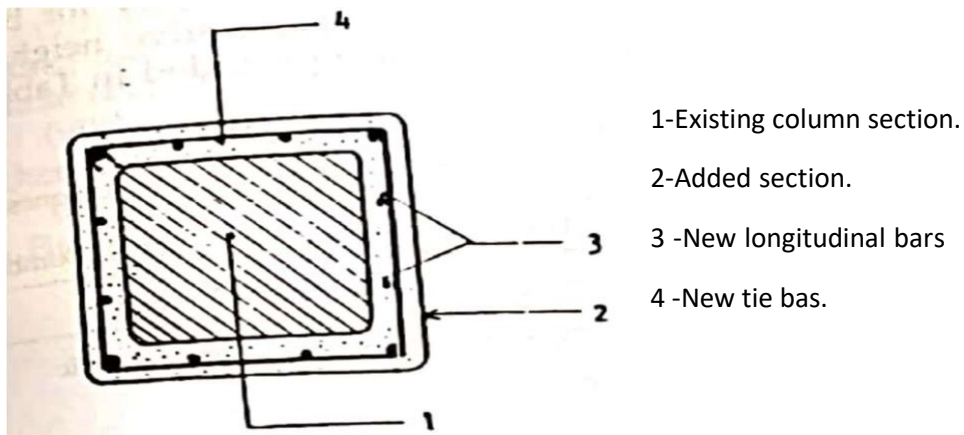
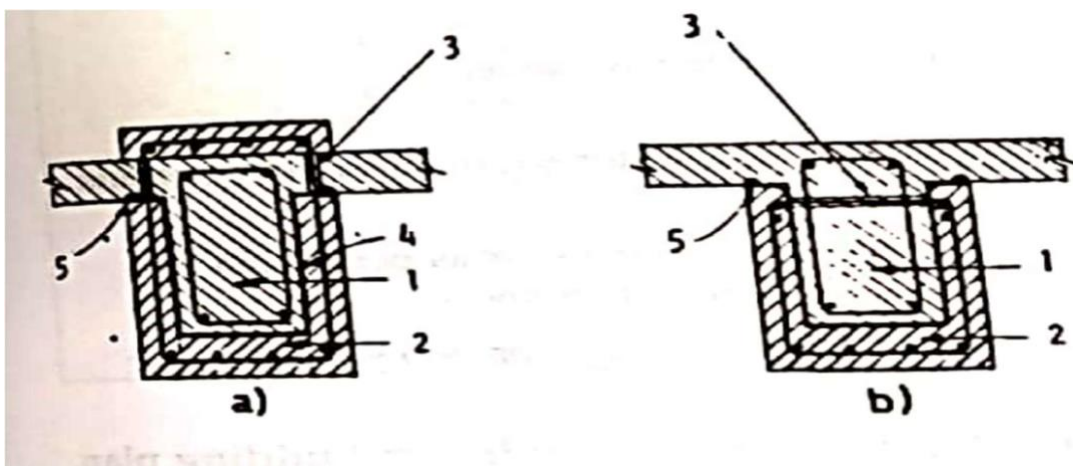


Fig. 9.7 Jacketing a concrete column (after Bibl. 152)



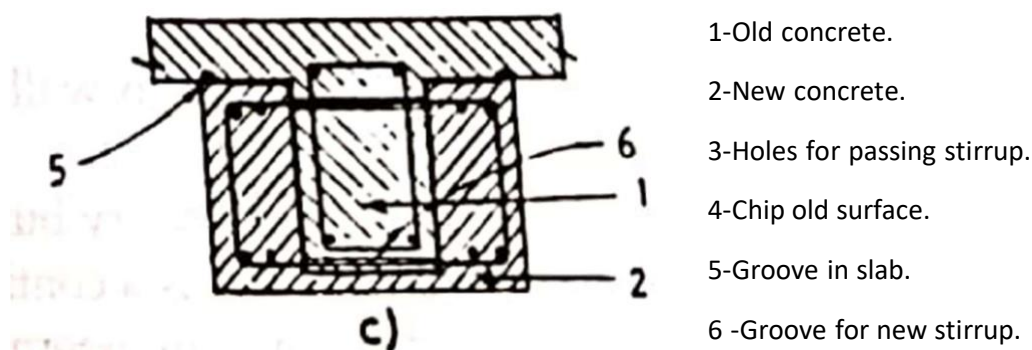


Fig. 9.8 Increasing the section and reinforcement of existing beams (after Bibl. 152)

- 2) Inadequate sections of the RC columns and beams can also be strengthened by removing the cover to the old steel, welding new steel to old steel and replacing the cover. In all cases of adding new concrete to old concrete, the original surface should be roughened, grooves made in the appropriate direction for proving shear transfer. The ends of the additional steel are to be anchored in adjacent beams or columns as the case may be.

Recommendations for Future Construction

Low Strength Masonry (LSM)

a) Burnt Brick Buildings

For earthquake resistance the following measures have been found effective not only in preventing collapses but also in controlling the propagation and widening of cracks in the burnt brick buildings.

- I. Restriction of height of load bearing wall is necessary for better seismic performance. The guidelines in respect of suggestive height for various types of low strength masonry buildings are provided in Table 3.

Table 3

Suggested height restrictions on buildings in moderate and severe seismic zones

(Zones III, IV and V of India)

(after Bibl. 7-reproduced with permission from The Indian Concrete Journal, Bombay)

Sr. No.	Building type	Suggested maximum height
1.	Adobe house	One storey, or one storey + attic

2.	Field stone (random rubble) masonry in caly mud mortar	Same as above
3.	Dressed stone masonry in cement mortar	Two storeys, or two storeys + attic
4.	Brick masonry in mud with critical sections in cement mortar	Same as above
5.	Brick or cement block masonry in good cement mortar	Three storeys, or three storeys + attic
6.	Reinforced masonry	Five storeys, or as per design by a qualified engineer
7.	Wood frame	Two storeys, or two storeys + attic

- II. There should be symmetry and rectangularity in the building plan.
- III. There should be symmetry in the location of opening in masonry walls.
- IV. Width of piers between openings or from opening to wall corner should not be less than half the height of opening.
- V. The most important concept in strengthening of masonry buildings in the provision of horizontal seismic bands. A seismic band is a continuous runner (beam) of reinforced concrete or wood running into all external and internal walls with proper connections at the corners and T-junctions of the wall. The seismic bands are generally provided at the plinth, lintel and roof levels of the building.

Plinth band should be provided in these situations wherein the soil is soft or uneven in its properties as it usually happens in the hilly tract.

Lintel band is the most important band and incorporates in itself all door and window lintels, the reinforcement of which should be extra to the lintel band steel. From earthquake stability considerations, it is necessary that such band is provided in all the storeys in the building.

Roof band should be provided at the eave level of trussed roofs or where flexible wood joist roof or precast roofing units are used. Such band may not be provided where concrete slabs having full bearing on all four walls in used for roof or floor.

- VI. Vertical reinforcement should be provided in the walls in the shape of vertical steel bars or bamboo/cane at the critical sections of the walls. The critical sections for vertical reinforcements are the corner walls and jambs of window and door openings.
- VII. Steel or wooden dowels should be provided at the corners or T-junctions to provide effective bonding between them, say every fourth course (Fig. 9.9 to 9.12).
- VIII. Use of stronger mortar in masonry, say cement-sand 1:6 or even richer mortar should be made in construction of walls.

IX. overall arrangement of reinforcing a masonry building is shown in Fig. 9.13.

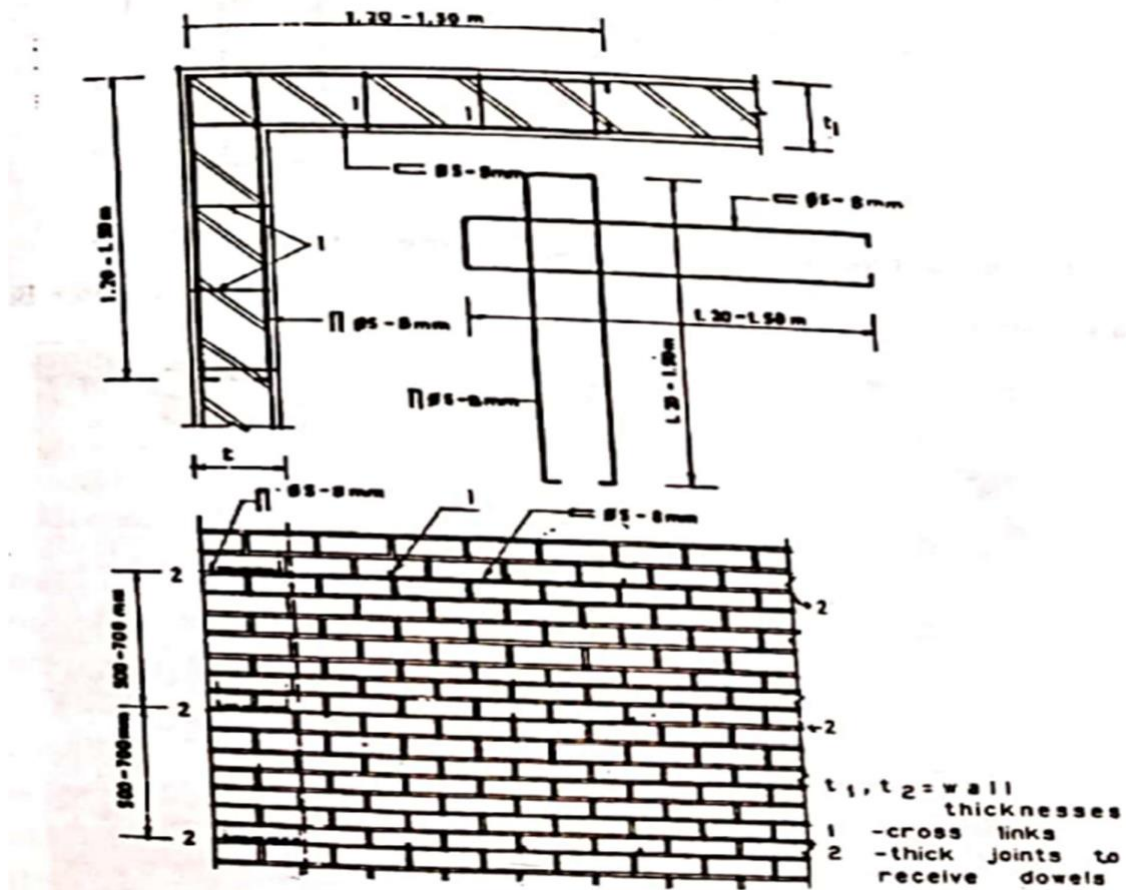


Fig. 9.9 Corner-strengthening by dowel reinforcement placed in one joint (after Bibl. 152)

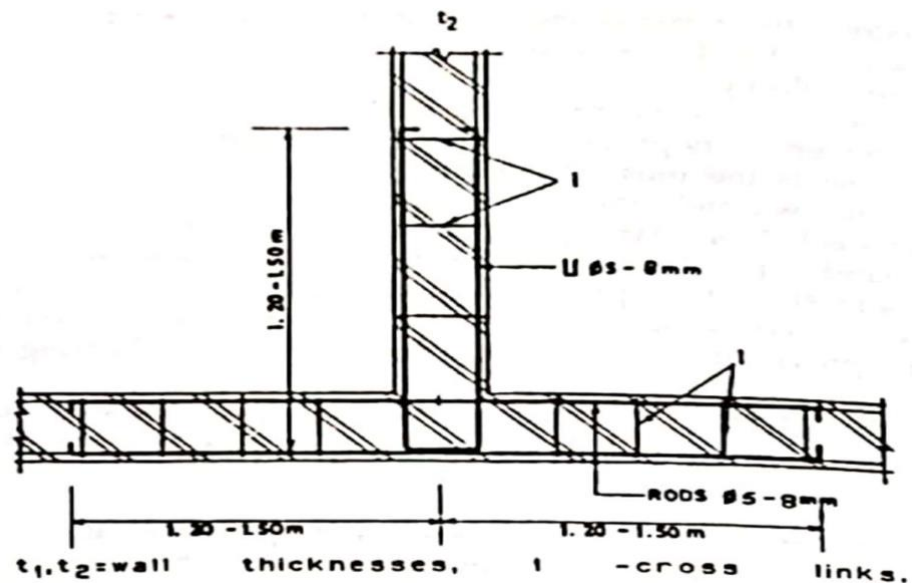


Fig. 9.10 T-junction strengthening by dowel reinforcement (after Bibl. 152)

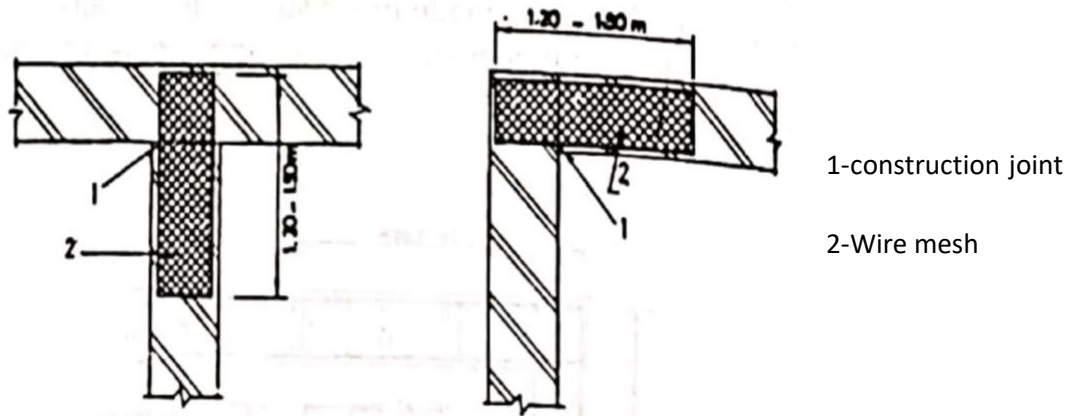


Fig. 9.11 Strengthening by wire fabric at junction and corner (after Bibl. 152)



Fig. 9.12 Reinforcement for lintel band and placement of corner steel

a) Stone Masonry Buildings

For stone masonry buildings, all the protective measures as for brick buildings, are found useful. For random rubble and half-dressed masonry, the following additional measures are absolutely necessary.

- I. Provision of 'through stones or bonding' elements (Fig. 9.14 & 9.15) along the wall thickness in every 60 cm lift at not more than 1.2 m apart horizontally.
- II. Restriction of the thickness to not more than 45 cm since larger thickness encourages undesirable filling material, inside, adding to mass but reducing the strength.

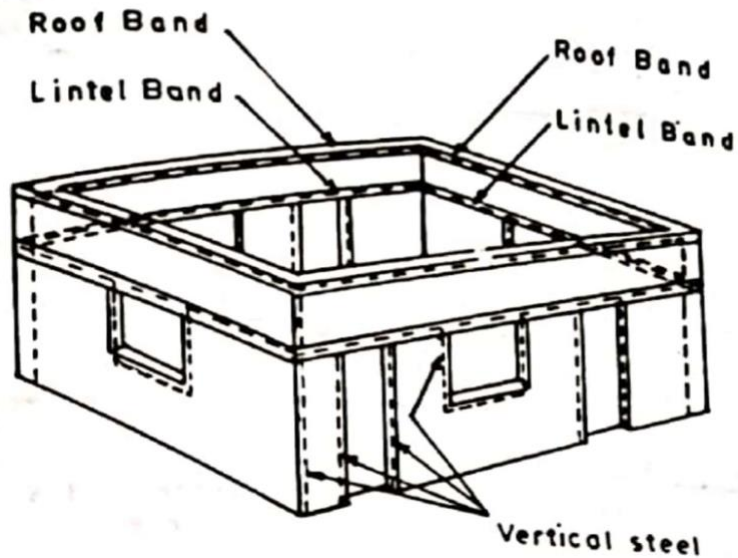


Fig. 9.13(a) Overall arrangement of reinforcing masonry building

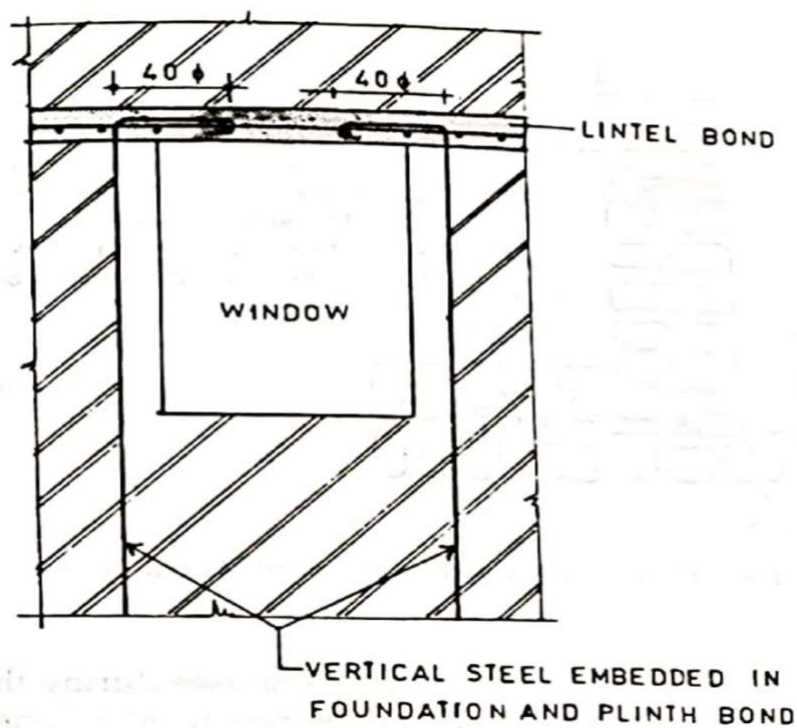


Fig. 9.13(b) Strengthening of masonry around openings

b) Earthen Houses

- I. In view of poor performance of earthen houses during the past earthquakes, the height of such houses should be restricted to one storey only or one storey plus attic (Table 3).
- II. Length of rooms should be restricted to 4m maximum and storey height to about 2.8 m.

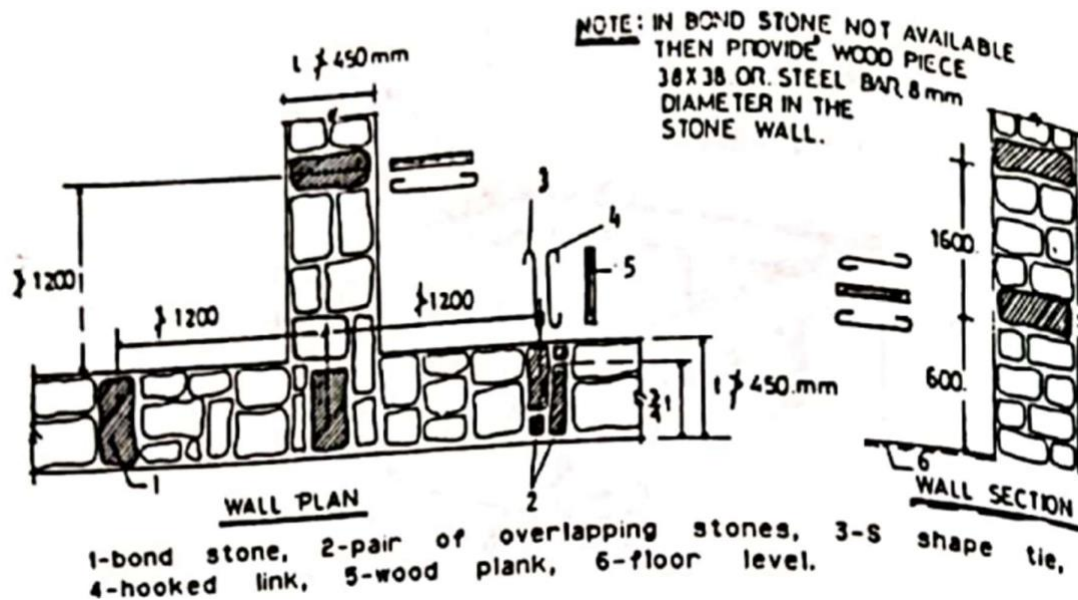


Fig. 9.14 "Through" stones or "Band" elements (after Bibl. 152)

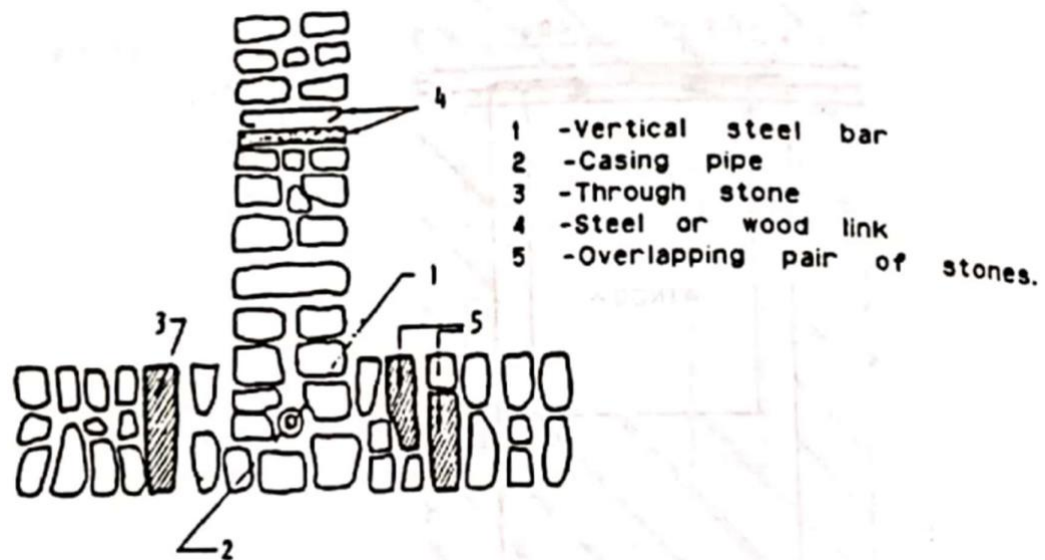


Fig. 9.15 Vertical steel in random rubble masonry

- III. As a thumb rule, openings' width should be restricted to one-third of the wall length, and these should be away from the corners by about three times the wall thickness.
- IV. Soil should be adequately stabilised using suitable stabilisers like cement, lime, asphalt, etc. to improve the structural performance of earthen houses in the event of earthquakes. For adobe units and mortar, clayey soil with about 20% clay content and adequate fibrous admixtures should be used.

- V. Earthen (adobe) houses should be strengthened by using lintel bands and roof bands made of wood. The roof rafters should be held to the roof band through spikes.
- VI. Mud walls are made stronger for better performance during an earthquake by using buttresses or pilasters at the corners and wall junctions.
- VII. Roof should be as light as possible in order to reduce inertial forces generated by its mass. It is also important that the roof acts as a stiff diaphragm capable of transmitting lateral forces to the stronger walls in each direction. Instead of burnt clay tiles and slates which are brittle and dislodged, corrugated iron, asbestos or asphaltic sheets should be preferred for roofing.
- VIII. Earth should be protected from deterioration by weathering effect, except in zones of very arid climate or when earth has been stabilised with stabilising agents like cement, lime or asphalt. The exterior walls should be adequately covered with waterproof mud plaster. Mixing bitumen cut back in mud mortar is a preferred alternative. Cut back is prepared by mixing bitumen 80/100 grade, kerosene oil and paraffin wax in ratio 100:20:1. For 1.8 kg cut back, 1.5 kg bitumen is melted with 15 gm of 100:20:1. For 1.8 kg cut back, 1.5 kg bitumen is melted with 15 gm of wax and this mixture is poured into a container having 0.3 litre kerosene oil followed by constant stirring till all the ingredients are properly mixed. This mixture is now ready for mixing with 0.03 cu.m. of mud plaster prepared by mixing chopped rice /paddy husk, soil, cow dung and water. A combination thus prepared is both water repellent for walls and fire retardant for thatch.

Requirements of Structural Safety of Thin Precast Roofing Units against Earthquake Forces

The Armenian earthquake had exposed the vulnerability of precast roofing components to the forces of earthquake. Precast concrete frame buildings suffered extensive damage during the said earthquake. On the contrary, precast large panel buildings gave better performance during the earthquake due to proper bracings. Different parts of such buildings were better tied together than the previously quoted type of buildings. As such, it becomes necessary to take adequate precautions in design and construction of roof using precast elements. The following measures may be adopted for the structural safety of precast roofing units against earthquake forces.

- a) Precast roofing/flooring, viz., RC channel units, RC planks and joists, etc., may be cambered to counteract the elastic as well as creep deflection.
- b) The units should be transversely connected with reinforcing bars to avoid opening of joists (Fig. 9.16).
- c) Use of such units may be avoided in wet areas, for e.g., baths, WCs and kitchens unless treated with reinforced topping concrete. Use of topping concrete could also eliminate the problem of transmission of sound inherent in thinner sections of precast units.

- d) All elements of an earthquake resistant structure must be tied together to ensure box-type action i.e. walls act as shear wall and roof as diaphragm.

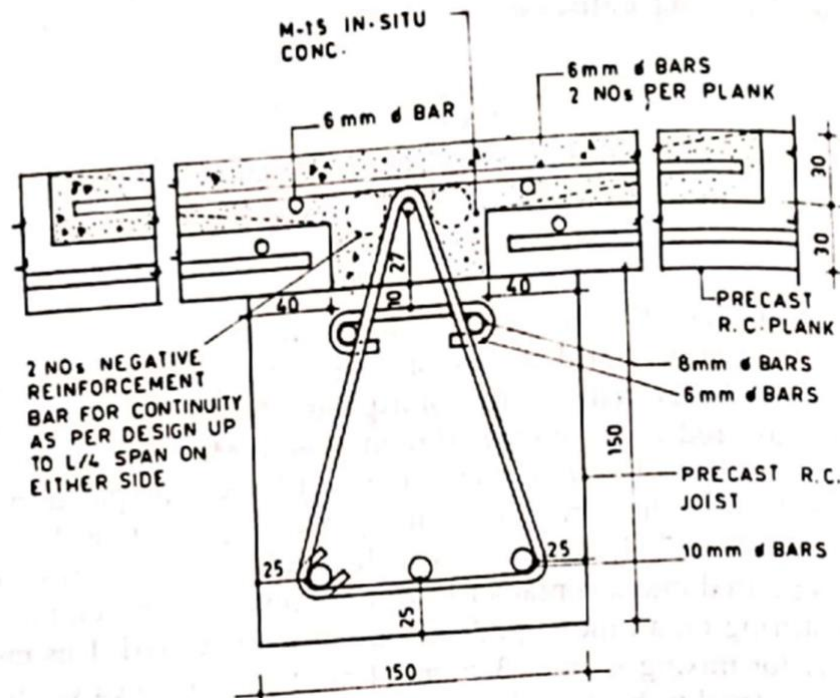


Fig. 9.16 Joint at beam support near wall

- e) Field inspection and quality control during construction are important for stronger construction which could sustain the damaging effects of earthquake.
- f) In case of precast flooring/roofing units, due attention should be given to the proper detailing of joints and for ensuring diaphragm action of such floor/roof. The revised edition of IS: 4326 incorporates all the precautionary measures required to be adopted in respect of design and construction of precast building in earthquake prone areas.

Status of R and D in Earthquake Strengthening Measures

Provision of proper earthquake strengthening measures in buildings to be constructed in earthquake prone areas has assumed greater significance from the point of view of providing structural stability to buildings and security to the lines of inmates. Considerable research has been undertaken in this direction. The Department of Earthquake Engineering, University of Roorkee, has done pioneering research in evolving measures for suitable strengthening of buildings against seismic forces of variable magnitude. Based on the research and development work done in India and abroad, the Bureau of Indian Standards (BIS) has formulated number of Codes of Practice for construction of earthquake resistant buildings in different seismic zones of the country. It will be noteworthy to mention here that research in earthquake engineering has been undertaken in India since 1960, and it was the first country in the world to formulate the Code on Earthquake Resistant Building Construction in 1967 which has since been adopted by several countries.

In the wake of large-scale damage of buildings in the earthquakes of recent past namely Kangra (Himachal, 1986), North Bihar (1988) and Uttarkashi (U.F., 1991), the BIS has recently revised IS:4326. The IS:4326-1993 (Second Revision) covers wide-ranging issues concerning earthquake strengthening measures. It deals with the selection of material for earthquake resistant construction, special features of design and construction of earthquake resistant building including masonry construction, timber construction and buildings with prefabricated flooring/roofing elements. Other useful IS codes on earthquake resistant construction brought out by BIS are IS: 13828:1993, 'Improving earthquake resistance of earthen buildings'; IS:13828:1993, 'Improving earthquake resistance of low strength masonry buildings-Guidelines,' IS: 13920:1993, 'Code of Practice for ductile detailing of reinforced concrete structures subject to seismic forces', and IS: 13935:1993, 'Repair and seismic strengthening of buildings - Guidelines.'

In recent years, 'Base Isolation' technique is being increasingly used in Japan, USA, New Zealand and Australia to safeguard structures from catastrophic effects of earthquake forces. Base Isolation is a concept whereby the motion of a structure during an earthquake is decoupled from the ground motion by the introduction of a low stiffness layer between the superstructure and the ground. Of late, combination of layers of natural rubber and mild steelplates has been successfully tried out as a viable base isolator. Today, more than 50 patented base isolators have been developed the world over. The additional cost likely to be incurred over provision of base isolator in a building is only to the extent of 5% of the cost of the structure. Provision of base isolators and shear walls in important structures such as school buildings, hospitals, and high-rise residential apartments and commercial buildings will help provide adequate safety and stability to such structure in the event of earthquakes. Shear walls help in resisting lateral force induced on structures due to earthquakes a great deal. They are usually provided between column lines, in stair wells, lift wells and in shafts that house other utilities. Buildings mounted on base isolators and incorporating provision of shear walls have been observed to sustain strong earthquakes force admirably.

Floods

Among several natural disasters, floods are more recurrent in the South Asian region. The recurring floods in the developing countries like India and Bangladesh cause extensive damage to crops, livestock's, communication, irrigation works, houses and property, not to mention loss of a large number of human lives every year.

Data available from the United Nations Disaster Relief Coordinator (UNDRC) indicates that the major impact of natural disaster is concentrated among the developing countries, mainly located in Asia, where agriculture is the principal economic activity. It is estimated that over 90% of disaster losses in these countries are attributable to floods, indicating the importance of flood hazard assessment and mitigation planning in the background of their impact on agricultural economy.

Magnitude of Losses in India

Among several natural disasters that India experiences every year, flood ranks the uppermost in terms of losses it causes to housing/building and other property.

Flood is a perennial problem in India. It is an annual feature in majority of the states in the country. A large number of States like Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal, Orissa, Assam, Tripura, Kerala, Karnataka, Andhra Pradesh, Gujarat and Maharashtra are flood prone. Off and on floods also occur in other States due to swollen rivers, resulting from cloud burst and unusual rains. Cloud burst caused untold misery in terms of loss of life and property in the otherwise dry and arid Rajasthan 1986, and in Himachal Pradesh in 1994.

As per the estimates of the National Commission of Floods, 40 million hectares (MHA) out of 329 MHA of the country's geographical area are prone to flood. Relatively, problem of floods causing heavy devastation is more frequent in the Brahmaputra basin in Assam; the Ganga basin in Uttar Pradesh, Bihar and West Bengal, and the Brahmani, Baitarani, Subarnarekha and Mahanadi basins in Orissa.

Extensive efforts have been made to mitigate flood damages through several schemes under the five-year plans. Nevertheless, floods still continue to cause extensive damage to life and property every year in the country. In 1987, floods were responsible for the loss of 1,612 human lives and property damage worth Rs 15,870 million. An area of 16.90 million hectares was affected by the 1987 floods throughout India. Floods of 1988 had a much higher tally. As many as 3065 human lives and 0.14 million cattle heads perished in the 1989 floods that caused havoc in a large portion of the country. The people belonging to the EWS category are the worst sufferers in the event of floods as their dwellings, which are generally of non-durable materials and construction, are easily damaged or washed away by the water currents.

The national policy on flood control was launched in 1954, but floods still, on an average, claim over 1,500 human lives and 10,000 cattle heads every year, besides damages to thousands of houses of the common people.

Types of Floods

Flood is a situation created when the river enters a flow in excess of its transporting capacity. It may also be caused by an overflow of storage tank or breach of a reservoir. Depending on its nature a flood can be categorised as follows:

a) Local Flood

It is caused by intense local rainfall. The inadequacy of the drainage system to carry away the water quickly results in flooding of the locality and low-lying areas.

b) Tidal Flood

Cyclonic winds cause high tidal waves which spill over land flooding low-lying areas in the interior.

c) Flash Floods

Flood flow develops gradually in many large rivers. In contrast, flash flood, as the name suggests is a sudden phenomenon. In this type the time lag between start of flood and peak discharge is much smaller in comparison. It is caused by isolated and local intense rainfall as from a thunderstorm (cloudburst) and by steep catchment slope conducive to fast run off. They are particularly dangerous because of the suddenness and speed with which they occur.

Damage to Houses by Floods

Floods constitute a hazard to houses and buildings when the flood plains which perform the natural function of carrying away excess water are used for human settlements. In spite of some awareness of the hazard, more and more people are moving to and building in flood prone areas due to pressure of population and economic compulsions.

Depending on their location the houses can suffer damage by different kinds of floods.

The type and extent of damage to houses depend on the particular type of flood they are exposed to. Flash floods, riverine floods, local floods type coastal waves have their own dynamic characteristics and leave distinct marks of distress on the structure after they recede. Damage of different degrees takes place is described below.

Severe Damage

Houses subjected to a large mass of water striking them with force collapse totally. This is typical of coastal waves following cyclonic winds (Tsunamis which are also in this category do not occur India) which may reach several meters above the sea-shore level. In the case of coastal flood, it might be possible to retrieve some of the building materials from the collapsed structures.

Flash floods also bring upon sudden pressure on the structure due to the high velocity of water and cause complete destruction. Even strong buildings of RCC construction are badly damaged. In locations where the houses are built on loose soil, severe inundation takes place leading to collapse of the structure. The swift water currents take with them most of the debris. In the case of 'kutchra' structures built of thatch, mud etc., not a trace of the house is left. Very little of building materials, if any, can be retrieved in the case of structures lying directly in the path of flash floods.

Partial Damage

Riverine floods are characterised by low velocity and longer duration. Prolonged inundation is expected from riverine floods which gives rise to two types of damages to houses.

- a) The soil gets softened leading to sinking of the foundation. This will result in uneven sagging of roof with consequent damages to roof covering and supporting members,

cracks in the wall and sinking of floors. Scouring of foundation by a slow and steady flow of flood water can further aggravate the problem. Repairs and restoration can be done with considerable expense depending on the extent of damage.

- b) The standing water causes deterioration of the building finishes like painting, distempering, plastering, etc. Timber works in doors, windows, storage shelves, pelmets are soaked leading to their decay. Flooding of low-lying areas by local heavy rainfall also damages houses in the manner described above.

Other Effects

Floods can also have a devastating effect on land by eroding the upper strata leaving behind a rugged surface which is unsuitable for construction. The reverse process is also a common phenomenon when the flood water brings with it large volumes of fine silt, and or other materials and deposits them on an existing settlement. This is very typical of flash floods which have the requisite momentum to carry soil and boulders in their way. In this case the clearing operations and restoration become difficult tasks.

Vulnerability of Mud Houses

By and large damages to houses and buildings depend on the topography of the area soil erosion characteristics, wall area exposed to the water pressure, foundation type and depth and the strength of the structure. However, houses of the 'kutcha' type, especially those constructed with mud, sundried bricks, thatch, bamboos, etc., are damaged extensively. Poor people who are able to afford only such type of houses are the worst sufferers.

Disaster Mitigation in Flood Prone Areas

Habitation in river valleys, around lakes and along coastal belts, will continue due to compulsions of occupation, in spite of their being prone to flood hazard. A total evacuation of the population and their resettlement at a safer site appears to be in practicable proposition. It leads to unfavourable socio-economic consequences. The flood problem, therefore, ought to be tackled with two-fold objectives of prevention of exposure of habitation to floods and mitigating the effects of floods.

Land Use and Building Regulations

Three aspects namely a) urban and rural land use b) flood plain occupation control, and c) building regulation have to be taken into account for a planned development to mitigate the effect of flood disasters.

Land Use

Changes in the land use can have an effect on the hydrology of the area thereby increasing the flood potential of catchments, hydrology areas cause attributed are i) reduction in the surface absorption causing higher volumes of run off, and ii) reduction of the concentration giving higher volume of run off. These aspects have to be carefully considered.

In rural areas, vegetation and afforestation have the effect of altering the catchment flood response. Studies in the USA have demonstrated that for an increase in afforestation on a small watershed, equivalent peak discharges are reduced gradually over a period of approximately of 25 years by an average value of about 60 per cent. The effects of rainfall induced floods have not been clearly understood. Different types of vegetation and rural land management cause differing effects in soil water consumption and retardance of run-off.

Settlements of Flood Plain

The height of land surface gradually increases above the river bank level as one goes farther away. Consequently, the flood hazard gets reduced as the distance of land from the river increases. However, for socio-economic reasons, the occupation of flood plains cannot be avoided. It is, therefore, advisable to exercise some control on building houses in such cases by confining the constructions to zones involving least risk. The concept of Flood Plain Zoning incorporates this idea.

By taking into account the variability of the flood hazard in river, zones are demarcated depending on the degree of risk and potential damage. Through legislation, the type and density of occupation could be controlled. Three zones are identified which are discussed below.

a) Prohibited Zone

This is the essential part of the flood way, the velocity and discharge of which contribute significantly to prohibited zone in the flood plain. Development in this zone of any kind should be totally disallowed to avoid damage to the property and also to avoid backlash flood effects up-stream, posing hazard to other settlements. The prohibited zone can be put to uses like cattle grazing, etc.

b) Restricted Zone

The area in the river bed where inundation is not too frequent and contributes small volume to the total flood discharge can be used as a restricted zone where the velocity of flow is low. Limited building-development and planned agricultural activity are feasible. However, restriction should be stipulated not only on the density and use but also on the design criteria. Minimum ground floor level, flood proofing arrangements, etc., should be considered for buildings.

c) Warning Zone

The inundation beyond the design flood level and up to the estimated maximum flood level is rare and, therefore, the potential disaster is negligible.

People who wish to settle in this zone should be warned and advised of the risk involved regarding the safe floor levels, etc. The choice of settling and developing will be their own and little or no restriction is imposed from the point of view of flood disaster prevention.

Building Regulations

Damage to individual buildings and structures may be prevented, to some extent, by incorporating in their design, the ability to withstand inundation and effect of high-water velocity. It is desirable to enforce some building regulations in this regard in order to see that the welfare of the individual as well as the community is protected. If such a control is exercised at regional and national levels, it would contribute substantially to mitigate flood disasters and save life and property in a large measure.

Flood Resistant House Construction

Floods affect the whole community and the related problems can be tackled adequately and satisfactorily only at the community level. Nevertheless, each aspect of flood disaster mitigation warrants its own special consideration and a solution which will at least partially meet the situation. The strengthening and protection of individual houses is one issue which should be given due priority and importance. Total flood proofing of dwelling would necessitate incorporation of strong reinforcing measures which will entail considerable cost. Such a type of construction is generally not within the means of majority of the people who are vulnerable to flood disaster.

However, the following steps could be taken to provide some protective measures to the houses which may at the least help to save lives:

- a) Good siting of houses should be aimed at to minimise risk of damage. The buildings should be constructed on the best bearing soil on the highest ground available.
- b) Ground drainage and escape lanes should be incorporated in the settlement layout.
- c) The houses may be raised on individual mounds which are prepared by spreading and thoroughly compacting soil. The mound should be constructed with suitable locally available soil. The roof level which is a critical factor during rising of flood water is raised adding a good measure of protection.
- d) The buildings may be designed in such a way that their roof spaces are above the designed flood level. The choice of roof type and area should be such that it gives protection for a period of at least 24 hours to the people who take shelter.
- e) It is a common practice to raise houses on stilts. However, such type of construction increases considerably the risk of damage/overturn in the event of a flood. Such buildings should therefore have a rigid frame construction for strength or adequately braced.

- f) The plinth should be preferably raised to a height of 0.75 m (minimum) so that the occupants of the houses get some breathing time in the event of flood waters surrounding the house and rising gradually.
- g) The mud houses should be preferably provided with pucca founda-

areas and most of them occupy non-engineered buildings which are built of low-quality materials. Such buildings are constructed by the people of through masons, carpenters and petty contractors having limited construction skills and knowledge. Even in the urban centres, many people live in 'kutchas' (temporary) houses, not so much as a matter of choice, but as a matter of economic compulsion. All these constructions are vulnerable to the hazards of cyclone. Based on the experience and analysis of the structural failure of non-engineered buildings during the past cyclones, the following guidelines are suggested for making common man's house cyclone resistant.

a) Location

Natural windbreaks such as trees or hedges should be taken advantage of when deciding on a site for building. Such a location can reduce the impact of prevailing winds, (Fig. 9.17).



Fig. 9.17 Protection to house by hillock/trees

b) Shape of Building

Based on the research done on the effect of geometry of a structure in withstanding wind forces, the following ratios are recommended (Fig. 9.18).

- I. Length Width = 1.5
- II. Height: Width = 1

Square houses are safer against winds. Pyramid-shaped deep steep roofs defy winds best. The following shapes should be avoided:

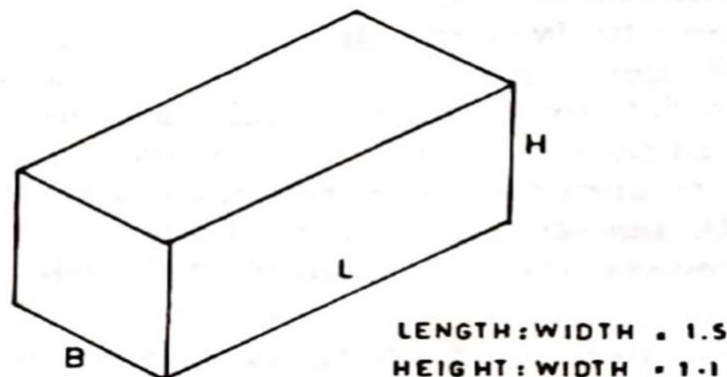


Fig. 9.18 Preferred shape of building in cyclone prone areas

- I. Very long dwellings
- II. 'L' shaped and zig-zag shapes
- III. Extra wings added to the main unit.

c) Orientation of Buildings

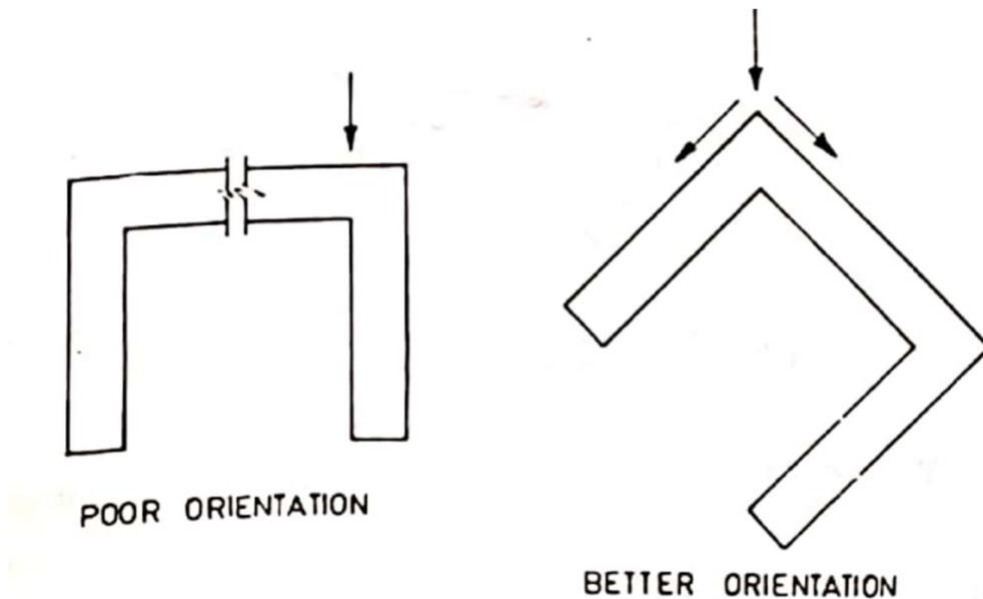


Fig. 9.19 Effect or orientation of building

The lay-out of building in relation to the direction from which the tidal wave is expected is an important factor. A wave striking at right angles can break a wall; at an oblique angle its force is dissipated (Fig. 9.19).

d) *Construction Details*

- I. It should be ensured that every part of the building is securely tied together; all roof elements, roof-to-wall, wall-to-wall, wall-to-floor and floor-to-foundation (Fig. 9.20).
- II. Reinforced masonry should be encouraged in cyclone prone areas. Where necessary, the walls should be reinforced with vertical steel reinforcement (Fig. 9.21)
- III. Horizontal reinforcement should be provided around corners between the intersecting walls and between columns and infill walls (Fig. 9.22).
- IV. Providing reinforced concrete band (ring beam on all external and internal load bearing walls) will greatly improve their individual stability as well as integral box like action and increase the stability of buildings to make them cyclone resistant to a great extent. Joint reinforcement should alternatively be provided at wall junctions, say in every fourth layer.

Fig. 9.22 Proving horizontal reinforcement around corners and between intersecting walls

- V. If a suspended floor construction is used, it should be ensured that the entire sub-structure is traced and positively anchored to the foundation (Fig. 9.23).
- VI. Large openings near the roof lines or near the corners of wall should be avoided (Fig. 9.24). These tend to weaken the structure; they are located in areas of greatest loading.
- VII. Bracing provides lateral stability to masonry walls (Fig. 9.25).

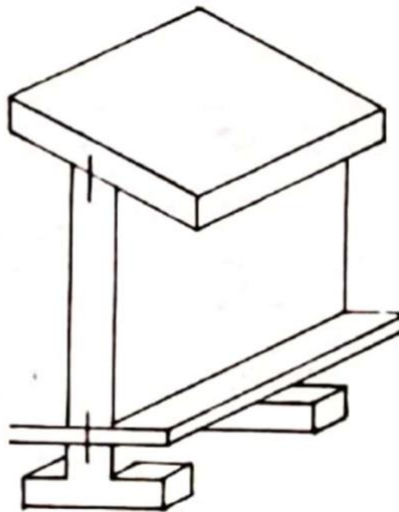


Fig. 9.20 Ensuring that every part of the building is securely tied together

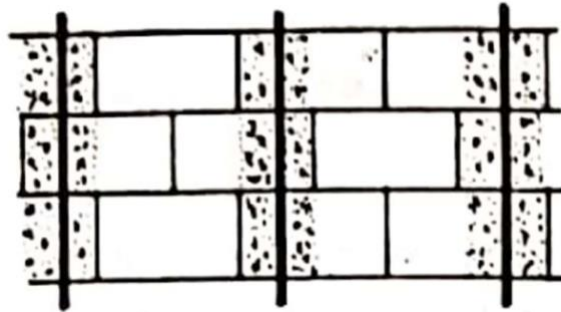


Fig. 9.21 Reinforcing masonry wall

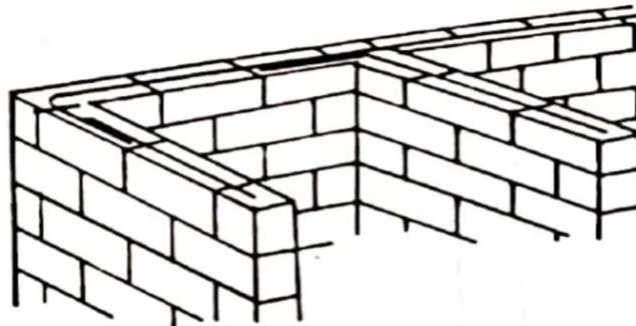


Fig. 9.22 Proving horizontal reinforcement around corners and between intersecting walls

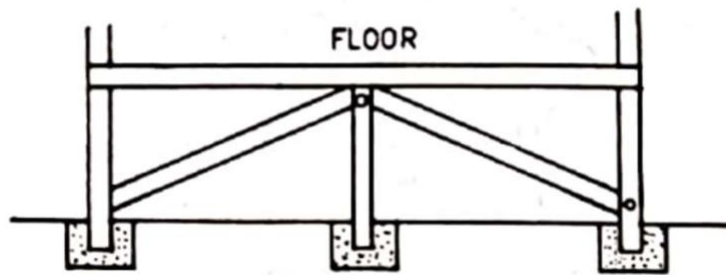


Fig. 9.23 Bracing the sub-structure of suspended floor

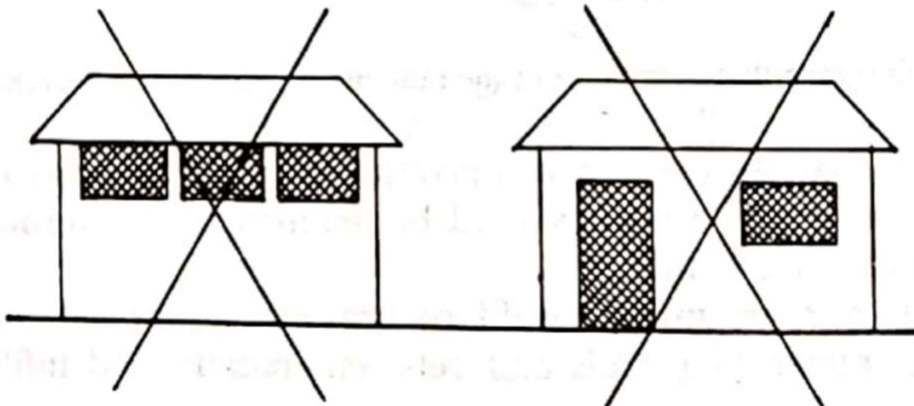


Fig. 9.24 Faulty placing of opening near roof lines and near the corners of walls

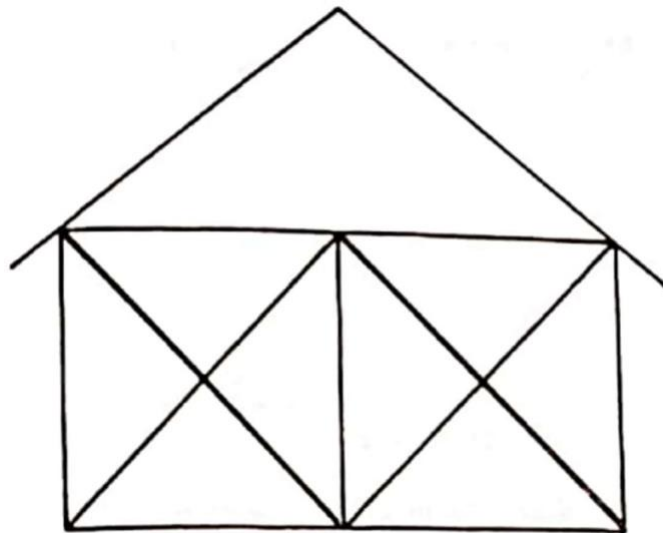


Fig. 9.25 Bracing of walls

e) Roof Construction

1. Whatever the form of construction it should be ensured to tie it securely to its supporting walls or posts (Fig. 9.26). Ignoring this caution is the greatest single cause of damage due to hurricane winds.

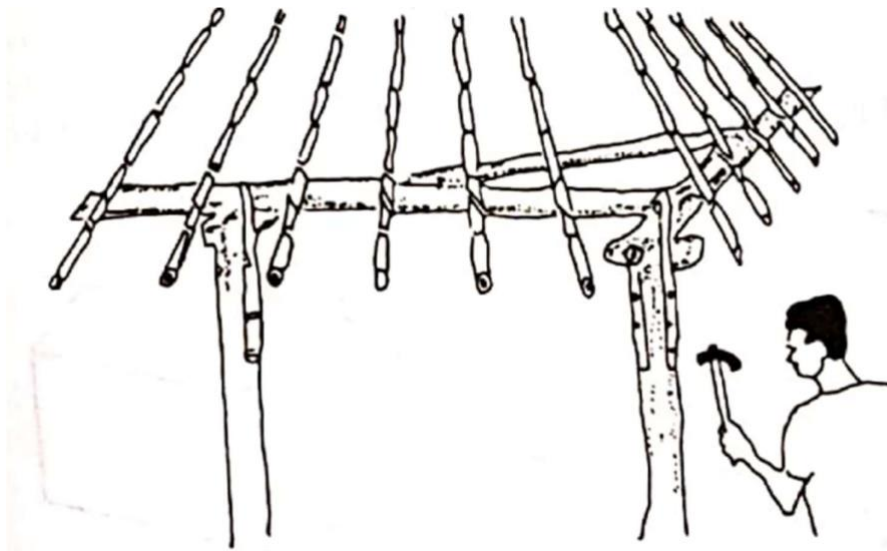


Fig. 9.26 Securing roof to supporting walls and proving cross pieces at corners

- II. The angle of roof is important in case of sloping roofs. The angle less than 30 produces suction of the wind on the roof and it can be blown away. An angle of more than 30 deflects the force of the wind (Fig. 9.27 and 9.28).
- III. A hip-angle roof is preferred to gable ends. Roof suctions are less in hip-angle roof (Fig. 9.29).
- IV. When connecting a timber roof to a masonry wall, a fastening strap a reinforcing bar should be used which is firmly embedded in the concrete or masonry (Fig. 9.30).

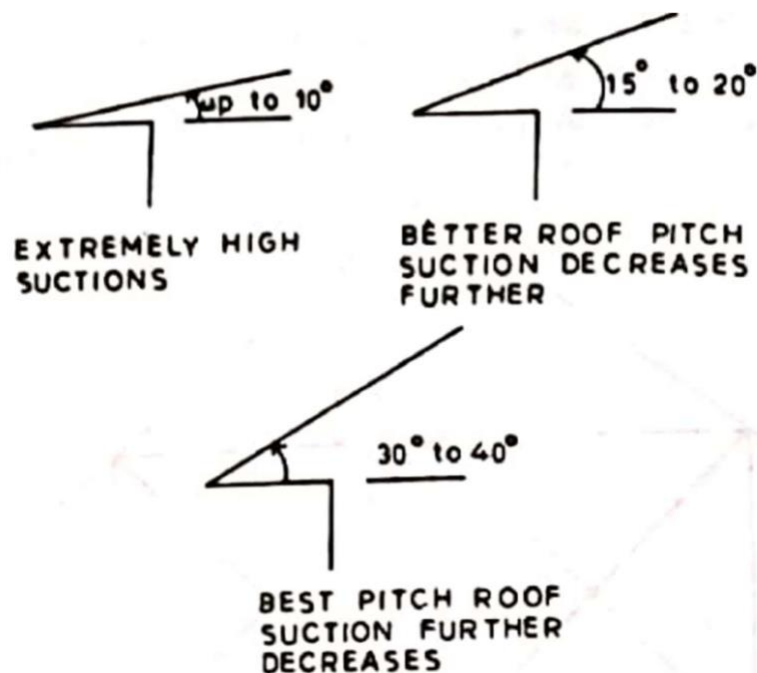


Fig. 9.27 Preferred roof slopes

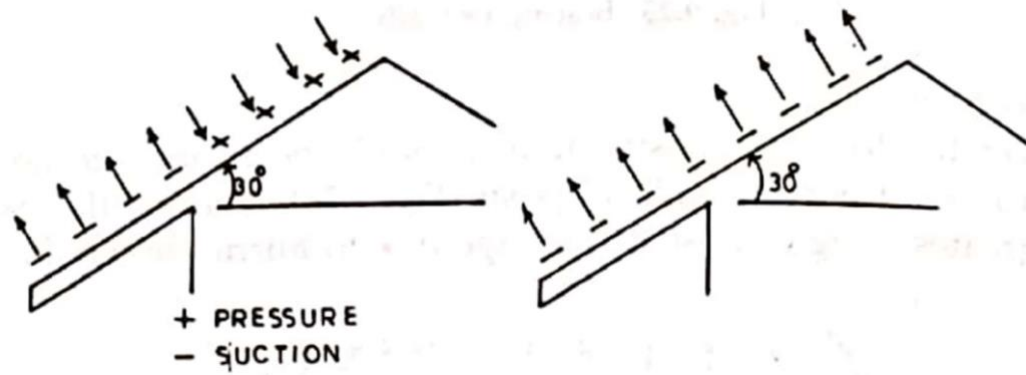


Fig. 9.28 Effects of roof slope on wind pressure

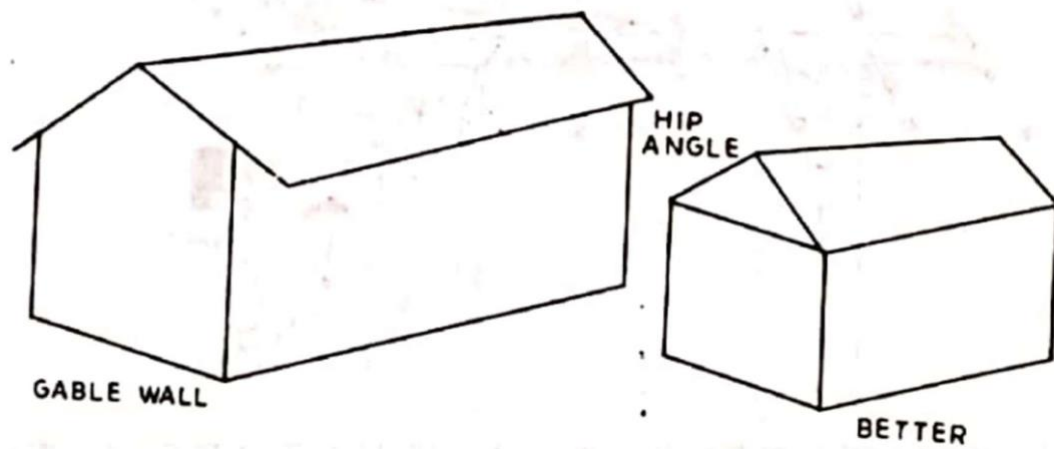


Fig. 9.29 HIP angle roof preferred to gable end

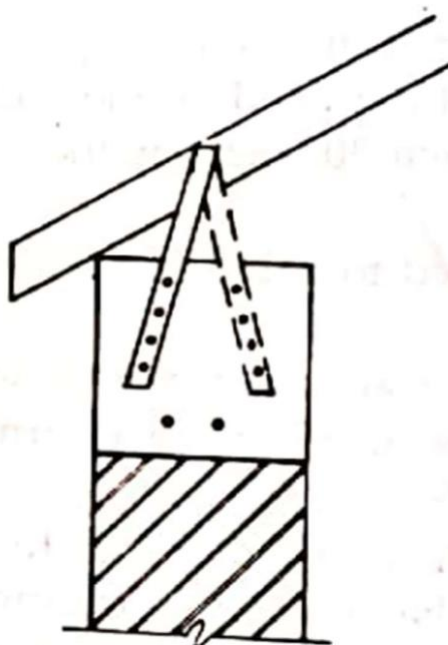


Fig. 9.30 Connecting timber roof to wall

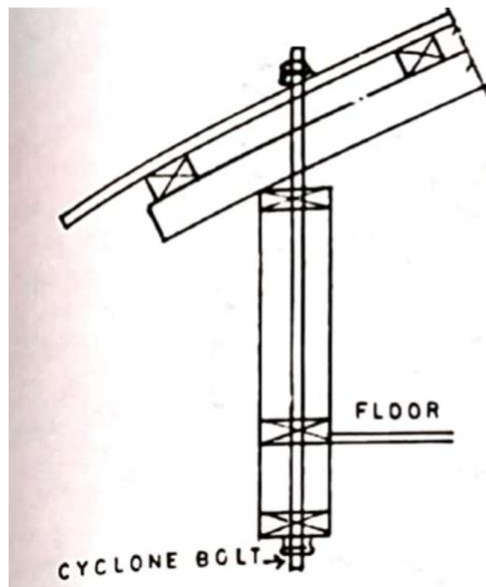


Fig. 9.31 Cyclone bolting

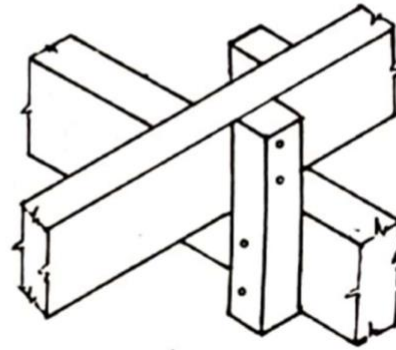


Fig. 9.32 Wooden Cleats nailed to purlin and rafter

- V. Failures by separation of the entire roof structure from the walls due to uplift and lateral forces have occurred during the past cyclones. It is, therefore, necessary to properly anchor the roof to the walls and then to the foundation. Cyclone bolts (Fig. 9.31) are provided to achieve structural continuity from the roof to the foundation.
- VI. All purlins should be tied to rafters by at least one and preferably two connectors. It should be ensured that nails are driven laterally, not in direct pull (Fig. 9.32 & 9.33). Various cyclone resistant joints and connections as shown in Fig. 9.33 to 9.36.

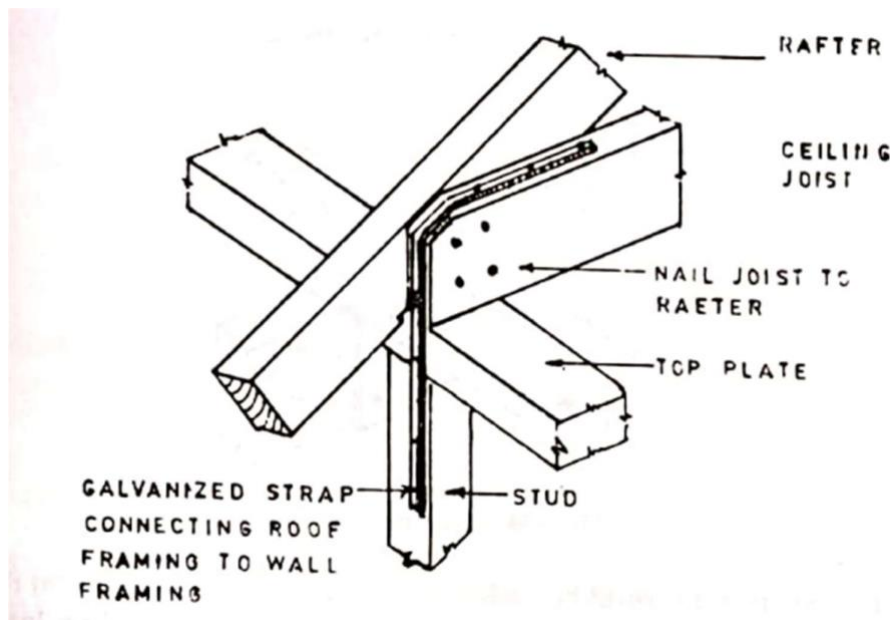


Fig. 9.33 Connection of roof frame

- VII. When nailing corrugated roof sheeting's to purlins, nailing should be done through the top of corrugations and a washer at least $3\frac{1}{2}$ " (18 mm) should be used (Fig. 9.37).

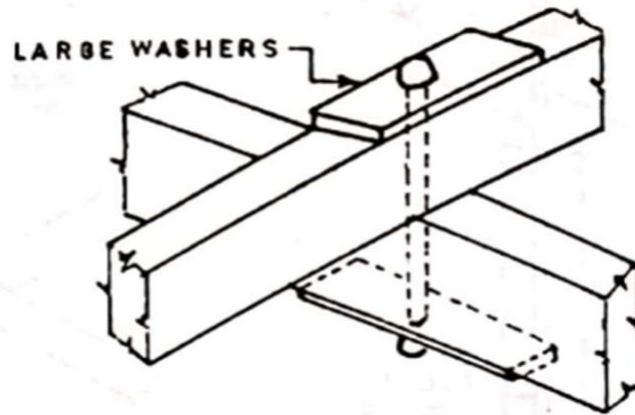


Fig. 9.34 Steel bolt fixing

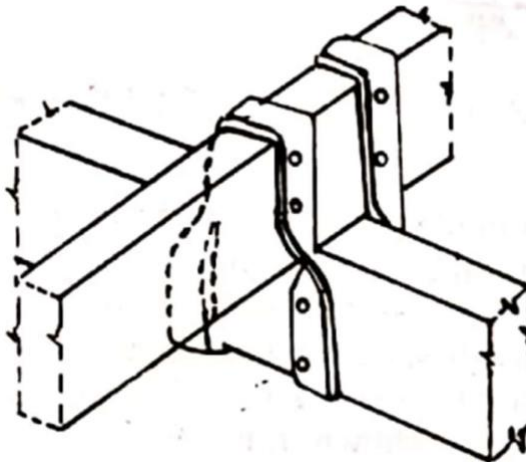


Fig. 9.35 Steel strap fixing

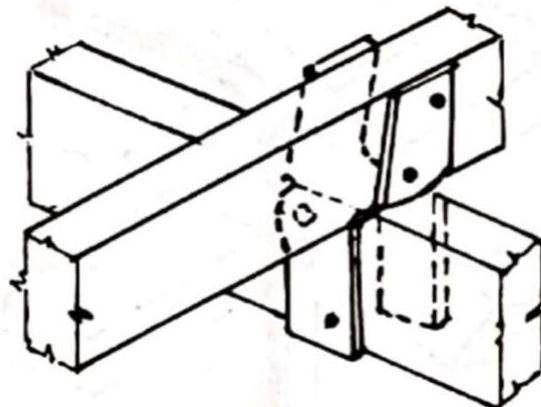


Fig. 9.36 Metal bracket fixing

- VIII. Every corrugation along the bottom purlin (at the lanes) should be nailed, and it should be nailed along the top of the purlin (at the edge) (Fig. 9.38). Ignoring these nailing recommendations often lead to the blowing away of the sheeting in the hurricane winds.

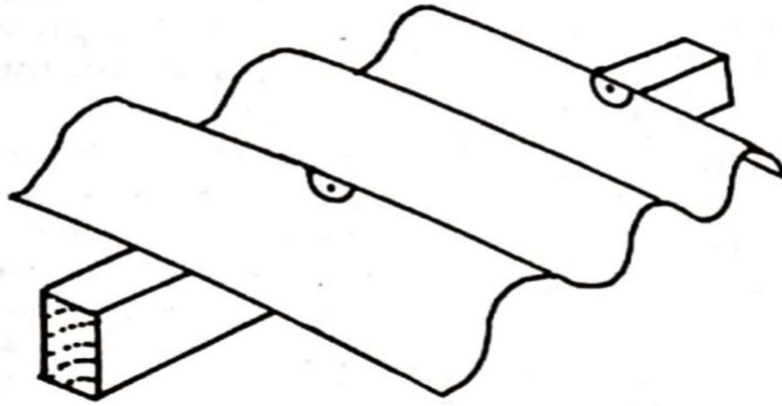


Fig. 9.37 Use of washer in nailing corrugated roof sheeting to purlins

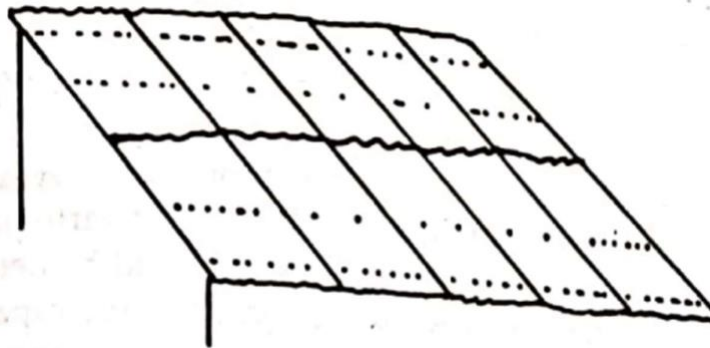


Fig. 9.38 Proper nailing of corrugated sheets

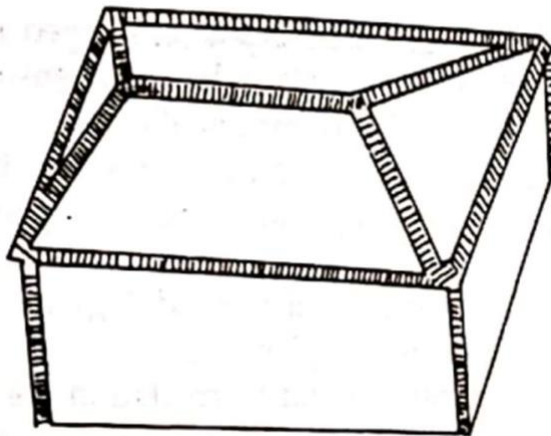


Fig.9.39 Provision of mortar bands on roof top

- IX. The probability of bamboo roofs covered with straw or palmyrah leaves being blown away could be greatly reduced by covering them with a form of netting made from customary local materials such as coconut fibre rope.
- X. A roof of clay tiles could be considerably strengthened and tiles prevented from blowing away by weighing it down with bands of mortar (Fig. 9.39).

The above measures could go a long way in making common man's house cyclone resistant.

Future Strategy

1. The Municipal Bye-laws and Building Regulations should incorporate mandatory provisions for strengthening of houses and buildings in areas prone to natural disasters. Strict enforcement of such provision's should be ensured. Existing Building By-laws and Regulations should be examined in depth to identify shortcomings in the provisions for adequate and satisfactory strengthening arrangements taking into account the type and the intensity of disaster experienced locally.
2. North India faces a serious threat of a devastating earthquake of probable magnitude eight or nine on Richter scale in near future as per the recent forecast of the groups of American and Indian scientists based on the detailed study of past India earthquakes and analysis of plate tectonics of the region backed up by satellite data. The region being a zone of high seismic risk and large number of buildings (around 1,00,000) in the Kobe, Japan earthquake (7.2 on Richter scale) in January, 1995 which claimed more than 5,400 human lives, provides a grim warning to the safety of houses and buildings in the event of occurrence of earthquake on the predicted lines in the northern part of India. With a view to minimising damage to structure during future earthquakes, all buildings in seismic prone zones should be designed and constructed for earthquake resistance as per the codal provisions of IS: 4326 and other IS codes recently brought out by the BIS on earthquake resistant construction.
3. Real seismic forces are much greater than provided in the design codes as revealed by the analysis of the Armenian earthquake. As such, all members of earthquake resistant structures should be detailed to perform in the inelastic range, well beyond the normal elastic capacity.
4. Like unreinforced masonry, precast constructions run high risk in the event of earthquakes. In view of the large-scale collapse of prefabricated buildings in the Armenian earthquake, future precast constructions in the country must take adequate precautions to ensure the integrity of precast elements. All precautionary measures as envisaged in the second revised edition of IS:4326 should be strictly followed to ensure structural stability of precast constructions in future earthquakes.
5. Micro-zoning pattern should be evolved for disaster prone areas to provide guidelines for planning and siting of housing and human settlements.
6. Guidelines for working out methodology of vulnerability analysis should be prepared for land-use planning.
7. Cyclone shelter belts should be created in the coastal areas by tree plantation. Emergency cyclone shelters should be constructed on a large scale in the coastal areas.

8. Much attention is required to be paid to drainage in forming new settlements, and escape lanes should be incorporated in the lay-out for rapid flow-out of water.
9. The importance of flood plain zoning should be recognised for orderly settlement free from undue hazards. Survey should be undertaken in flood prone areas to demarcate prohibited restricted and warning zones and the inhabitants should be made aware of these zones with proper demarcation. The concerned authorities should develop flood-plain zoning maps which should be implemented in establishing human settlements.
10. Well-designed and maintained drainage system should be provided in towns to avoid local flooding.
11. Growing of regular plantation in the catchment area on the banks of the rivers as well as coastal areas, should be encouraged to retard run-off.
12. Plants and shrubs should be grown around the house as they provide a measure of protection by breaking the force of gushing water.
13. People should be educated about the importance of keeping the drains clean in order to avoid choking.
14. With a view to creating awareness among masses more TV and video films on natural calamities should be prepared on the lines of the one prepared by the NBO on earthquake, and shown on the national network of Doordarshan. Natural disaster should form a part of the syllabi of school and college level education. In areas prone to severe earthquakes, mock exercises should be carried out as is regularly done in the quake prone areas of Japan

Social Housing Programmes

Introduction

With a view to meet the housing needs of the masses in the country, the erstwhile Union Ministry of Works and Housing had introduced nine social housing schemes (Table 1) between 1952 and 1971. All the schemes except Subsidised Housing Scheme for Plantation Workers are in the State sector, and are implemented by the respective State governments and Union Territories (UTs) which have been invested with full powers to formulate, sanction and execute projects under these schemes and to release financial assistance to the concerned construction agencies. In the August 1982, the said Ministry under the Union Government in consultation with the Planning Commission, had re-classified (Annexure II) the then existing social housing schemes into 4 categories namely, EWS, LIG, MIG and Rental Housing Schemes. The scheme for EWS which was initiated in 1952 as a part of the LIG Housing Scheme was integrated with the Subsidised Housing Scheme for Industrial Workers (SIHS) in April 1966. The same was, however, delinked from SIHS, and given an independent status under the reclassified pattern. The Rental Housing Scheme was introduced in 1959 with a view to assist the State governments to provide adequate housing accommodation to their own employees. Income levels for first three categories of housing schemes were fixed at upto Rs. 350 per month (p.m.), Rs. 350-600 p.m. and Rs. 601-1500 p.m. respectively. There has been revision in the income eligibility for the EWS, LIG and MIG categories from time to time. The income ceiling during the Eighth Plan (1992-97) for the EWS category is Rs 1250 p.m. whereas for the LIG and MIG categories the income eligibility is in the range of Rs 1251-2650 p.m. and Rs 2651-4450 p.m. respectively.

The HUDCO provides bulk of the finance for the social housing schemes. Over 88% of the housing units promoted by the Corporation are meant for the LIG and EWS categories. The current loan ceiling for the EWS category is Rs 19,500 for house construction, and Rs 9,500 for undertaking repairs. The corresponding figures for the LIG category are 85% of the cost of house-construction or Rs 42,000 whichever is less, and Rs 21,000 respectively. For the MIG category, the loan ceiling for house-construction is 75% of the cost of construction or Rs 1,15,000 whichever is less, and Rs 55,000 for repairs.

In respect of the Central Sector Scheme of Subsidised Housing for Plantation Workers and the State Sector Scheme of 'Rural House Sites-cum-Hut Construction for Landless Workers' there has, however, been no change in their status till date (1994).

Table 1
Social Housing Schemes

S.NO	Name of the Scheme	Year of Introduction
1.	Integrated Subsidised Housing Scheme for Industrial Workers and Economically Weaker Section (EWS) of Community	1952
2.	Low Income Group (LIG) Housing Scheme	1954
3.	Subsidised Housing Scheme for Plantation Workers	1956
4.	Slum Clearance/Improvement Scheme	1956
5.	Village Housing Project scheme	1957
6.	Middle Income Group (MIG) Housing Scheme	1959
7.	Rental Housing Scheme for State Government Employees	1959
8.	Land Acquisition and Development Scheme	1959
9.	Rural House Sites-cum-Hut Construction Scheme for Landless workers	1971

At present only the following Social Housing Schemes are operational in the State sector with State Plan provisions and loan assistance from HUDCO and other financial institutions:

1. Housing Scheme for the EWS
2. Housing Scheme for the LIG
3. Housing Scheme for the MIG
4. Rental Housing Scheme for the State Government Employees.
5. Rural House Sites-cum-construction Assistance Scheme for the Landless Workers.

Implementing Agencies for Social Housing Programmes

In most of the States, Housing and Urban Development Departments of the respective State governments are responsible for the implementation of all the housing schemes under the reclassified categories. In case of the U.Ts, the implementing agencies are either Public Works Department or Revenue/Planning Departments which are also engaged in monitoring of these schemes. Besides, the Housing Development Boards of the respective State governments/U.T. Administrations also look after the progress in respect of implementation of these schemes.

Performance of Social Housing Schemes

A review of the performance of the social housing schemes during the Seventh Five-Year Plan (1985-90) undertaken by the NBO reveals that in case of LIG Housing Schemes, the achievements outnumbered the targets, whereas about 89% of targets under the EWS scheme, and 65% under the rental Housing Scheme were achieved during the aforesaid period.

Tamil Nadu and Andhra Pradesh were the leading States in respect of providing houses in the EWS category during the seventh plan-period. Six States and U.T. outnumbered the targets for the plan under the EWS category. Karnataka, Kerala and Gujarat were in the forefront in terms of providing houses to the people in the LIG category. Ironically, performance of Delhi U.T. has been far below the expectation during this period. Inability of the administration to provide housing to the people in the LIG and EWS categories is an important factor in the deteriorating housing situation in the national capital.

Typical Specifications for Low-cost Housing: Urban

For the guidance of prospective builders and the agencies entrusted with the execution of mass housing programmes in the urban areas in the country, typical alternative specifications for low-cost housing to be adopted, are provided in Annexure I.

Annexure I

Typical Alternative Specifications for Low-Cost Housing in Urban Areas

1. Foundation and Plinth

a) Base Concrete

The base concrete may be 100 to 150 mm thick consisting of:

- i. Lime Concrete 1:2:6 (1 lime: 2 fine aggregate i.e, surkhi, sand :6 coarse aggregate - i.e., stone or brick aggregate) (Please see note 1).
- ii. Cement Concrete 1:5:10 (1 cement: 5 fine aggregate: 10 coarse aggregate stone or brick) (Please see note 2).

b) Masonry

- i. Brick masonry in cement mortar 1:6 (Please see note 3) Cement lime mortar 1: 1: 8; Lime mortar 1: 2-3; Lime mortar 1:1:8; Lime mortar 1:2:3 (Please see notes 4 and 5).

2. Damp Proof Course

- a) 2 cm. thick Cement lime plaster (1 cement: 1 Lime: 5 Coarse Sand) with one or two coats of Bitumen. (Please see note 6).
- b) 2 ½ to 4 cm. thick coat of Cement concrete 1:2:4 with two coats of Bitumen (Please see note 7).
- c) 4 cm. thick coat of Cement concrete 1: 1 ½: 3 with integral waterproofing compound (Please see note 8).

3. Walls

- a) Brick masonry
- b) Coursed stone masonry/Precast stone masonry blocks/Random rubble stone masonry (Please see note 9 below).

Cement mortar 1:6;

Cement lime mortar 1: 1 :6-9/1 :2 :9-12/1 :3 :12-15:

Lime mortar 1: 2-3.

(Please see note 10).

Mud mortar with 2% lime.

(Please see note 11).

4. Arches/Lintels

- a) Flat or segmental arch in brick work up to 1.2 metre span.
- b) Precast RC thin lintels in M-150 concrete up to 1.8 metre span ensuring composite action between lintel and brick work above.
- c) Stone patties of suitable size (where available for small spans).
- d) RC Lintels suitably designed for any span length

5. Structural Floor/Roof

- a) Brick tiles on precast RC or timber battens/floor.
- b) Timber joists and planking.
- c) Trusses in Timber/ RC/ Steel with AC/ CGI sheets/ Corrugated asphaltic sheets /tiles /Slates /Shingles.
(Please see note 12).
- d) Reinforced brick or Reinforced brick concrete slabs.
(Please see note 13).
- e) Stone patties of required thickness for spans up to 3.2 m. where available at reasonable cost.
- f) Precast units like channel units, cellular units, doubly curved tiles, solid RC planks, reinforced brick panels on precast RC joists
(Please see note 14).
- g) Reinforced cement concrete slab.

6. Doors and Windows

- a) Frames
 - i. Secondary species of timbers locally available after proper seasoning and chemical treatment (Please see note 15).

- ii. Angle iron, tee iron or pressed steel.
- b) Shutters
 - i. Secondary species of locally available timbers after proper seasoning and chemical treatment, of thickness around 3.5 cm's.
 - ii. Pressed steel.
 - iii. MDF board shutters.
- C) Fittings
 - i. MS Black Japan painted fittings.
 - ii. Anodised aluminium fittings.

7. Flooring

- a) *Base for ground floor*
 Rammed earth followed by 10 to 15 cm of sand filling and 10 cm of lean cement or lime concrete.
- b) Wearing Course
 - i. 2.5 to 3.5 cm thick Cement concrete 1:2:4.
 - ii. Brick/brick tile flooring laid in cement/lime mortar.
 - iii. Stone slabs.

8. Finishing

- a) External rendering on walls
 - i. Cement lime mortar 1:1:6-8
 - ii. Cement mortar 1:6
 Rough cast cement plaster in two coats.
 (Please see notes 16 and 17).
- b) Internal rendering on walls
 - i. Cement lime mortar 1:1:8
 - ii. Lime Surkhi mortar 1: 1: 2 (1 lime: 1 surkhi: 2 sand)
 - iii. Cement mortar 1:6. (Please see notes 18 and 19).
- c) Finish on walls
 - i. Internal-White wash
 - ii. External Colour wash
- d) Finish on wood work and iron work
 - i. One coat of primer plus two coats of painting.

9. Water Proofing

Bitumen coating at 17 kg. per 10 m² impregnated with a coat of coarse sand at 60 dm³ per 10 m² with

- a) residual type petroleum bitumen of 80/100 grade or;

- b) with hot outback bitumen followed by:
 - i. Mud-phaska with tiles if the terrace is used or
 - ii. Mud-phaska finished with mud plaster and gobri leaping if the terrace is not to be used or
 - iii. Lime concrete
(Please see note 20).